

# **Causal Reasoning with Statistics**

*Design, Identification, and Reproducible Evidence in Psychology  
and the Social Sciences*

Nicholas Elliott Karlson

NEKpress Research: Proof-First Guides

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NEKpress Research

## **Causal Reasoning with Statistics: Design, Identification, and Reproducible Evidence in Psychology and the Social Sciences**

Nicholas Elliott Karlson

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The companion data and code are public and free at [github.com/nicholaskarlson/data](https://github.com/nicholaskarlson/data), in the directory `open-materials/causal-reasoning-with-statistics`.

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Version 1.0. All ten units of this open educational resource are complete and computationally verified: every printed number is reproduced by a primary NumPy path, an independent standard-library Python path, and a base-R path, and the public companion carries the data and the code needed to repeat them. This edition has not been through external methods review; its version DOI is 10.5281/zenodo.22965577. All participants, municipalities, responses, and results are synthetic. Results are demonstrations, not findings about people or places.

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## This book is free, and so is everything behind it

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A free copy of this edition is published on the internet alongside its data and code, so buying this one is a convenience rather than a toll gate. If you would rather read the free copy, take it; if you paid for this one, thank you, and everything below is yours to use either way.

Every number printed in this book can be recomputed on your own machine from files that are public and free. They live in the companion repository at <https://github.com/nicholaskarlson/data>, in the directory `open-materials/causal-reasoning-with-statistics`. There is nothing to register for, no account to create, and no commercial software to buy. The address is short enough to type from the page, and the chapter called “Reproduce and extend this book” near the end explains what each step proves. The short version follows.

Open that address in any browser, choose the green Code button and then Download ZIP, and unzip it. In the companion directory, `python3 run_all.py` regenerates the ten synthetic datasets from their seeds, recomputes every published result, and cross-checks the whole set against a second implementation written in plain Python with no libraries; `Rscript run_all.R` repeats the checks a third time in base R. Both commands leave the files exactly as you downloaded them. If you prefer to read rather than run, every dataset is a plain CSV and every verified result is a small JSON file you can open in a text editor.

If a number in this book ever disagrees with what the companion computes, the disagreement is a defect and I want to know about it. Open an issue in that repository with the file name, the command you ran and the output you saw.

## About this book

Statistics can describe patterns and produce accurate forecasts without answering what would happen under an intervention. Causal reasoning begins by naming the decision: *What changes, for whom, compared with what, at what time, and under which assumptions?* A design and its substantive assumptions determine whether observed outcomes identify that contrast. Statistical models then estimate it and express a limited form of uncertainty.

The ten units establish that workflow. Unit 1 separates descriptive, predictive, and causal questions in a randomized interface experiment. Unit 2 shows that intervention, assignment, treatment receipt, and target population determine whether an estimate is an ATE, ATT, ITT, or transported contrast. Unit 3 uses a causal diagram to distinguish a baseline confounder from a mediator and collider. Unit 4 follows a blocked cluster-randomized experiment through assignment, receipt, baseline adjustment, clustering, and outcome-specific attrition. Unit 5 standardizes an observational comparison with outcome regression and inverse-probability weighting, then uses overlap, balance, weights, and a negative-control outcome to expose what measured adjustment cannot prove. Unit 6 uses a phased municipal policy to compare changes over time, audit parallel-trends reasoning, and separate a policy contrast from a same-month regional interruption. Unit 7 pairs a diagnostic-score cutoff with a separate randomized encouragement to distinguish a local eligibility effect from a complier receipt effect. Unit 8 returns to a randomized cognitive-training trial to separate total effects, prespecified baseline heterogeneity, and post-assignment mediation claims. Unit 9 follows a binary outcome through observation, measurement, and transport to a differently composed target population. Unit 10 makes residual-confounding assumptions quantitative, distinguishes triangulation from repeated specifications, separates reproducibility from replication, and plans cluster-aware precision.

Every completed case uses entirely synthetic data. The printed numerical results can be independently reconstructed from the included CSV files by a primary NumPy path, a separate standard-library Python calculation, and base R scripts required by continuous integration. Agreement among programs verifies arithmetic on these files; it does not prove a causal graph, valid measurement, transportability, or relevance to real people. Hidden simulator parameters test the teaching machinery and are never treated as evidence available to the analyst.

## The recurring causal workflow

1. **Question.** Separate description, prediction, and intervention. Define the decision and time zero.
2. **Estimand.** Name the intervention, comparison, unit, outcome time, target population, and averaging rule.
3. **Identification.** Use assignment, design knowledge, timing, and explicit assumptions to connect observed data to the estimand.
4. **Estimation.** Choose an estimator and uncertainty calculation suited to the identified comparison.
5. **Claim Audit.** State the strongest supported conclusion, exclusions, vulnerabilities, and next informative study.

An impressive coefficient cannot rescue a vague intervention. Randomization does not transport itself. Regression adjustment does not prove exchangeability. A narrow interval does not contain

uncertainty about the graph or construct. The workflow keeps these tasks separate so that each can be criticized and improved.

## How to use the book

Read the scenario and write your own scientific question before accepting the printed one. At every **Pause and Decide**, record **answer, reason, confidence (1–5), and revision**. The revision is the cue you missed or the next comparison you would check. Attempt the six principal problems before reading the worked solutions. Each solution explains a decision and its claim ceiling, not merely the arithmetic.

Readers new to causality should follow Units 1–10 in order. Readers evaluating a study can begin with the recurring workflow, then use the unit matching its design. Instructors may use one case at a time, but should retain the Claim Audit and Study in the Larger Evidence Program so that favorable results are not mistaken for completed scientific evidence.

## Contents and completion status

### Part I. Asking causal questions

1. From Association to Intervention — **complete**.
2. Estimands, Populations, and Claim Ceilings — **complete**.
3. Causal Diagrams and Adjustment Decisions — **complete and reader-calibrated**.

### Part II. Designs that support causal inference

4. Randomized Experiments — **complete blocked cluster-randomized case**.
5. Observational Adjustment and Standardization — **complete voluntary public-health case**.
6. Changes Over Time: Difference-in-Differences and Interrupted Time Series — **complete phased municipal-policy case**.
7. Thresholds and Encouragements: Regression Discontinuity and Instrumental Variables — **complete two-caselet local-design unit**.

### Part III. From one estimate to trustworthy evidence

8. Mechanisms, Mediation, and Effect Heterogeneity — **complete stratified cognitive-training case**.
9. Selection, Missingness, Measurement, and Generalization — **complete multisite binary-outcome replication**.
10. Sensitivity, Triangulation, and Reproducible Causal Analysis — **complete cross-case synthesis**.

A method chooser and a glossary precede Unit 1; an assumptions index, scope-and-omissions, references, and reproducibility guidance conclude the book. The ten-unit manuscript is complete in scope.

## Working notation

Let  $A$  denote an intervention or assigned condition and  $Y(a)$  the outcome the same unit would have under condition  $a$ . The individual contrast  $Y(1) - Y(0)$  is generally unobservable because a unit cannot

occupy both conditions at the same time. An average treatment effect,  $E[Y(1)-Y(0)]$ , is incomplete until its intervention versions, outcome time, and target population are named. Later units introduce  $Z$  for randomized assignment or encouragement,  $D$  for treatment received,  $X$  for pre-exposure covariates, and  $S$  for site or selection.

Notation organizes the question; it does not identify the answer. The meaning of a symbol comes from the study protocol, timing, and assumptions.

## Method chooser: from design to defensible claim

This list locates the unit that matches a design you are facing. It does not license a claim. Each entry names the estimand a design can support, the assumption that carries the identification, the estimator this book applies, and the diagnostic that could embarrass the result.

Read an entry through before choosing an estimator. The estimator is defensible only when the assumption named beside it is defensible, and the diagnostic can only fail to discredit that assumption. None of them can establish one.

**Randomized assignment, outcomes observed for nearly everyone.** Estimand: Average treatment effect in the randomized sample; Identification rests on: Randomization, consistency, no interference; Estimator here: Difference in means with HC1 standard errors; Diagnostic: Balance table; attrition audit. See Unit 1, 4.

**Randomized assignment, incomplete receipt.** Estimand: Intention to treat for assignment; local effect for compliers; Identification rests on: Randomization for ITT; exclusion and monotonicity for the complier effect; Estimator here: ITT contrast; Wald ratio with influence-function standard error; Diagnostic: Receipt rates by arm; first-stage strength. See Unit 4, 7.

**Randomization at the level of groups.** Estimand: Average effect with group-level dependence; Identification rests on: Randomization performed on the clusters; Estimator here: Cluster-level means with HC1 and a  $t$  multiplier on cluster degrees of freedom; Diagnostic: Floor-level versus individual-row uncertainty; cluster-size balance. See Unit 4.

**Observational, confounders measured, arms overlap.** Estimand: Standardized average effect in the cohort or among participants; Identification rests on: Conditional exchangeability, positivity, consistency; Estimator here:  $g$ -computation with the delta method; Hajek weighting with a stacked sandwich; Diagnostic: Overlap and weight diagnostics; balance; negative control. See Unit 5.

**Observational, the target population differs from the sample.** Estimand: Transported average effect in the named target; Identification rests on: The above, plus transportability of conditional effects given the covariates used; Estimator here: Target standardization with target cell weights; Diagnostic: Sample-versus-target composition table. See Unit 2, 9.

**Panel data, policy begins on one common date.** Estimand: Average effect on adopters over the post-period; Identification rests on: Parallel untreated trends; no anticipation; Estimator here: Two-way fixed effects with cluster-robust variance and a  $t$  multiplier; Diagnostic: Placebo adoption date; pre-period differential slope. See Unit 6.

**One series interrupted at a known date.** Estimand: Level and slope change at the interruption; Identification rests on: Correct interruption timing; no co-occurring change; Estimator here: Segmented regression with Newey-West standard errors; Diagnostic: Shared-interruption reasoning; lag sensitivity. See Unit 6.

**Treatment assigned by a threshold on a running variable.** Estimand: Local effect at the cutoff; Identification rests on: Continuity of potential outcomes through the threshold; Estimator here: Local linear regression fitted on each side; Diagnostic: Bandwidth sensitivity; density or bunching count at the cutoff. See Unit 7.

**Randomized encouragement to take up a treatment.** Estimand: Local effect for compliers; Identification rests on: Relevance, exclusion, monotonicity; Estimator here: Wald estimator with influence-function standard error; Diagnostic: First-stage strength; implied principal-stratum shares. See Unit 7.

**A mechanism question inside a randomized trial.** Estimand: Total effect and prespecified subgroup contrasts; direct and indirect effects only under further assumptions; Identification rests on: Randomization for total effects; sequential ignorability and cross-world conditions for mediation; Estimator here: Prespecified interaction models; mediation reported as not identified; Diagnostic: Mediator-outcome confounding audit; collider warning. See Unit 8.

**Binary outcome where the scale is a choice.** Estimand: Risk difference or risk ratio, each standardized; Identification rests on: Whatever the design already supplies; Estimator here: Standardized marginal risks, then the contrast; Diagnostic: Non-collapsibility check across scales. See Unit 9.

**Outcomes missing or measured with error.** Estimand: Bounds first; assumption-based estimates second; Identification rests on: Nothing for the bounds; missing at random given the conditioning set for the estimates; Estimator here: Cell-weighted complete case; Manski-type bounds; Diagnostic: Observation-rate table by cell; sensitivity and specificity reasoning. See Unit 9.

**Unmeasured confounding is plausible and cannot be removed.** Estimand: A bias-adjusted effect conditional on stated parameters; Identification rests on: The assumed confounder gap between arms and its outcome association; Estimator here: Two-parameter bias function; negative-control association; Diagnostic: Calibrated versus structural comparison; provenance of the parameters. See Unit 10.

**Planning a study that does not exist yet.** Estimand: Minimum detectable effect for a proposed design; Identification rests on: Assumed standard deviation, intraclass correlation, allocation, analysis model; Estimator here: Design-effect formula for the minimum detectable effect; Diagnostic: Intraclass-correlation sensitivity; what the formula omits. See Unit 10.

# Glossary of recurring terms

Definitions here are working definitions for this book, phrased to match the way each term is used in the units rather than to survey every usage in the literature. Terms appear in alphabetical order.

**Assignment mechanism.** The process determining which units receive which condition; randomization makes it known by design rather than assumed.

**Average treatment effect (ATE).** The mean contrast of potential outcomes over a stated population.

**Average effect on the treated (ATT).** The same contrast averaged only over units that received the treatment.

**Backdoor path.** A non-causal path from treatment to outcome beginning with an arrow into the treatment; blocking one is what adjustment is for.

**Bandwidth.** The window of the running variable a local estimator uses; it is part of the estimand, not a tuning detail.

**Bias function.** An explicit formula giving how far an estimate would be off under stated unmeasured-confounding parameters.

**Claim ceiling.** The strongest statement a design can support even when the estimation is flawless.

**Cluster-robust variance (CR1).** A variance estimator treating each group as the unit of dependence, with a finite-sample correction.

**Collider.** A variable with arrows into it from two others; conditioning on it can create an association that is otherwise absent.

**Complier.** A unit that takes the treatment when encouraged and not otherwise; the group a local average treatment effect describes.

**Conditional exchangeability.** Within levels of the measured covariates, treated and untreated units have the same distribution of potential outcomes.

**Consistency.** A unit's observed outcome equals its potential outcome under the condition it received; this requires a well-defined intervention.

**Controlled direct effect.** The effect of treatment when the mediator is fixed at the same value for everyone.

**Delta method.** A first-order approximation used here to obtain a standard error for a function of fitted coefficients.

**Design effect.** The factor by which clustering inflates variance relative to an independent sample; one plus  $(m - 1)$  times the intraclass correlation for equal clusters of size  $m$ .

**Differential misclassification.** Measurement error whose rates differ by arm; it can move a contrast in either direction and is not averaged away by sample size.

**Estimand.** The quantity a study is trying to learn, stated before any estimator is chosen.

**Estimate.** The number an estimator returns in one sample.

**Estimator.** The rule applied to data to approximate an estimand.

**E-value.** The minimum strength of unmeasured confounding, on the risk-ratio scale, that could explain away an observed association.

**Exclusion restriction.** The assumption that an instrument affects the outcome only through treatment receipt.

**First stage.** The estimated effect of the instrument on receipt; a weak one makes a ratio estimator unstable.

**g-computation.** Standardization performed by fitting an outcome model and averaging its predictions under each intervention.

**Hajek weighting.** Inverse-probability weighting normalized by the sum of weights within each arm.

**HC1.** A heteroskedasticity-consistent variance estimator with a degrees-of-freedom correction.

**Identification.** The argument that a target quantity is recoverable under stated assumptions, settled before any estimation.

**Influence function.** A per-unit contribution to an estimator, used here to obtain standard errors for ratio estimators.

**Instrument.** A variable that shifts receipt, is unrelated to the outcome's other causes, and acts only through receipt.

**Intention to treat (ITT).** The contrast defined by assigned condition regardless of what was received.

**Intraclass correlation (ICC).** The share of outcome variance lying between groups rather than within them.

**Local average treatment effect (LATE).** The average effect among compliers, and among no one else.

**Manski-type bounds.** The range a contrast could take when missing outcomes are filled with their most extreme possible values.

**Minimum detectable effect (MDE).** The smallest true effect a design would detect with stated power and significance level.

**Monotonicity.** The assumption that encouragement never moves a unit away from treatment.

**Natural direct and indirect effects.** A decomposition using the mediator value a unit would have had; identified only under cross-world assumptions.

**Negative control.** An outcome or exposure that should show no effect if the design is sound; an association there signals residual bias.

**Newey-West.** A standard-error estimator robust to heteroskedasticity and autocorrelation up to a chosen lag.

**Non-collapsibility.** The property that a marginal risk ratio or odds ratio need not equal a common conditional one even without confounding.

**Parallel trends.** The assumption that treated and comparison groups would have changed identically without the intervention.

**Positivity (overlap).** Every covariate stratum used has a non-trivial probability of both conditions; without it, standardization extrapolates.

**Potential outcome.** The outcome a unit would have under a specified intervention, only one of which is ever observed.

**Principal strata.** Groups defined by receipt under both assignments: always-takers, compliers, never-takers, defiers.

**Propensity score.** The probability of treatment given measured covariates.

**Running variable.** The continuous score whose threshold determines eligibility in a discontinuity design.

**Sandwich estimator.** A variance estimator built from a derivative matrix and a matrix of squared contributions; used here to carry propensity-model uncertainty into the effect.

**Sensitivity parameters.** In Unit 10, the standardized confounder gap between arms and the confounder's association with the outcome.

**Sequential ignorability.** The pair of no-confounding conditions, for treatment and for the mediator, that mediation analysis requires.

**No interference.** One unit's outcome does not depend on other units' assignments.

**Target standardization.** Reweighting covariate-specific effects to a target population's composition.

**Transportability.** The assumption that conditional effects estimated in the sample apply in the target given the covariates used.

**Triangulation.** Comparing evidence whose biases differ in structure; repeating a specification is not triangulation.

**Two-way fixed effects.** A regression including unit indicators and time indicators.

# Unit 1 From Association to Intervention

## Purpose and objectives

This unit establishes the question discipline used throughout the book. By the end, you should be able to distinguish descriptive, predictive, and causal questions; define a manipulable intervention and its comparison; state a potential-outcome estimand; explain why one person's two potential outcomes cannot both be observed; use random assignment to justify an intention-to-treat comparison; and keep post-assignment behavior from silently replacing the intervention of interest.

## The synthetic library-interface study

A fictional university library tests two versions of its search-results interface. Eligible students complete one standardized research task in a controlled session. The standard interface and redesigned interface use the same collection, task instructions, time limit, and scoring rubric. The redesign changes result ordering and adds a query-refinement panel. Before the task, each student completes a search-skill assessment and is classified by whether the assigned task is in the more difficult tier. Software then randomly assigns the interface. During the task it records clicks, and afterward an independent rubric produces a task score.

The CSV contains 1800 fictional students. Assignment placed 921 on the redesigned interface and the remainder on the standard version. There is no attrition or missing field in this teaching file. No student or task response is real.

**Scientific question.** For these eligible students under this task protocol, what would assignment to the specified redesigned interface do to the task score compared with assignment to the specified standard interface?

**Causal estimand.** The average effect of assignment in the randomized cohort,  $E[Y(1) - Y(0)]$ , where  $A=1$  is assignment to the redesigned interface,  $A=0$  is assignment to the standard interface, and  $Y(a)$  is the task score the same student would receive under assignment  $a$ . This is an intention-to-treat effect of interface assignment. It is not the effect of using the refinement panel, making fewer clicks, or adopting every future interface called “redesigned.”

**Unit, time zero, and treatment versions.** One row is one fictional student. Random assignment is time zero. Baseline search skill and task difficulty precede assignment; click count occurs afterward; task score is the outcome. The two interface versions, setting, task, scoring rule, and immediate follow-up define the intervention contrast. If instructors can override the assigned interface, if students use different devices, or if the redesign changes over time, consistency requires more careful versions than the labels 0 and 1 reveal.

## Three questions that use the same columns

The presence of  $A$ , clicks, and  $Y$  in one file does not make every question causal.

| Question                                                           | Type        | Quantity answered        |
|--------------------------------------------------------------------|-------------|--------------------------|
| What was the average score among students assigned each interface? | Descriptive | Two observed group means |

| Question                                                                     | Type       | Quantity answered                                                                 |
|------------------------------------------------------------------------------|------------|-----------------------------------------------------------------------------------|
| Which recorded features best predict a student's task score?                 | Predictive | Out-of-sample predictive performance for a specified population and loss function |
| What would assigning this redesign instead of the standard interface change? | Causal     | A contrast between potential outcomes under two interventions                     |

A descriptive difference can estimate a causal effect when a design justifies the comparison. Prediction can be useful without identifying any intervention. Conversely, a causal estimate need not be the best individual-level prediction. The question determines which success criterion matters.

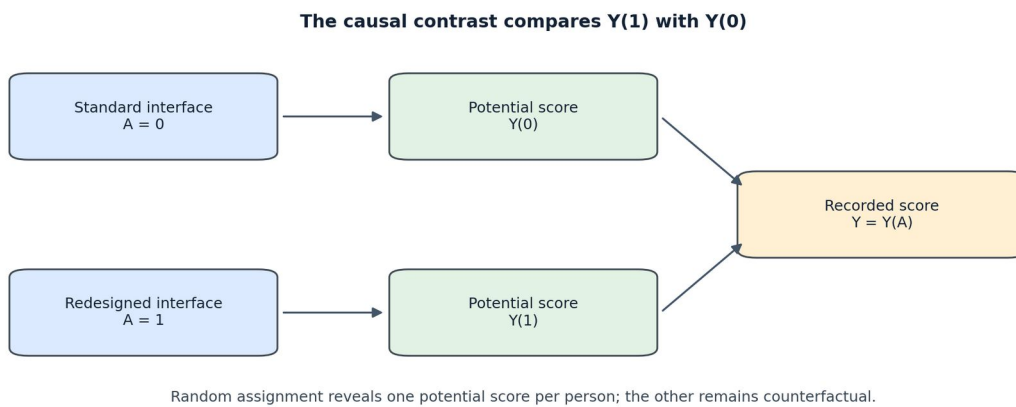


Figure 1. Potential-outcome map for the library-interface experiment: standard assignment defines  $Y(0)$ , redesigned assignment defines  $Y(1)$ , and random assignment reveals only  $Y(A)$  for each student.

Figure 1 is a missing-data representation, not a claim that both potential scores exist in the CSV. For any student assigned  $A=1$ ,  $Y(1)$  is recorded and  $Y(0)$  is counterfactual; assignment  $A=0$  reverses those roles. Randomization does not reveal both outcomes for anyone. It makes the groups exchangeable in expectation, allowing group averages to stand in for the two unobservable average potential outcomes.

## From a topic to an intervention

“Do interfaces matter?” is a topic, not an estimand. A defensible question names at least five elements: the intervention, comparison, unit, outcome and timing, and target population. It also specifies assignment versus receipt. Here the intervention is assignment to a particular software version. Click count is a consequence of the assigned interface, prior skill, task difficulty, and idiosyncratic navigation. Forcing every student to make exactly ten clicks would be a different—and poorly defined—intervention.

A useful intervention test is to ask whether another team could implement both conditions from the description. “More usable,” “engaged,” and “received treatment” are not sufficiently precise on their

own. If implementation varies, the causal contrast may be a mixture of versions. That is a design problem even when the statistical calculation is flawless.

### Pause and Decide

Before seeing results, classify each question as descriptive, predictive, or causal: (1) Which interface group had the higher observed average? (2) How accurately can baseline skill, task difficulty, and clicks predict task score for similar students? (3) What would assigning the redesign rather than the standard interface change? Then write the intervention, comparison, population, outcome time, and estimand for question 3. Decide whether click count belongs in the primary causal adjustment set and record an answer, reason, confidence rating, and revision cue.

## Identification assumptions, separate from the statistical model

The randomized comparison requires successful random assignment, adherence to the recorded assignment for an assignment-effect interpretation, consistent outcomes under the two specified versions, no relevant interference between sessions, complete or appropriately handled outcome ascertainment, and a scoring process unaffected by knowledge of assignment except through genuine performance. Randomization supports exchangeability for the assigned interface in this cohort; it does not guarantee balance in every realized baseline variable, prevent attrition, define the intervention, or transport the result elsewhere.

The primary estimator is the difference in mean task score, represented as the coefficient on assignment in an OLS regression with HC1 standard error. A second prespecified regression adds pre-assignment search skill and task difficulty. Because assignment is randomized, these baseline variables are included for precision rather than to repair confounding. The additive model is an estimation choice. With treatment-effect heterogeneity, its coefficient and the marginal intention-to-treat contrast need not coincide without appropriate interactions or standardization.

Clicks are post-assignment. Adjusting for them would change the comparison and could introduce bias because clicks are also affected by skill and task difficulty. The primary effect of assigning the interface includes all pathways through navigation behavior. A regression of score on clicks is reported only to expose the distinction between an observed association and an intervention effect.

## Worked analysis from the CSV

The mean score is 67.684 under the standard interface and 71.616 under the redesign. Their difference is the unadjusted randomized estimate shown below.

| Prespecified comparison                                            | Estimate | HC1 SE | Approximate 95% interval |
|--------------------------------------------------------------------|----------|--------|--------------------------|
| Assignment effect, unadjusted                                      | 3.933    | 0.346  | [3.256, 4.610]           |
| Assignment effect, adjusted for baseline skill and task difficulty | 3.687    | 0.280  | [3.139, 4.235]           |
| Score association per additional observed click                    | -1.362   | 0.064  | [-1.487, -1.237]         |

The unadjusted intention-to-treat estimate is 3.933 score points. The baseline-adjusted estimate is 3.687 points and has the smaller standard error. The two point estimates need not match exactly: randomization balances baseline causes in expectation, not by arithmetic identity in a realized sample. The observed baseline-skill difference, redesign minus standard, is 0.407 points. Adjustment was prespecified and increases precision here; selecting covariates after examining which specification is most favorable would weaken the design-based interpretation.

Click count and score have correlation -0.452, and the simple score-on-click regression associates each additional click with -1.362 score points. That coefficient does not estimate what would happen if software forced one fewer click. More skilled students tend to navigate with fewer clicks, difficult tasks require more clicks, and the interface itself changes clicking. The slope is a valid descriptive association for this CSV and an invalid substitute for the interface-assignment estimand.

The generator gives every fictional student a 4.000-point interface-assignment effect. That hidden parameter checks the simulation; it is not available to the analyst and does not turn proximity between 4.000 and either estimate into a general diagnostic of causal validity. In a real experiment, the design—not access to a hidden answer—supports the causal interpretation.

### Diagnostics and falsification

Begin with the assignment record: confirm allocation counts, no duplicate identifiers, time order, and the absence of impossible values. Compare pre-assignment skill and task difficulty across arms without treating a nonsignificant difference as proof of randomization. The 0.407-point skill difference is compatible with chance; a large systematic collection of discrepancies could nevertheless expose implementation trouble.

Next examine missing outcomes, assignment overrides, repeated participation, session contamination, device or browser incompatibility, ceiling effects in the rubric, and whether graders knew the assigned interface. Report outcomes by randomized assignment even when students did not use a new feature. A manipulation measure can describe implementation, but conditioning the primary estimate on successful use generally sacrifices the randomized comparison.

A useful falsification check is a pre-assignment measure the interface could not have caused. Detecting an apparent “effect” on that variable could indicate assignment, joining, or coding problems. A replication should prespecify the primary outcome and analysis, audit software logs, vary task types, and assess whether any benefit persists after the immediate session.

### Claim Audit

**Supports:** a reproducible intention-to-treat estimate for assignment to two specified interfaces in this complete fictional randomized cohort, conditional on correct implementation, measurement, and no relevant interference.

**Does not establish:** the effect of forcing fewer clicks; the effect of every refinement panel; benefits for other institutions, tasks, devices, or long-term learning; that every student benefits; or that randomization repairs an ill-defined intervention or biased outcome measurement.

**Strongest defensible conclusion:** “In this synthetic randomized study, assignment to the specified redesigned interface increased the mean immediate task score by 3.933 points, approximate 95% HC1 interval [3.256, 4.610], relative to the specified standard interface. The estimate concerns assignment in this fictional cohort and provides no evidence about real students.”

## Study in the Larger Evidence Program

This A/B experiment is a focused responsiveness test: changing the assigned interface changes the immediate score under controlled conditions. A next study should test generalization and mechanism without weakening randomization—for example, block assignment by institution and task type, measure longer-term independent search performance, and preregister whether the refinement panel changes query quality rather than merely click count. Qualitative usability work could reveal treatment versions and failure modes the score misses. Agreement across outcomes and settings is informative only when the measures and biases are not redundant.

## Six principal problems

For each, write your answer, reason, confidence from 1 to 5, and one revision cue before reading the solutions.

**Problem 1.1 — Classify the question.** Classify these as descriptive, predictive, or causal: the observed mean score by interface; a model forecasting score from baseline skill and clicks; the effect of assigning the redesign. Explain why the same CSV can support all three questions without making them interchangeable.

**Problem 1.2 — Define the contrast.** State  $A$ ,  $Y$ , the target cohort, outcome time, and  $E[Y(1)-Y(0)]$ . Name two interface details that must remain stable enough for consistency.

**Problem 1.3 — Explain the missing pair.** For a student assigned the redesign, which potential score is observed and which is counterfactual? Explain what randomization contributes when it cannot reveal both.

**Problem 1.4 — Calculate and interpret.** Use the unadjusted estimate and HC1 standard error to calculate estimate  $\pm 1.96 \times SE$ . Compare your result with the reported interval and identify one uncertainty it omits.

**Problem 1.5 — Choose and reject adjustments.** Explain why baseline skill and task difficulty may improve precision without being required to remove confounding, and why observed clicks should not be inserted into the primary total assignment-effect regression merely because they predict score.

**Problem 1.6 — Repair the claim.** Repair: “Each avoided click raises performance by -1.362 points, and the experiment proves the redesign works everywhere.” Write two defensible sentences and name the next study that would address the largest generalization gap.

## Fully worked solutions

**Solution 1.1. Answer:** observed means are descriptive; forecasting score is predictive; changing assignment is causal. **Reason:** description summarizes realized data, prediction evaluates performance on new cases, and causation compares outcomes under interventions. Randomization can make a descriptive arm difference estimate a causal assignment effect, but that design fact does not transform every association in the file. **Confidence and revision:** rate 1–5; rewrite each question with its success criterion—accurate summary, accurate forecast, or intervention contrast.

**Solution 1.2. Answer:**  $A$  is assignment to the redesigned (1) or standard (0) interface;  $Y$  is the immediate rubric score; the target is the fictional eligible randomized cohort; the estimand is  $E[Y(1)-Y(0)]$ . Stable details include result-ordering rules, refinement-panel functions, collection, task instructions, device environment, time limit, and scoring. **Reason:** potential outcomes require

conditions specific enough that  $Y(1)$  and  $Y(0)$  are interpretable. **Confidence and revision:** rate 1–5; if your treatment was simply “better interface,” replace the adjective with implementable features.

**Solution 1.3. Answer:**  $Y(1)$  is observed and  $Y(0)$  is counterfactual for a student assigned the redesign. **Reason:** the recorded score obeys  $Y=Y(A)$ ; no student completes the same task at the same time under both assignments. Randomization makes assignment independent of the potential outcomes in expectation, so arm means can estimate their averages. **Confidence and revision:** rate 1–5; draw two potential outcomes and circle only the one selected by  $A$ .

**Solution 1.4. Answer:**  $3.933 \pm 1.96 \times 0.346$  gives  $[3.256, 4.610]$ , subject to rounding of the displayed inputs. **Reason:** the HC1 interval describes sampling/model-based uncertainty for the prespecified estimator. It omits uncertainty from treatment-version ambiguity, outcome validity, interference, attrition not represented in this complete CSV, and transport to other settings. **Confidence and revision:** rate 1–5; name the source of the standard error separately from identification and measurement assumptions.

**Solution 1.5. Answer:** baseline skill and task difficulty are pre-assignment and predictive, so prespecified adjustment can reduce residual variation. Randomization—not the regression—supports exchangeability. Clicks occur after assignment and partly transmit how the interface operates; they are also related to skill and difficulty. **Reason:** adjusting for a post-assignment variable changes the randomized comparison and may induce selection bias. Its predictive value is not a causal selection rule. **Confidence and revision:** rate 1–5; put every candidate covariate on a timeline before deciding its role.

**Solution 1.6. Answer:** “In the synthetic randomized cohort, assignment to the redesigned interface increased the mean immediate task score by 3.933 points relative to the standard interface, approximate 95% HC1 interval  $[3.256, 4.610]$ . The negative click–score association is descriptive and does not identify an intervention on clicks or effects in other institutions.” A multisite randomized replication with varied tasks and delayed independent outcomes would address generalization. **Reason:** the experiment manipulated interface assignment, not click count, and sampled no real target population. **Confidence and revision:** rate 1–5; name the randomized intervention and remove claims about untested settings.

## Unit 2 Estimands Populations and Claim Ceilings

### Purpose and objectives

This unit makes the target quantity explicit before choosing an estimator. By the end, you should be able to distinguish the average treatment effect, average treatment effect among the treated, and intention-to-treat effect; separate assignment from treatment received; identify the population over which an average is taken; standardize site-specific effects to a stated target composition; describe the additional assumptions required for transport; and set a claim ceiling that respects design, measurement, and sampling.

### The synthetic multisite psychology study

Three fictional campus counseling centers—Cedar, Lake, and Ridge—recruit students for a study of a short planning workshop. Within each site, software independently offers each student immediate access with probability 0.50; students not offered access receive the existing information sheet. Not everyone offered the workshop attends, and nobody in this one-sided-compliance example attends without an offer. Baseline stress is measured before the offer. Well-being is scored immediately after the workshop period.

The recruited sample contains 2400 fictional students: 1200 at Cedar, 720 at Lake, and 480 at Ridge. There are 1184 offers and 585 attenders. Uptake among students offered access is 0.494. The file is complete and contains no real person.

The program planner is not asking only about this recruited sample. The stated future target population has site composition 25% Cedar, 35% Lake, and 40% Ridge, whereas the recruited sample is 50%, 30%, and 20%. Ridge therefore contributes twice as much to the target average as to the sample average. A transport calculation must make that change visible.

**Scientific question.** What would a policy of offering immediate workshop access rather than the existing information sheet change in immediate well-being for the stated target mixture of eligible students across the three sites?

**Primary causal estimand.** The target-standardized intention-to-treat effect of the offer,

$$\tau_{\text{target}} = 0.25\tau_{\text{Cedar}} + 0.35\tau_{\text{Lake}} + 0.40\tau_{\text{Ridge}},$$

where  $\tau_s = E[Y(1) - Y(0) \mid \text{site}=s]$  is the site-specific effect of randomized offer assignment. The intervention is an offer, not compelled attendance. The target weights are fixed features of the planning question, not estimated from this convenience recruitment.

**Unit and timeline.** One row is one fictional recruited student. Site and baseline stress precede randomized offer  $Z$ . Workshop attendance  $D$  occurs after the offer. Immediate well-being  $Y$  follows the workshop period. The distinction  $Z \rightarrow D \rightarrow Y$  matters: randomization directly identifies contrasts by  $Z$ , while a contrast by observed  $D$  compares self-selected attenders with a mixture of other students.

### ATE ATT and ITT name different averages

**Average treatment effect (ATE).**  $E[Y(1) - Y(0)]$  averages a named intervention contrast over a named population. “The ATE” is incomplete until both the intervention and population are supplied.

In this experiment, the sample ATE of *offer assignment* is also the sample intention-to-treat effect because all randomized students are analyzed by assigned offer.

**Average treatment effect among the treated (ATT).**  $E[Y(1)-Y(0) | D=1]$  asks about the treatment-receipt contrast among people who actually received treatment. Here attendance is post-offer and selective. Comparing observed attenders with nonattenders does not identify this ATT. The randomized offer could later support an instrumental-variable estimand for compliers under additional assumptions, but it does not turn the attendance coefficient into an ATT; Unit 7 develops that distinction.

**Intention-to-treat effect (ITT).**  $E[Y(Z=1)-Y(Z=0)]$  compares assignment to the offer regardless of attendance. It answers a policy-relevant question: what happens when the center implements an offer under the observed uptake behavior? Its magnitude can reflect both workshop effectiveness and participation. It is not a diluted or defective estimate of receipt; it is the effect of a different intervention.



Figure 2. Recruited and target site shares: Cedar changes from 50% to 25%, Lake from 30% to 35%, and Ridge from 20% to 40%, so target standardization gives greater weight to Ridge.

Figure 2 shows why “overall effect” is not self-defining. The same site-specific estimates produce different averages when the population mixture changes. Standardization makes the weights explicit; it does not itself establish that the within-site effects will transport.

## The claim ceiling

A claim ceiling is the strongest conclusion supported jointly by the question, design, measurement, data collection, and analysis. Random assignment can support a causal offer effect in the recruited cohort. It does not make attendance randomized, guarantee that the well-being scale captures the intended construct, or make the recruited site mixture representative. Target standardization can address a known difference in site composition if site is sufficient for transport, but it cannot repair unmeasured differences in implementation, eligibility, baseline composition, or follow-up.

The ceiling is not determined by whether a p value crosses a threshold. A precise estimate of the wrong intervention or population remains the wrong estimand. Conversely, a wide interval around

the correct estimand is an honest result rather than a reason to substitute a more favorable comparison.

### Pause and Decide

The planner will implement an *offer* across the stated future site mixture. Before examining results, choose among five candidate quantities: the pooled recruited-sample offer ITT, the recruited-site-share standardized offer ITT, a baseline-and-site-adjusted offer coefficient, the observed attendance association, and the target-standardized offer ITT. State which directly answers the planning question, which are alternative estimators for the recruited sample, and which fails because attendance is not randomized. Then write the additional transport assumption required to move from recruited site-specific estimates to the target mixture, along with your confidence and revision cue.

### Identification assumptions, separate from the statistical model

For the recruited-sample ITT, the working assumptions are successful offer randomization within site; consistency for the defined offer and information-sheet conditions; analysis by assignment; complete or appropriately handled outcome ascertainment; no relevant interference between students; and valid, assignment-neutral well-being measurement. The randomized design supports exchangeability of  $Z$  within the enrolled cohort. Baseline balance checks can detect some implementation problems but do not prove the randomization record is correct.

For the target-standardized ITT, add conditional transportability: within each named site, the offer effect estimated among recruits must apply to eligible members of the stated target population, or all effect modifiers that differ between recruits and target must be measured and standardized. Every target site needs recruited observations under both assignments. The fixed target weights must correspond to the policy population. Sites also need sufficiently comparable intervention versions for  $\tau_s$  to retain a common policy meaning.

The estimators use OLS coefficients and HC1 standard errors with normal 1.96 intervals. The two standardized estimates are fixed-weight sums of the three site-specific randomized differences. One uses the recruited site shares and the other uses the stated target shares; each variance is the sum of squared fixed weights times the independent site-specific variances. The target interval treats the target weights as fixed and does not include uncertainty about transportability, outcome validity, or future implementation.

### Worked analysis from the CSV

| Comparison                                              | Estimate | HC1 SE | Approximate 95% interval |
|---------------------------------------------------------|----------|--------|--------------------------|
| Pooled recruited-sample offer ITT                       | 1.724    | 0.269  | [1.197, 2.250]           |
| Offer coefficient adjusted for site                     | 1.923    | 0.261  | [1.410, 2.435]           |
| Offer coefficient adjusted for baseline stress and site | 1.719    | 0.229  | [1.270, 2.168]           |

| Comparison                                  | Estimate | HC1 SE | Approximate 95% interval |
|---------------------------------------------|----------|--------|--------------------------|
| Naive observed attendance association       | 3.165    | 0.304  | [2.570, 3.760]           |
| Recruited-site-share standardized offer ITT | 1.925    | 0.261  | [1.414, 2.436]           |
| Target-standardized offer ITT               | 2.438    | 0.313  | [1.825, 3.051]           |

The pooled recruited-sample offer estimate is 1.724 well-being points. Adjusting only for site gives 1.923; adding baseline stress gives 1.719 with a smaller standard error. The near agreement between 1.724 and 1.719 is not evidence that the covariates were balanced or irrelevant. In this realized assignment, accounting for site raises the coefficient, while adding baseline stress lowers it by nearly the same amount. These chance site and baseline imbalances offset. Randomization, not that cancellation or the adjustment model, supports the within-sample causal interpretation.

The attendance association is 3.165 points. It is numerically larger, but attendance depends on response to the offer, baseline stress, and site implementation. Its apparent attractiveness does not make it an identified treatment-receipt effect.

The site-specific offer estimates are:

| Site  | Sample n | Offer ITT | HC1 SE | Approximate 95% interval | Target weight |
|-------|----------|-----------|--------|--------------------------|---------------|
| Cedar | 1200     | 1.391     | 0.357  | [0.691, 2.090]           | 0.250         |
| Lake  | 720      | 1.474     | 0.480  | [0.533, 2.414]           | 0.350         |
| Ridge | 480      | 3.936     | 0.621  | [2.719, 5.152]           | 0.400         |

Applying the recruited shares 0.50, 0.30, and 0.20 to the site estimates gives 1.925 points. This differs from the pooled 1.724 because independent Bernoulli assignment produced different realized site mixtures in the offer and comparison groups. Moving from the pooled estimate to the recruited-site-share standardized estimate adds 0.201 points. These are alternative estimators of the recruited-sample offer effect. Their difference reflects how they combine sites in this realized assignment, not a change in the causal population.

Replacing the recruited shares with target weights 0.25, 0.35, and 0.40 gives 2.438 points, a further increase of 0.513 points. Ridge receives more weight in the target and has the largest observed site estimate. Thus the full 0.714-point gap contains two steps: 0.201 from pooled to recruited-site standardization and 0.513 from changing the population weights. The site intervals are wide, and the Cedar–Lake point difference should not be promoted into a verified interaction. The calculation illustrates how a target changes an average; it does not prove the observed heterogeneity will recur.

The generator contains hidden attendance response types and site-specific structural effects. Those values validate the simulation but are unavailable to the analyst. The target estimate is justified from randomized within-site comparisons plus the stated transport assumption—not by selecting a weighting scheme that happens to approach hidden truth.

## Diagnostics and falsification

Audit randomization within each site, not only overall. Confirm assignment counts, baseline timing, one-sided attendance coding, unique identifiers, and complete outcomes. The realized offer rates are 46.4% at Cedar, 50.7% at Lake, and 54.6% at Ridge. Those deviations from 50% are possible under independent probability-0.50 assignment and explain why the pooled offer and comparison groups have different site mixtures. Compare baseline stress distributions by offer within site, inspect outcome ranges and missingness, and check whether the offer and information-sheet versions were delivered consistently. Report uptake by site because implementation is part of the offer effect, but do not condition the primary comparison on uptake.

For transport, compare recruited and target distributions of plausible effect modifiers rather than site label alone. The hypothetical target supplies only site shares; a real planner would need age, baseline severity, language, delivery mode, and eligibility information if those alter effects. A sensitivity analysis could recalculate the target effect across plausible site weights and within-site shifts. A future target sample can falsify transport predictions by comparing prespecified site-specific outcomes.

Negative-control outcomes measured before the offer can expose coding or enrollment errors. They cannot validate all transport assumptions. Likewise, statistical equality of site coefficients would not prove no effect modification; low power can hide important differences.

## Claim Audit

**Supports:** reproducible pooled and recruited-site-share standardized intention-to-treat estimates from randomized offer assignment and, under explicit within-site transport assumptions and fixed target weights, a target-standardized offer estimate for the fictional planning population.

**Does not establish:** the causal effect of attending the workshop; an ATT among observed attenders; a complier effect; effects outside the three sites; that site captures every transport-relevant difference; that the well-being score measures durable benefit; or that the largest site point estimate represents a real site interaction.

**Strongest defensible conclusion:** “In this synthetic multisite randomized study, the target-standardized effect of offering immediate workshop access is 2.438 well-being points, approximate 95% HC1 interval [1.825, 3.051], for the stated 25% Cedar, 35% Lake, and 40% Ridge target mixture, if the site-specific recruited effects transport within site. This is an offer effect, not an identified effect of attendance, and it is not evidence about real students.”

## Study in the Larger Evidence Program

The randomized offer provides a responsiveness test for an implementable policy. The next evidence should separate transport and mechanism. A target-representative multisite trial could sample from the intended population, standardize richer baseline effect modifiers, measure attendance and intervention fidelity, and use a delayed well-being outcome. A later encouragement or instrumental-variable analysis could address receipt among compliers only after its relevance, exclusion, independence, and monotonicity assumptions are stated. Qualitative work can determine whether “the same workshop” means comparable treatment across sites.

## Six principal problems

For each, write your answer, reason, confidence from 1 to 5, and one revision cue before reading the solutions.

**Problem 2.1 — Name the estimand.** Define the recruited-sample ITT, target-standardized ITT, and ATT among attenders. Which intervention and population does each use?

**Problem 2.2 — Keep assignment and receipt separate.** Explain why random assignment of the offer identifies an offer effect but does not make the observed attendance comparison causal.

**Problem 2.3 — Decompose the sample-to-target contrast.** First use recruited-sample weights 0.50, 0.30, and 0.20 with the three site point estimates. Then use target weights 0.25, 0.35, and 0.40. Compare both standardized estimates with the pooled offer estimate and decompose the full pooled-to-target difference into its two steps.

**Problem 2.4 — Interpret uncertainty.** Read the target-standardized interval. Name one source of uncertainty it includes and two important sources it omits.

**Problem 2.5 — Set the claim ceiling.** A planner writes, “The workshop improves every student’s well-being by 2.438 points.” Identify at least four errors and write the strongest defensible replacement.

**Problem 2.6 — Design the next study.** Propose a study that would improve transport to the stated population while preserving a clear offer estimand. Explain which additional question would require an instrumental-variable design rather than a naive attendance regression.

## Fully worked solutions

**Solution 2.1. Answer:** the sample ITT is  $E[Y(Z=1) - Y(Z=0) \mid \text{recruited cohort}]$ ; the target ITT averages the same offer contrast over the stated target site mixture; the attendance ATT is  $E[Y(D=1) - Y(D=0) \mid D=1]$  for a sufficiently defined receipt intervention among attenders. **Reason:** estimands change when intervention, subgroup, or population changes. In this design, ITT is an ATE of assignment, while observed receipt is post-assignment. **Confidence and revision:** rate 1–5; attach an intervention and population to every acronym before using it.

**Solution 2.2. Answer:** within site,  $Z$  was randomized;  $D$  was not. Students attend only after an offer and according to characteristics and response types. **Reason:** comparing  $Z$  groups preserves the randomized exchangeability of assignment. Comparing  $D$  groups selects on post-assignment behavior and mixes baseline differences, site differences, and treatment response. **Confidence and revision:** rate 1–5; draw the timeline site/baseline  $\rightarrow Z \rightarrow D \rightarrow Y$  and circle the randomized variable.

**Solution 2.3. Answer:** recruited-site standardization gives  $0.50 \times 1.391 + 0.30 \times 1.474 + 0.20 \times 3.936 = 1.925$  points. Target standardization gives  $0.25 \times 1.391 + 0.35 \times 1.474 + 0.40 \times 3.936 = 2.438$  points, subject to rounding of displayed site estimates. Starting from the pooled 1.724, the first step adds 0.201 points and the target reweighting adds another 0.513 points, for a total difference of 0.714. **Reason:** independent assignment with probability 0.50 produced unequal realized offer rates by site, so the pooled contrast is not numerically identical to the recruited-share standardization. The target then gives Ridge 40% weight instead of its 20% recruited share, and Ridge has the largest observed offer estimate. The first gap comes from how the two sample estimators combine sites in this realized

assignment; the second changes the population weights. **Confidence and revision:** rate 1–5; write each set of weights beside the population it represents before multiplying.

**Solution 2.4. Answer:** the interval is [1.825, 3.051]. It includes sampling/model-based uncertainty from the three site-specific randomized contrasts under the HC1 calculation. It omits uncertainty about whether effects transport within site, whether target weights will remain fixed, treatment-version differences, outcome validity, interference, and future implementation. **Reason:** a standard error reflects the estimator’s stated sampling calculation, not every threat to the causal claim.

**Confidence and revision:** rate 1–5; separate estimation uncertainty from identification, measurement, and transport uncertainty.

**Solution 2.5. Answer:** the claim confuses an average with an individual effect, offer with attendance, a synthetic illustration with human evidence, and conditional transport with an established fact; it also ignores the interval and immediate outcome timing. A defensible replacement is: “In this synthetic randomized study, the target-standardized offer estimate is 2.438 immediate well-being points, approximate 95% HC1 interval [1.825, 3.051], under the stated site weights and within-site transport assumption; it provides no evidence about real students.” **Confidence and revision:** rate 1–5; circle every population, intervention, time, and averaging word in the original claim.

**Solution 2.6. Answer:** recruit or probability-sample eligible students in the intended site mixture, randomize the same offer within site, prespecify baseline effect modifiers and delayed outcomes, audit delivery and attrition, and standardize only variables justified before analysis. The effect of receipt among compliers would require an instrumental-variable analysis using randomized offer plus relevance, exclusion, independence, and monotonicity assumptions. **Reason:** better sampling and effect-modifier measurement address transport; random assignment preserves the offer estimand. A naive D comparison does neither. **Confidence and revision:** rate 1–5; name which design feature addresses identification and which addresses generalization.

## Unit 3 Causal Diagrams and Adjustment Decisions

### Purpose and objectives

This unit converts a familiar warning about regression into an actionable adjustment decision. By the end, you should be able to state a total-effect estimand; trace a backdoor path; distinguish a measured confounder, a mediator, and a collider; estimate four clearly different comparisons; diagnose the measured-overlap problem; and write a claim that does not borrow authority from the simulator's hidden truth.

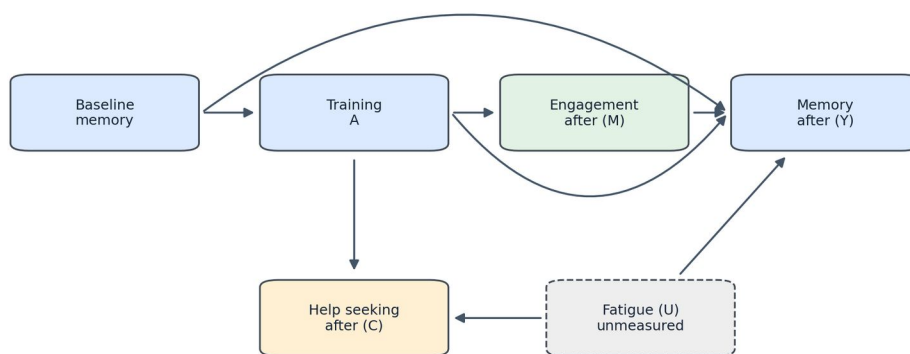
### The synthetic psychology study

A fictional campus invites eligible volunteers to a memory-training program. Some elect to take it; others do not. A baseline memory assessment happens *before* that choice, followed by a training period and an immediate follow-up memory score. The program may also change engagement during practice. Help-seeking intensity during the period can respond both to being in the program and to latent fatigue. Those later variables are recorded, but their existence does not make them suitable controls for every question. The CSV contains 2400 fictional individuals, of whom 1089 chose training. There are no missing fields. No person in this example is real.

**Scientific question.** What would access to this specified memory-training program do to the immediate follow-up memory score among eligible volunteers, relative to no access?

**Causal estimand.** The total average effect among eligible recruited volunteers,  $E[Y(1) - Y(0) \mid \text{eligible recruited volunteers}]$ , in memory-score points. The intervention version, period, outcome scale, comparison, and target group belong in the definition. A different intervention or delayed outcome gives a different estimand. The observed data do not supply both outcomes for any one individual.

**Unit and mechanisms.** One row is one fictional person. Training is chosen, not assigned. The cohort is constructed as an illustrative volunteer sample, not randomly sampled from a human population. Baseline memory affects both choice and later memory. Unrecorded fatigue affects later memory and help seeking; the generator has arranged fatigue to be independent of training conditional on baseline. This last fact is a property of the synthetic mechanism, not a guarantee obtainable from an observed CSV. Engagement is caused by training and contributes to the outcome. Help seeking is caused by training and fatigue.



Adjust for baseline memory B. Do not adjust for post-training M or collider C when estimating the total effect.

Figure 3. Memory-training DAG: baseline confounder B, mediator M, and collider C with latent fatigue U.

The solid nodes are recorded; dashed U is latent. Arrows encode a data-generating hypothesis, not conclusions learned by fitting regression. B is a common cause of A and Y; C is a collider on  $A \rightarrow C \leftarrow U \rightarrow Y$ ; M is on a pathway from A to Y.

The backdoor path  $A \leftarrow B \rightarrow Y$  remains open when we compare unadjusted group means. Adjusting for pre-choice baseline memory B blocks it *if the graph is correct*.  $A \rightarrow C \leftarrow U \rightarrow Y$  is initially closed at C. Adjusting for C opens this otherwise closed path and can produce selection bias. Adjusting for M, a descendant of A on a causal path to Y, discards part of the total effect; a direct-effect question would require its own estimand and stronger assumptions. A graph can show the paths, but it cannot establish that the proposed arrows, omissions, and timing match the world. See Greenland, Pearl, and Robins (1999), and Elwert and Winship (2014).

### A slower adjustment routine

First, state the estimand without naming a regression. Second, place every candidate variable before treatment, after treatment, or outside the recorded timeline. Third, trace paths between A and Y, asking which are causal and which enter A through an incoming arrow. Fourth, choose variables that block the relevant noncausal paths without blocking the causal pathways or opening a collider. Only then write the statistical model.

For this total-effect question, baseline B belongs in the adjustment set because it is a pre-choice common cause. M is rejected because it is part of the program's pathway. C is rejected because it is a common effect of A and U. "Associated with Y," "improves fit," and "changes the coefficient" are not sufficient causal roles. This routine deliberately slows the transition from a variable list to a model.

### Pause and Decide

Before looking at the estimates: choose an adjustment set for the total effect from {baseline memory, engagement after training, help seeking after training}. Say which two variables you reject and why. Predict the direction of the naive distortion, then commit to an answer, a reason, a confidence rating, and a revision cue. Do not choose the regression with the smallest standard error or the result nearest a hidden simulated answer.

## Identification assumptions, separate from the statistical model

The working analysis requires a well-defined training and no-training contrast, consistent outcome measurement, and conditional exchangeability of training given *accurately measured* baseline memory. Exchangeability includes the claim that no other pre-choice cause of both choice and outcome remains. It is not a synonym for including many covariates.

The analysis also requires a positive chance of each condition at the baseline levels being compared; no relevant interference between volunteers; an appropriate study inclusion mechanism for the within-cohort target; and the stated graph, especially the status and timing of C and M. These are identification and design assumptions. Some have testable implications, but their conjunction is not proven by a fitted model. Random sampling, had it occurred, would address a different question about a wider population.

The estimate uses additive ordinary least squares with baseline as a covariate and heteroskedasticity-consistent HC1 standard errors; 95% limits use a normal multiplier of 1.96. A model for a conditional mean is an estimation assumption. The simple additive regression recovers a marginal total ATE only with appropriate effect homogeneity or correct averaging/standardization. This simulator has a constant treatment effect, so the two coincide by construction. In real data, report the precise population contrast and examine interactions or standardize explicitly when effect heterogeneity matters; Unit 5 develops that step.

## Worked analysis from the CSV

Each row in the following table is the training coefficient from a separate prespecified OLS model. All values come from the packaged CSV and are rounded to three decimal places. The confidence intervals describe sampling/model-based uncertainty conditional on the stated estimators; they do **not** encompass unidentified confounding or errors in the graph.

| Model and adjustment              | Training estimate, points | HC1 SE | Approximate 95% interval |
|-----------------------------------|---------------------------|--------|--------------------------|
| No adjustment                     | 8.352                     | 0.305  | [7.755, 8.949]           |
| Pre-choice baseline memory B      | 2.000                     | 0.195  | [1.617, 2.382]           |
| B plus post-choice help seeking C | 4.691                     | 0.168  | [4.362, 5.020]           |
| B plus post-choice engagement M   | 0.901                     | 0.245  | [0.421, 1.380]           |

### Read the table in two passes

On the first pass, ask what each row conditions on and which estimand it could represent. Do not compare magnitudes yet. On the second pass, read the estimate and interval only for the rows whose conditioning set matches a defensible identification strategy. This prevents the most precise or favorable coefficient from silently choosing the scientific question.

The naive difference, 8.352 points, answers an associational question about volunteers who did or did not choose training. Their baseline memory differs by 9.101 points on average; stronger pre-choice memory predicts both participation and later performance. The B-adjusted estimate, 2.000 points, is the intended total-effect estimate **under the stated identification and model assumptions**. It is not the “true effect.” Conditioning on C changes it to 4.691 points by opening a path through fatigue. The C-adjusted result can appear precise and still answer the wrong causal comparison. Conditioning on M produces 0.901 points: this conditions on part of the program’s pathway and is not a total-effect estimate; it also does not become a validated natural direct effect simply by naming it one.

The generator’s total effect is **2.000** points and its controlled direct effect at fixed M is **0.800** points. These parameters validate a teaching simulation, not a claim the analyst could know from the observed data. A real analyst cannot compare an observational coefficient with the hidden answer, and near agreement would not verify exchangeability.

## Diagnostics and falsification

Training rates by baseline-memory quintile are 0.142, 0.288, 0.463, 0.585, and 0.792. Every displayed stratum contains both choices, while the gradient cautions against extrapolating at the extremes. Inspect raw baseline distributions, nonlinear outcome–baseline patterns, model residuals, influential observations, outcome scale validity, and dropout in a real implementation. As a *pre-choice* balance diagnostic, compare baseline memory by treatment; the observed difference of 9.101 points warns against the naive comparison. No in-sample graph test can prove absence of an unmeasured common cause. The included Unit 3 dataset is complete; attrition, measurement error, different treatment versions, and interference are untested real-world threats rather than defects simulated in this file.

Useful future falsification checks would measure a pre-training outcome that access could not have changed, use an exposure-inert negative-control outcome chosen on substantive grounds, compare alternative baseline specifications, and quantify sensitivity to an omitted common cause. Such checks can challenge an explanation, but passing them cannot certify the DAG. Estimation robustness across regressions is different from robustness to unmeasured bias. The graphical rules for conditioning on a collider do not dictate the *sign* of its bias; the sign here arises from the specific generator.

## Claim Audit

**Supports:** a documented synthetic association and a model-based total-effect estimate for the fictional eligible cohort if the proposed graph, exposure definition, and identification assumptions are granted. The data-and-code audit establishes numerical reproducibility of that conditional statement.

**Does not establish:** that the same adjustment set is correct for actual human memory research; that there is no unmeasured common cause in a real voluntary program; that the effect applies to all university students; that help-seeking is always a collider; that controlling engagement estimates an unqualified direct effect; or that an R/Python rerun independently validates a causal mechanism.

**Strongest defensible conclusion:** “In this fully synthetic volunteer cohort, the baseline-adjusted training coefficient is 2.000 memory-score points, approximate 95% HC1 interval [1.617, 2.382]. Under the simulator’s stated graph and assumptions, it estimates the total effect of program access for the eligible cohort. This numerical example provides no causal evidence about real people.” For an analogous human observational study, the language would remain explicitly conditional on exchangeability, selection, measurement, and overlap; the claim ceiling would depend on independent evidence for those assumptions.

## Study in the Larger Evidence Program

The pilot teaches responsiveness reasoning by specifying an intervention, but its observed exposure is self-selected and is **not** a randomized responsiveness test. A next psychology study could randomize an *offer* to eligible volunteers, prespecify the memory outcome and follow-up, measure pre-choice baseline and attrition, and report the intention-to-treat effect of the offer. That experiment would address a related but distinct intervention and estimand. A complementary observational replication with better fatigue measurement, different recruitment sites, and a preregistered negative-control prediction would test the proposed graph from another angle. Consistency across biased studies is not strong triangulation; designs should have meaningfully different vulnerabilities. Mechanism work should test a prediction about engagement with independent measures and timing, not retrofit a story to a favorable coefficient.

## Six principal problems

For each, write your answer, reason, confidence from 1 to 5, and one revision cue before reading the solutions.

**Problem 3.1 — Recall and recognize.** Identify B, M, C, U, A, and Y from the figure. Which recorded variable is a common cause of A and Y? Which recorded variable is a post-choice collider, and what two arrows meet there?

**Problem 3.2 — Predict before calculating.** Given that higher baseline memory makes training more likely and also predicts later memory, predict whether the naive training comparison will be above or below a B-adjusted comparison. State one reason that the difference between estimates alone would not prove the DAG correct in a human study.

**Problem 3.3 — Calculate and explain.** Use the reported baseline-adjusted estimate and HC1 standard error to calculate a normal-approximation 95% interval as estimate  $\pm 1.96 \times \text{SE}$ . Round to three decimals. Describe one form of uncertainty the interval omits.

**Problem 3.4 — Read output.** Compare the baseline-only, baseline-plus-C, and baseline-plus-M coefficients. Which is aimed at the total-effect estimand? Why does adding C alter the first comparison? Why does adding M alter the scientific question even if the coefficient is smaller?

**Problem 3.5 — Choose and reject methods.** A colleague proposes “control for baseline, engagement and help seeking, because all three predict memory.” Choose the pilot’s prespecified adjustment set for the total effect, reject both additions by causal role, and name the extra assumptions needed before a direct-effect analysis could be defended.

**Problem 3.6 — Report and critique.** Repair: “Regression proves training improves human memory by 2.000 points; the narrow interval and cross-language code rule out bias.” Write two defensible sentences about the *synthetic* study, name an unresolved assumption for analogous human work, and identify the next study that would most reduce uncertainty.

## Fully worked solutions

**Solution 3.1. Answer:** A is chosen training; Y is follow-up memory; B is pre-choice baseline memory; M is post-choice engagement; C is post-choice help seeking; U is unrecorded fatigue. B is a common cause because  $B \rightarrow A$  and  $B \rightarrow Y$ . C is a collider because  $A \rightarrow C \leftarrow U$ . **Reason:** arrow direction and timing, not correlation magnitude, determine each role. Conditioning on B closes  $A \leftarrow B \rightarrow Y$ ; conditioning on C opens  $A \rightarrow C \leftarrow U \rightarrow Y$ . **Confidence and revision:** assign 1–5; if you called every predictor a confounder, redraw the two arrows into C before choosing controls.

**Solution 3.2. Answer:** Above. The naive estimate is 8.352, while the B-adjusted estimate is 2.000. **Reason:** selection favors volunteers with higher baseline memory, which also improves follow-up memory, biasing the unadjusted training comparison upward in this simulator. The observed difference between these two coefficients is consistent with the assumed path but does not by itself establish that no other backdoor path exists in real people. **Confidence and revision:** rate 1–5; trace B’s arrows and check the direction of each before reading a coefficient.

**Solution 3.3. Answer:**  $2.000 \pm 1.96 \times 0.195$  gives [1.617, 2.382], subject to rounding of inputs shown in the table; the unrounded record reproduces the endpoints. **Reason:** the calculation is a model-based normal interval with HC1 SE. It does not include uncertainty about unmeasured

confounding, the DAG, measurement validity, or selection. **Confidence and revision:** rate 1–5; identify what generated the SE and what was assumed before interpreting its interval causally.

**Solution 3.4. Answer:** B only targets the total effect under the stated graph: 2.000. Adding C gives 4.691 because conditioning on  $A \rightarrow C \leftarrow U$  opens an association between A and fatigue U that influences Y. Adding M gives 0.901, which removes part of  $A \rightarrow M \rightarrow Y$  from the coefficient on A.

**Reason:** all three regressions are arithmetically valid conditional comparisons, but only the B-only specification was selected to estimate this total effect. The C and M coefficients are not alternate “best model” votes. **Confidence and revision:** rate 1–5; classify a variable by causal timing and incoming/outgoing arrows before interpreting model fit.

**Solution 3.5. Answer:** Adjust for B, and omit M and C for the total effect. **Reason:** B blocks the measured backdoor path; C opens a noncausal path; M is a mediator and conditioning on it changes the target. A direct-effect question would need a precise intervention on M and assumptions about mediator–outcome confounding, including no treatment-induced common cause of M and Y, consistency, and positivity. The B-plus-M coefficient is not an automatically identified natural direct effect. **Confidence and revision:** rate 1–5; when someone says “control for everything,” draw the proposed intervention and trace each path first.

**Solution 3.6. Answer:** “In this synthetic voluntary-cohort example, the baseline-adjusted coefficient is 2.000 memory-score points with an approximate 95% HC1 interval [1.617, 2.382]. It represents the total-effect estimand for these fictional eligible volunteers only under the simulator’s graph, adequate measurement, exchangeability, positivity, and other specified assumptions; it is not evidence about human memory.” A relevant next study could randomize a prespecified *offer* among actual eligible volunteers and measure attrition, while noting that the offer’s ITT estimand differs from voluntary training’s effect. **Reason:** code reproduction checks arithmetic; it cannot establish exchangeability or generalize the sample. **Confidence and revision:** rate 1–5; replace “proves,” state population and intervention, name the remaining identification vulnerability, and specify a study that addresses it.

## Unit 4 Randomized Experiments

### Purpose and objectives

Random assignment is a design property, not a label attached to a regression. This unit follows a randomized experiment from assignment through analysis and claim. By the end, you should be able to distinguish the unit of observation from the unit of assignment; define an intention-to-treat effect before examining receipt; preserve blocking in an analysis; choose uncertainty that reflects the number of independently randomized units; use baseline adjustment for precision without conditioning on post-assignment variables; and explain how contamination, attrition, and repeated measurements affect different questions.

### The synthetic residence-floor sleep study

A fictional university tests an eight-week sleep-support program. Its 48 residence floors are grouped into four campus sectors—North, East, South, and West—with 12 floors per sector. Within every sector, software selects exactly 6 floors for the program and 6 for usual sleep information. All 50 enrolled students on a floor receive the floor’s assignment, giving 2400 student records and 48 randomized clusters.

Before assignment, the study records usual sleep hours and an attention score. Program floors then offer planning prompts, peer reminders, and a structured sleep plan; comparison floors retain the usual information. Some students on program floors do not use the plan, and some comparison students obtain similar planning tools elsewhere. At eight weeks, the primary attention score is recorded for every student. An optional sleep diary supplies a secondary follow-up sleep measure only when it is completed.

**Scientific question.** For the fictional enrolled cohort, what would assigning residence floors to offer the sleep-support program rather than usual information change in mean eight-week attention?

**Primary estimand.** The intention-to-treat effect of floor assignment,

$$\tau_{\text{ITT}} = E[Y(1) - Y(0)],$$

where  $Y(1)$  is a student’s eight-week attention score under assignment of their floor to the program offer and  $Y(0)$  is the score under assignment to usual information. This is an effect of the assignment policy under the study’s realized patterns of use and contamination. It is not the effect of personally using a sleep plan.

**Units and time zero.** A CSV row is a student, but a residence floor is the unit randomized. Campus sector is the randomization block. Baseline measures precede time zero; floor assignment occurs at time zero; plan use occurs during the intervention; and attention and optional diary outcomes follow eight weeks. The analysis must respect this order.

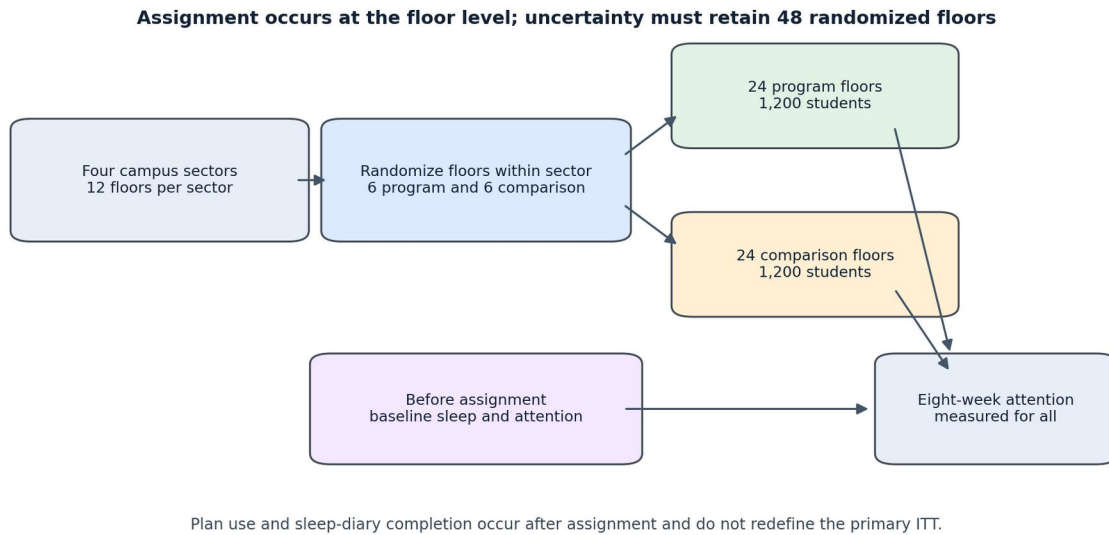


Figure 4. Blocked cluster-randomization timeline: 48 residence floors are randomized within four campus sectors, baseline measures precede assignment, and eight-week attention is measured for all 2,400 students.

### Four design choices that should not be blended

| Design choice                      | What is randomized or compared?                       | Main analytical consequence                                                     |
|------------------------------------|-------------------------------------------------------|---------------------------------------------------------------------------------|
| Simple individual randomization    | Individual people without blocks                      | Independent units are people if interference and other assumptions are credible |
| Blocked individual randomization   | People randomized within pre-assignment groups        | Preserve blocks to match the design and often gain precision                    |
| Cluster randomization              | Groups such as floors, schools, or clinics            | Assignment and uncertainty operate at the group level                           |
| Baseline and follow-up measurement | The same outcome measured before and after assignment | Baseline adjustment can improve precision; follow-up remains the outcome        |

This study combines blocking, cluster assignment, and repeated measurement. Those features answer different problems. Blocking guarantees equal program-floor counts within sector. Cluster assignment makes the floor—not each student—the independent randomized unit. Baseline attention supplies prognostic information measured before assignment. None of these features makes later program use random.

## Assignment is not receipt

The offer reaches floors, while students decide whether to use the plan. Among students on assigned floors, 67.4% used it; among comparison-floor students, 13.1% did so. That is nonadherence in one arm and contamination in the other. It changes the intervention represented by the ITT: assignment produces access and encouragement, not universal receipt.

Analyzing everyone by assigned floor preserves the randomized comparison. Restricting the program group to users or comparing all users with nonusers breaks it. Use can depend on motivation, baseline sleep, peer behavior, or early response. The user/nonuser coefficient is therefore an association after adjustment for the variables in its model—not an identified effect of receipt.

### Pause and Decide

Before reading the estimates, choose the primary row from five candidates: an individual-row assignment regression, a floor-level blocked assignment regression, a floor-level baseline-adjusted assignment regression, a floor-level change-score regression, and an individual user/nonuser regression. Write the intervention, population, unit of assignment, and source of independent information for each. Mark any post-assignment variable. Then state which analysis you would prespecify and why.

## Identification assumptions, separate from the statistical model

The blocked randomized design supports exchangeability of floor assignment within campus sector if the allocation record was generated and implemented as stated. A causal ITT interpretation also requires consistency for the program and usual-information versions; no relevant interference across floors; no post-assignment exclusion from the analysis; correct linkage of students to floors and assignments; and valid, assignment-neutral measurement of the primary attention outcome. With only 48 randomized floors, uncertainty must reflect floors rather than pretend that 2,400 assignments occurred.

The estimand can include within-floor spillovers because the policy assigns an entire floor. Cross-floor spillovers are different: if comparison students are materially affected by neighboring program floors, the contrast no longer represents isolated floor assignments. The exact treatment version also matters. A future program without peer reminders would not automatically inherit this estimand.

The complete primary outcome avoids an attrition problem for attention. The optional diary does not. Completion is 79.9% on program floors and 73.6% on comparison floors, a 6.3% difference. A complete-case comparison of follow-up sleep would additionally require assumptions about post-assignment observation. Randomization does not by itself repair outcome missingness created after assignment.

## Estimation and uncertainty

**Orientation.** The design randomizes floors, not students, so the arithmetic that follows answers a question about floors and then reports it on a per-student scale. Read the estimate as a contrast between two sets of floor means, and read the standard error as a statement about how those floor means would vary on repetition. Unit 1's difference in means is still the target; only the unit of dependence has changed.

The preferred unadjusted analysis first averages follow-up attention within each floor, then regresses the 48 floor means on assignment and campus-sector indicators. Because all floors contain 50 students, this unweighted floor analysis targets the student-average ITT in this balanced study. Its HC1 standard error is computed across 48 floor rows, and the interval uses the residual degrees of freedom, 43.

A prespecified precision analysis adds the floor mean of baseline attention. Baseline is safe to use as a precision covariate because it precedes assignment. The adjusted coefficient targets the same assignment effect if the working model is used as planned, but it should not replace the design definition of the estimand. A change-score analysis subtracts baseline floor means from follow-up floor means and is included as a transparent sensitivity analysis; it need not be more precise than covariance adjustment.

The individual-row assignment regression includes sector and produces the same point estimate here because every floor is the same size. Its ordinary HC1 standard error treats students as independent assignments. That standard error is a deliberate negative example. A model can print the correct coefficient while representing the wrong randomization uncertainty.

## Worked analysis from the CSV

| Analysis                              | Estimate | SE    | Approximate 95% interval |
|---------------------------------------|----------|-------|--------------------------|
| Preferred floor-level blocked ITT     | 3.201    | 1.050 | [1.082, 5.319]           |
| Floor-level baseline-adjusted ITT     | 3.313    | 0.293 | [2.721, 3.905]           |
| Floor-level change-score ITT          | 3.271    | 0.480 | [2.302, 4.239]           |
| Individual-row assignment coefficient | 3.201    | 0.303 | [2.607, 3.795]           |
| Naive observed-use association        | 1.779    | 0.248 | [1.294, 2.264]           |

The preferred floor-level estimate is 3.201 attention points, with approximate 95% interval [1.082, 5.319]. The baseline floor means differ by only -0.070 points in this realization, but balance is not the justification for adjustment. The prespecified baseline-adjusted estimate is 3.313, with a smaller standard error because baseline attention predicts follow-up attention. The change-score result, 3.271, is compatible but less precise than the covariance-adjusted row.

The numerical proximity of the baseline-adjusted floor-level SE 0.293 and the invalid individual-row SE 0.303 is coincidental. The former is small because the prespecified floor-level baseline adjustment accounts for substantial between-floor outcome variation. The latter is small because it incorrectly treats student rows as independent assignments. Similar numerical values do not make the two uncertainty calculations equivalent.

The individual-row coefficient is also 3.201, yet its HC1 standard error is 0.303 rather than 1.050. The former is only 0.288 of the latter. Forty-eight randomized floors, not 2,400 student rows, determine the assignment information. The narrow individual-row interval is not supporting evidence; it is a demonstration of unit-of-analysis error.

The observed-use association is 1.779 after adjustment for baseline attention and sector. It is smaller than the ITT here, but neither its size nor direction repairs selection into use. The simulator contains hidden motivation that affects use and attention. An analyst does not observe that variable, and the book never treats the generator's hidden structure as empirical knowledge.

## Contamination and attrition belong to named outcomes

Receipt rates of 67.4% and 13.1% show that assignment strongly changes access to the plan without perfectly determining use. Those rates describe implementation and help interpret the policy effect. They do not authorize an as-treated causal claim. A complier estimand would require a separate instrumental-variable argument, developed later in Unit 7, including relevance, independence, exclusion, and monotonicity.

Diary completion differs by 6.3%. That finding threatens a causal comparison of follow-up sleep among observed diaries because completion follows assignment and may depend on student characteristics. It does not create missingness in the complete attention outcome. Avoid the vague claim that “the study has attrition.” Name the outcome, observation process, and estimand affected.

## Diagnostics and falsification

Reconstruct the assignment table before fitting an outcome model. Each sector must contain 6 program and 6 comparison floors, every student on a floor must share its assignment, every floor must contain 50 students, and baseline times must precede allocation. Check unique student identifiers, plausible ranges, complete primary outcomes, and the declared rule linking blank follow-up sleep values to diary noncompletion.

Inspect assignment implementation and spillover opportunities by floor. Compare prespecified baseline summaries as diagnostics for coding or allocation failures, not as significance tests that decide whether randomization “worked.” Report floor counts with every clustered result. Reanalyze with the baseline-adjusted and change-score specifications chosen before examining outcomes. If cluster sizes differed, document the population weighting implied by floor-level versus individual-level estimators.

Negative-control outcomes measured before assignment can reveal impossible treatment effects caused by linkage errors. They cannot prove absence of interference or validate the attention construct. Likewise, agreement among three estimators does not prove their assumptions; disagreement is a cue to examine their targets, weights, and precision models.

### Claim Audit

**Supports:** a reproducible intention-to-treat estimate for offering the fictional sleep-support program to residence floors in this enrolled cohort, under the blocked cluster-randomized design and the stated consistency, interference, implementation, and measurement assumptions.

**Does not establish:** the causal effect of personally using a sleep plan; an effect of assignment in real universities; an individual treatment effect for every student; an effect on sleep among diary

completers without missingness assumptions; absence of cross-floor interference; or that the attention scale measures lasting educational benefit.

**Strongest defensible conclusion:** “In this synthetic blocked cluster-randomized study, assignment of residence floors to offer the sleep-support program is associated by design with an estimated 3.201-point increase in mean eight-week attention, approximate floor-level 95% interval [1.082, 5.319]. This is an intention-to-treat effect of assignment in fictional data, not evidence about real students or the causal effect of individual plan use.”

### Study in the Larger Evidence Program

This experiment clarifies whether an implementable floor-level offer can move a complete primary outcome. A next study should prespecify cluster-aware power, monitor cross-floor exposure, standardize delivery, and collect a delayed outcome. If sleep is a primary mechanism, make follow-up sleep ascertainment equally intensive in both arms or plan a transparent missing-data analysis. If the scientific target is receipt rather than assignment, design the encouragement and measurement needed for a credible instrumental-variable analysis instead of relabeling an as-treated regression.

## Six principal problems

For each, write your answer, reason, confidence from 1 to 5, and one revision cue before reading the solutions.

**Problem 4.1 — Identify the randomized unit.** Explain why 2,400 rows do not represent 2,400 independent assignments. Name the block, cluster, observation unit, and primary outcome.

**Problem 4.2 — Define assignment and receipt effects.** State the ITT in potential-outcome notation. Explain what the observed-use coefficient estimates descriptively and why it does not identify an effect of use.

**Problem 4.3 — Reconstruct the preferred interval.** Using estimate 3.201, HC1 SE 1.050, and  $t$  critical value 2.017, calculate the approximate 95% interval and interpret it.

**Problem 4.4 — Diagnose false precision.** Compare the individual-row and floor-level standard errors. Why can their point estimates agree while their intervals differ so much?

**Problem 4.5 — Separate baseline adjustment from attrition.** Explain why baseline attention may be used for precision, why the complete attention outcome has no attrition problem, and why a diary-completer analysis of follow-up sleep needs extra assumptions.

**Problem 4.6 — Repair an overclaim.** Rewrite: “Students who use the sleep plan gain 3.201 attention points, and the narrow student-level interval proves the program works.”

## Fully worked solutions

**Solution 4.1. Answer:** campus sector is the block; residence floor is the randomized cluster; student is the observation unit; eight-week attention is the complete primary outcome. There are 48 independent assignment units, with 6 program and 6 comparison floors in each sector. **Reason:** all students on a floor inherit one assignment and share contextual shocks, so adding students within a floor does not create new randomizations. **Confidence and revision:** rate 1–5; annotate every design with separate rows for assignment, observation, and analysis units.

**Solution 4.2. Answer:**  $\tau_{ITT} = E[Y(1) - Y(0)]$ , where 1 and 0 denote the floor's assignment to program offer or usual information. The 1.779 coefficient describes the adjusted association between observed plan use and attention among individual rows. **Reason:** use occurs after assignment and depends on student decisions and characteristics. Only assignment was randomized. **Confidence and revision:** rate 1–5; put a time-zero line in the DAG and reject any primary adjustment variable to its right.

**Solution 4.3. Answer:**  $3.201 \pm 2.017 \times 1.050$  gives [1.082, 5.319], subject to rounding. The interval summarizes repeated-sampling uncertainty for the specified floor-level blocked estimator; it does not include uncertainty about interference, measurement validity, or transport to real universities.

**Reason:** the t critical value uses 43 residual degrees of freedom from 48 floor rows and four sector indicators plus assignment. **Confidence and revision:** rate 1–5; write the randomized-unit count beside every standard error.

**Solution 4.4. Answer:** the individual-row SE is 0.303, only 0.288 of the floor-level SE 1.050. Equal cluster sizes make the two assignment coefficients equal, but the individual HC1 calculation acts as if 2,400 assignments supplied information. **Reason:** students on the same floor share assignment and a floor context. Correct point estimation does not guarantee correct uncertainty. **Confidence and revision:** rate 1–5; trace the randomization mechanism before selecting a variance estimator.

**Solution 4.5. Answer:** baseline attention precedes assignment and can improve precision when prespecified. Follow-up attention is present for all 2,400 students, so the primary ITT is not selected by outcome completion. Follow-up sleep is observed only among diary completers, whose rates are 79.9% and 73.6% by arm. **Reason:** diary completion is post-assignment and can be related to unobserved sleep or motivation, so a complete-case sleep contrast needs missingness assumptions or a justified sensitivity analysis. **Confidence and revision:** rate 1–5; describe missingness separately for every outcome rather than for “the dataset.”

**Solution 4.6. Answer:** “In this synthetic trial, assigning a residence floor to offer the sleep-support program produced an estimated 3.201-point change in mean eight-week attention, floor-level approximate 95% interval [1.082, 5.319], under the stated assumptions.” **Reason:** the original substitutes use for assignment, assigns an average to every student, uses an invalid student-level standard error, treats an interval as proof, and implies evidence about real people. **Confidence and revision:** rate 1–5; underline intervention, averaging population, uncertainty unit, time, and evidence source in every conclusion.

## Unit 5 Observational Adjustment and Standardization

### Purpose and objectives

An observational analysis can make unlike groups more comparable on measured baseline variables, but it cannot manufacture randomization. This unit uses two standardization methods to estimate the same eligible-cohort contrast. By the end, you should be able to define the target population before choosing a model; distinguish outcome-regression standardization from inverse-probability weighting; explain what a propensity score does and does not represent; inspect overlap, weights, effective sample size, and measured-covariate balance; use a pre-exposure negative-control outcome as a falsification diagnostic; and write a claim that reflects residual-confounding risk even when two estimators agree.

### The synthetic community hypertension-coaching study

Three fictional community clinics offer a 12-week hypertension-coaching program to 3200 eligible adults. The program combines weekly planning, blood-pressure self-monitoring, and medication-adherence support. Joining is voluntary. 1308 adults join, a participation rate of 40.9%. Everyone remains in the study whether or not they join, and follow-up systolic blood pressure is complete.

At baseline, the clinics record systolic blood pressure, age, current smoking, blood-pressure medication, transport barriers, and clinic. The CSV also contains a prior-year preventive-visit score measured before coaching was offered. That score is reserved as a negative-control outcome: a program offered at time zero cannot cause visits that already occurred.

**Scientific question.** Among all adults eligible at the three clinics during the enrollment window, what would mean 12-week systolic blood pressure be if everyone joined coaching compared with if no one joined?

**Primary estimand.** The eligible-cohort average treatment effect,

$$\tau = E[Y(1) - Y(0)],$$

where  $Y(1)$  is 12-week systolic blood pressure if an eligible adult joins this coaching program and  $Y(0)$  is the outcome if that adult does not join. Lower values favor coaching. This is not an ATT among the people who chose to join and not an effect of an offer policy.

**Units and time zero.** A row is one eligible adult. Prior-year visits and all baseline covariates precede time zero. Voluntary participation is defined at time zero. Follow-up systolic blood pressure is measured 12 weeks later. No variable caused by participation belongs in the primary adjustment set.

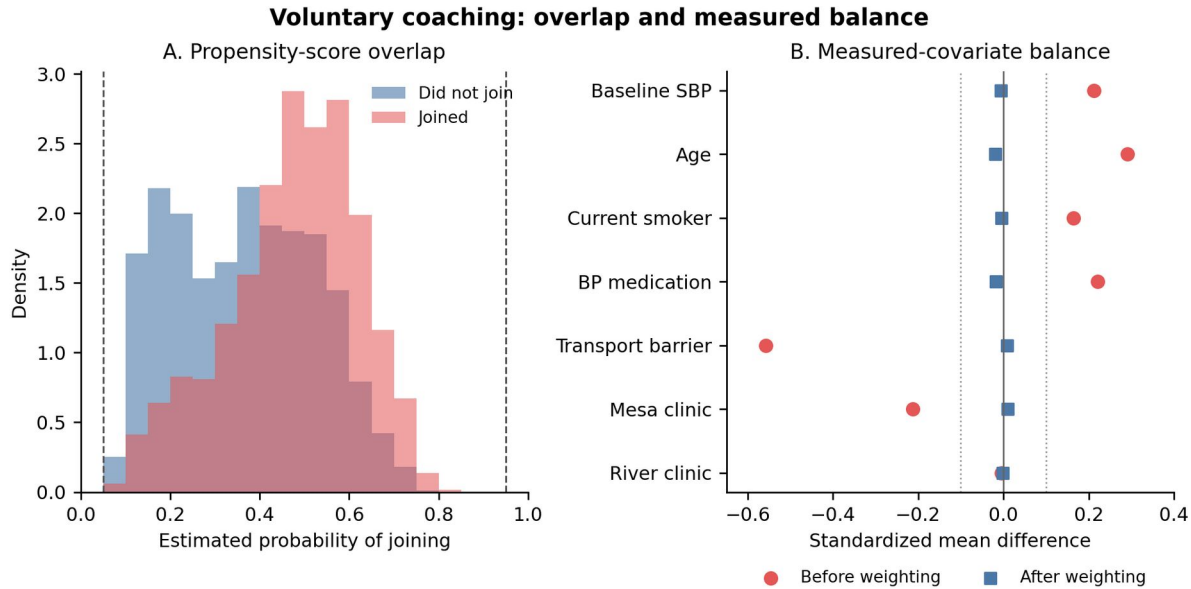


Figure 5. Voluntary hypertension-coaching diagnostics: participant and nonparticipant propensity-score distributions overlap from about 0.06 to 0.81, while inverse-probability weighting reduces the largest absolute standardized mean difference across seven measured baseline covariates from 0.557 to 0.018.

## What the adjustment set is trying to accomplish

Design knowledge supplies the adjustment set. Baseline systolic blood pressure, age, smoking, medication, transport barriers, and clinic can affect participation and follow-up blood pressure. Call these measured covariates  $X$ . The intended backdoor strategy is to compare people with the same  $X$  values and then average those conditional comparisons over the full eligible cohort.

The required graph is compact:  $X$  points to participation  $A$  and outcome  $Y$ ; participation points to  $Y$ ; and unmeasured health engagement  $U$  may also point to  $A$  and  $Y$ . Prior preventive visits  $V$  occur before  $A$  and may respond to  $U$ . Adjustment closes paths through measured  $X$  only. It cannot close  $A \leftarrow U \rightarrow Y$  when  $U$  is not adequately represented by  $X$ .

| Quantity                                   | Role in this unit                                                       | What it cannot establish                                                   |
|--------------------------------------------|-------------------------------------------------------------------------|----------------------------------------------------------------------------|
| Outcome model $E[Y \text{ given } A, X]$   | Predict each eligible adult's outcome under both exposure levels        | That its functional form is correct or $U$ is absent                       |
| Propensity score $P(A=1 \text{ given } X)$ | Describe measured participation patterns and construct weights          | The probability of a beneficial response or a guarantee of exchangeability |
| Weighted balance on $X$                    | Check whether the fitted weights align measured covariate distributions | Balance on unmeasured causes or correct outcome measurement                |
| Prior-year visits $V$                      | Negative-control outcome that participation cannot cause                | Which omitted variable caused an association or how large the bias is      |

## Pause and Decide

Before seeing the estimates, write the target population, treatment versions, outcome time, and averaging rule. List the six prespecified measured covariates and draw the remaining U path. Then choose whether you would trust an unadjusted difference, outcome-regression standardization, weighting, or none as a causal answer. State what result would make you revise that choice: poor overlap, extreme weights, residual imbalance, a pre-exposure association, estimator disagreement, or a combination.

## Identification assumptions, separate from the estimators

A causal interpretation requires consistency: joining and not joining must refer to treatment versions sufficiently well defined for the question. Conditional exchangeability requires  $Y(a)$  to be independent of actual participation after conditioning on the measured baseline covariates. Positivity requires a nonzero chance of both participation states for every covariate pattern in the target population. The analysis also assumes no relevant interference among adults, accurate timing and measurement, and complete follow-up under a defensible observation process.

These are scientific assumptions. A high C statistic, small p value, or balanced weighted table cannot verify them. Measured balance is necessary for this weighting implementation but not sufficient for conditional exchangeability. The negative-control analysis is valuable because it can contradict a simple causal story, not because passing it would certify the story.

## Two routes to the same standardized target

**Orientation.** Two estimators follow, and they are not a robustness check on one another in the usual sense. They rest on the same identification assumptions and differ only in what gets modeled: one models the outcome, the other models participation. Agreement between them shows that the modeling choice did not drive the answer. It does not show that the adjustment set was sufficient, which is the question Unit 10 returns to with explicit parameters.

### Outcome-regression standardization

The primary outcome model regresses follow-up systolic blood pressure on participation, the six baseline covariates, and a participation-by-baseline-pressure interaction. The interaction allows the conditional contrast to differ across baseline pressure. The analysis then performs three explicit steps:

1. Set participation to 1 for every eligible adult and predict 12-week pressure.
2. Set participation to 0 for those same adults and predict again.
3. Average each prediction column over all 3200 adults and subtract the two averages.

The standardized means are 131.130 mm Hg under universal participation and 137.272 mm Hg under no participation. Their difference is -6.142 mm Hg. Because both prediction columns use the same eligible adults, the target distribution is visible rather than hidden inside a regression coefficient.

### Inverse-probability weighting

The propensity model estimates each adult's probability of joining from the same six baseline covariates. A participant receives weight  $1/e(X)$ ; a nonparticipant receives weight  $1/[1-e(X)]$ . Within each exposure group, normalized weights reconstruct the measured covariate distribution of the

eligible cohort. The weighted means are 131.116 and 137.264 mm Hg, giving a Hájek contrast of -6.149 mm Hg.

Outcome regression models the outcome surface and standardizes predictions. IPW models participation and standardizes the covariate distribution. Neither estimator is automatically safer. Both depend on the covariates being sufficient, their measurements being credible, and the target population being represented where both exposure states are possible.

## Worked analysis from the CSV

| Contrast or diagnostic                                         | Estimate | SE    | Approximate 95% interval |
|----------------------------------------------------------------|----------|-------|--------------------------|
| Unadjusted participant minus nonparticipant difference, mm Hg  | -5.322   | 0.352 | [-6.012, -4.632]         |
| Outcome-regression standardized ATE, mm Hg                     | -6.142   | 0.268 | [-6.668, -5.616]         |
| IPW Hájek standardized ATE, mm Hg                              | -6.149   | 0.288 | [-6.713, -5.584]         |
| Adjusted association with prior-year visits, visit-score units | 0.309    | 0.033 | [0.246, 0.373]           |

Participants' observed mean follow-up pressure is 131.578 mm Hg; the nonparticipant mean is 136.900 mm Hg. The raw difference is therefore -5.322 mm Hg. Adjustment makes the estimated contrast more negative because participants in this realization also have higher baseline pressure, are older, smoke more often, use medication more often, and report fewer transport barriers. These measured differences do not all bias in the same direction.

Outcome-regression standardization estimates -6.142 mm Hg, approximate 95% interval [-6.668, -5.616]. IPW estimates -6.149, interval [-6.713, -5.584]. Their close agreement shows that two different implementations of measured-covariate standardization give similar answers in this dataset. It does not make the study randomized, and it does not show that both models omitted nothing important.

The intervals use HC1 and sandwich calculations for sampling uncertainty under the fitted procedures. They do not include uncertainty about the causal graph, treatment versions, unmeasured health engagement, or the meaning of systolic pressure as a longer-term health outcome.

**Simulator truth, for teaching audit only.** The generator sets each adult's individual effect to  $-4.2 - 0.85 \times (\text{baseline systolic pressure} - 137)/10$  mm Hg. Averaging that hidden formula over the realized eligible cohort gives a sample ATE of -4.227 mm Hg; among the adults who participated, the hidden ATT is -4.328 mm Hg. The naive, outcome-regression, and IPW estimates differ from the hidden sample ATE by -1.095, -1.915, and -1.922 mm Hg, respectively. In this simulated realization, measured adjustment removed offsetting positive confounding while latent engagement continued to bias the contrast downward; consequently, both adjusted estimates are farther from the hidden sample ATE than the naive difference. This accidental bias cancellation does not make the naive estimator preferable, and the hidden answer would not be available in a real observational study. The negative control warned that residual confounding remained plausible, but it could not calculate either the direction or the magnitude of the bias.

## Overlap, weights, and balance

The fitted participation probabilities range from 0.062 to 0.815; the first and 99th percentiles are 0.102 and 0.726. No fitted score lies below 0.05 or above 0.95, so the diagnostic does not show an immediate empirical positivity failure. It also cannot prove structural positivity for covariate combinations absent from this finite cohort.

The largest observed inverse-probability weight is 16.067. Weighting reduces the participant effective sample size from 1308 to 1001.798 and the nonparticipant effective sample size from 1892 to 1741.615. Those losses are the precision cost of asking some less common covariate patterns to represent more people. Report them; do not hide them behind the original row count.

Before weighting, the largest absolute standardized mean difference is 0.557, driven by transport barriers. After weighting, the maximum is 0.018 across the seven encoded baseline terms. That is strong measured balance. The word *measured* is doing essential work.

## Diagnostics and falsification

Start by confirming eligibility, complete follow-up, unique identifiers, covariate timing, and plausible ranges. Inspect propensity-score distributions by participation group, not only a single minimum. Tabulate weights, their tails, and arm-specific effective sample sizes. Plot every prespecified measured-covariate standardized mean difference before and after weighting. If balance remains poor, revise the propensity specification without consulting the outcome and explain the change.

Next, compare the outcome-regression and IPW answers. Large disagreement can reveal extrapolation or model dependence. Close agreement is reassuring about implementation but weak as a falsification test because both analyses use the same measured covariates and share the same exchangeability assumption.

The prior-year preventive-visit score provides a sharper check. After adjustment for the prespecified X set, later coaching participation remains associated with 0.309 more prior visit-score units, interval [0.246, 0.373]. Coaching cannot cause an outcome measured before it began. The association could reflect residual health-seeking differences, measurement structure, or model misspecification. It does not identify a specific omitted cause or quantify the bias in blood pressure, but it prevents measured balance from being presented as proof that confounding is solved.

## Claim Audit

**Supports:** reproducible descriptions of participation; measured-covariate standardized outcome contrasts for the complete fictional eligible cohort; documented overlap, weights, effective sample sizes, and balance; and a negative-control warning about the credibility of conditional exchangeability.

**Does not establish:** random participation; balance on latent health engagement; that the propensity score is a health-risk score; that the negative-control association equals the bias in the blood-pressure contrast; a real clinical effect; transport beyond the three fictional clinics; or a causal effect for people with covariate patterns not represented in both groups.

**Strongest defensible conclusion:** “In this synthetic observational cohort, outcome-regression standardization and inverse-probability weighting both estimate about a 6.1 mm Hg lower mean 12-week systolic blood pressure under universal coaching participation than under no participation after adjustment for six measured baseline covariates. Participation was voluntary, and its adjusted association with a pre-exposure negative-control outcome leaves residual confounding plausible. The result is therefore an adjusted observational contrast whose causal interpretation depends on assumptions not established by model agreement or measured balance.”

## Study in the Larger Evidence Program

The next study should randomize an offer or encouragement while retaining complete follow-up for everyone eligible. Measure health engagement before time zero, record reasons for participation, and reduce transport barriers so relevant covariate patterns have both participation options. A randomized offer would identify an ITT for access; it would not automatically identify the effect of receipt. If voluntary participation remains the target, combine richer baseline measurement with prespecified negative controls and a sensitivity analysis for unmeasured confounding rather than treating another regression specification as independent evidence.

## Six principal problems

For each, write your answer, reason, confidence from 1 to 5, and one revision cue before reading the solutions.

**Problem 5.1 — Define the observational estimand.** Name the intervention, comparison, target population, outcome time, and averaging rule. Explain why the participant ATT and the eligible-cohort ATE are different questions.

**Problem 5.2 — Reconstruct standardization.** Use the outcome-regression standardized means 131.130 and 137.272 to calculate the ATE. Interpret the sign and unit.

**Problem 5.3 — Compare outcome regression and IPW.** Explain how each method standardizes to the eligible cohort. Does their numerical agreement establish exchangeability?

**Problem 5.4 — Audit overlap and precision.** Interpret the propensity range, maximum weight, and arm-specific effective sample sizes. Which findings are reassuring, and what remains uncertain?

**Problem 5.5 — Use the negative control.** Why can coaching not cause prior-year visits? What does the adjusted 0.309 association challenge, and what does it fail to quantify?

**Problem 5.6 — Repair an overclaim.** Rewrite: “After propensity-score weighting balanced the groups, coaching caused a -6.149 mm Hg reduction in blood pressure.” Include the target population, adjustment basis, uncertainty, negative-control result, and synthetic-data boundary.

## Fully worked solutions

**Solution 5.1. Answer:** The intervention is joining the defined 12-week coaching program rather than not joining; the target is all 3200 adults eligible at the three clinics; the outcome is 12-week systolic blood pressure; and the estimand averages  $Y(1) - Y(0)$  over that eligible cohort. An ATT would average over the covariate distribution of the 1308 people who actually joined. **Reason:** An acronym does not fix its population. Voluntary participants differ from all eligible adults, so changing the averaging distribution changes the question. **Confidence:** Give 5 only if every element and the time ordering are explicit. **Revision:** Add the missing treatment version, population, outcome time, or averaging rule.

**Solution 5.2. Answer:**  $131.130 - 137.272 = -6.142$  mm Hg after rounding. The negative sign means lower standardized 12-week pressure under universal participation. **Reason:** Standardization predicts both exposure states for the same eligible adults before averaging; it does not compare the observed participant mean with the observed nonparticipant mean. **Confidence:** Give 5 if you subtract in the declared joined-minus-not-joined order and retain mm Hg. **Revision:** Write both standardized means and the subtraction before interpreting the sign.

**Solution 5.3. Answer:** Outcome regression predicts  $Y$  under  $A=1$  and  $A=0$  for every eligible adult, then averages. IPW uses inverse participation probabilities to make each exposure group represent the measured covariate distribution of the eligible cohort. Their estimates, -6.142 and -6.149, are close, but agreement does not establish exchangeability. **Reason:** The methods emphasize different nuisance models yet use the same  $X$  set and share the assumption of no unmeasured confounding given  $X$ . They can agree while sharing an omission. **Confidence:** Give 5 only if the distinction between estimator agreement and identification is explicit. **Revision:** Name one shared assumption before calling agreement reassuring.

**Solution 5.4. Answer:** Scores from 0.062 to 0.815 show observed overlap without fitted values past 0.05 or 0.95. The maximum weight 16.067 and effective samples 1001.798 and 1741.615 show that some observations carry extra influence and precision is lower than the row counts imply. **Reason:** Positivity diagnostics concern where comparisons are supported; effective sample size describes weight concentration. Finite observed overlap cannot prove both exposure options exist for every substantively relevant pattern. **Confidence:** Give 5 if the answer is neither “overlap is perfect” nor “the weights invalidate everything.” **Revision:** Report the range, tail weight, and both effective sample sizes together.

**Solution 5.5. Answer:** Prior-year visits occur before coaching is offered, so later participation cannot cause them. The adjusted 0.309 association challenges the claim that the measured covariates fully remove pre-existing participant differences. It does not identify the omitted cause or equal the bias in the blood-pressure estimate. **Reason:** A negative-control outcome tests an impossible temporal effect. Failure is evidence against at least part of the working identification story, but mapping that failure to the primary outcome needs additional structure. **Confidence:** Give 5 if timing, warning, and limitation all appear. **Revision:** Remove any claim that the negative control directly corrects the ATE.

**Solution 5.6. Answer:** “Among 3200 eligible adults in this synthetic three-clinic cohort, normalized inverse-probability weighting based on six measured baseline covariates estimates a -6.149 mm Hg joined-minus-not-joined contrast in mean 12-week systolic blood pressure, approximate 95% interval [-6.713, -5.584]. Measured balance improved, but voluntary participation remained associated with a pre-exposure negative-control outcome. A causal interpretation therefore requires conditional exchangeability and the other stated assumptions; the result is not evidence about real patients.”

**Reason:** The repair retains the estimate while restoring population, method, uncertainty, falsification result, assumptions, and synthetic boundary. **Confidence:** Give 5 only if “caused” is removed or clearly made conditional. **Revision:** Add whichever claim limit the original sentence concealed.

## Unit 6 Changes Over Time

### Purpose and objectives

When an intervention begins at a known time, a before-after change is visible but rarely self-interpreting. Other events, seasonality, and trends move outcomes too. This unit combines a phased municipal policy with a contemporaneous region-wide intervention to teach difference-in-differences (DiD), municipality-clustered uncertainty, parallel-trends reasoning, pre-period diagnostics, and interrupted time series (ITS). By the end, you should be able to define the DiD estimand; reconstruct a difference in changes; explain why municipality and month effects belong in the model; align uncertainty with repeated observations within municipalities; use placebo-date and differential-trend diagnostics without treating them as proofs; interpret ITS level and slope changes with serial-correlation-aware uncertainty; and explain why a comparison series can remove a shared interruption that contaminates a treated-only ITS.

### The synthetic phased municipal policy

Twenty-four fictional municipalities are observed for 36 consecutive months. Twelve are early adopters of a mobile youth-crisis response policy; twelve are not-yet adopters during this study window. Early adopters begin the policy at month 19. In that same month, a separate regional hotline begins in every municipality. The outcome is monthly youth emergency transports per 10,000 residents. The panel is complete: 864 municipality-month rows with no missing outcomes.

Early adoption is not randomized. Municipal leaders chose the first-wave sites before month 1. The design therefore does not inherit exchangeability from an assignment mechanism. It asks whether the not-yet-adopter trajectory provides a credible counterfactual change for early adopters after accounting for stable municipality differences and shared calendar-month shocks.

**Scientific question.** Among the twelve early-adopter municipalities during months 19-36, how would the mean monthly youth-emergency-transport rate differ under the mobile response policy versus without it?

**Primary estimand.** The average policy effect for the early-adopter municipalities over the eighteen post-policy months. The comparison condition is the early adopters' counterfactual outcome trajectory without the policy, not simply their own pre-policy mean and not the observed not-yet-adopter level.

**Units and time zero.** A row is a municipality-month. The municipality is the repeated-observation and clustering unit. Month 19 is the first month of both the mobile policy in early adopters and the regional hotline everywhere. Not-yet adoption describes this 36-month observation window; it is not a claim that comparison municipalities can never adopt.

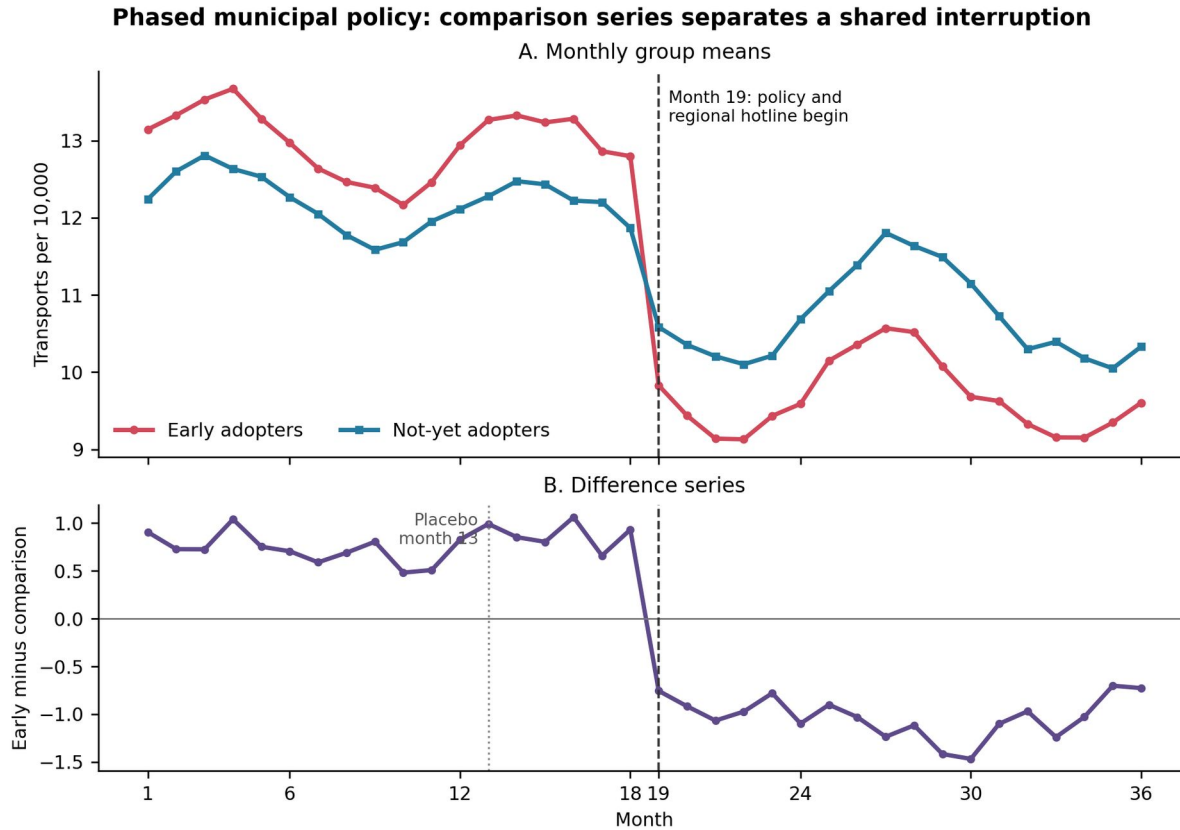


Figure 6. Monthly youth-emergency-transport rates for 12 early-adopter and 12 not-yet-adopter municipalities: the series move similarly before month 19, both fall when the regional hotline starts, and the early-adopter series falls farther after the mobile-team policy; the lower panel shows the early-minus-comparison gap and a placebo date at month 13.

## From two changes to one contrast

The early-adopter mean falls from 12.987 before month 19 to 9.674 afterward, a change of -3.313 transports per 10,000. That before-after contrast combines the mobile policy with the regional hotline, calendar changes, and anything else that shifts at month 19.

The not-yet-adopter mean falls from 12.207 to 10.702, a change of -1.505. Subtracting that comparison change gives

$$(9.674 - 12.987) - (10.702 - 12.207) = -1.809.$$

This is the two-by-two DiD calculation. The regression version uses every municipality-month and adds municipality fixed effects plus month fixed effects. Municipality effects remove stable level differences. Month effects remove shocks shared by early and not-yet adopters, including the region-wide hotline if its effect is common across the two groups. With one common adoption date and a balanced panel, the fixed-effects coefficient answers the same design question as the difference in changes, while its uncertainty respects the municipality-level dependence.

## Pause and Decide

Before looking at the model interval, write down the intervention, comparison, target municipalities, post-policy months, and sign convention. List which threats municipality effects remove, which threats month effects remove, and which threats neither removes. Decide whether the regional hotline invalidates a treated-only ITS, the comparative DiD, both, or neither. Finally, name one observation that would make the parallel-trends story less credible even if the DiD coefficient were precise.

## Identification assumptions, separate from the estimator

A causal DiD interpretation requires more than a regression with two sets of fixed effects. The central condition is parallel untreated trends: without the mobile policy, early adopters' outcomes would have changed like the not-yet adopters' outcomes over months 19-36. Stable level differences are allowed. Different untreated changes are not.

The interpretation also requires no anticipation before month 19; sufficiently consistent policy versions; no policy spillover that changes comparison outcomes; no group-specific contemporaneous event other than the policy; and an observation process that does not change differentially at adoption. The shared hotline is handled by month effects only if its outcome effect is suitably common. If the hotline works differently in early-adopter municipalities because of staffing, publicity, or service integration, the DiD can still mix the two interventions.

Pre-period similarity can challenge parallel trends, but cannot verify the unobserved post-period counterfactual. A null placebo or slope interaction may reflect genuine similarity, low power, offsetting deviations, or a misspecified diagnostic. The design argument must combine timing, institutional knowledge, graphs, and prespecified diagnostics.

## Worked difference-in-differences analysis

**Orientation.** The calculation subtracts two changes rather than two levels, which is why stable differences between municipalities never have to be measured. Everything then rests on what the early adopters' trajectory would have been without the policy, a quantity no data set contains. Read the estimate first, then the two diagnostics, and treat those diagnostics as tests that could have failed rather than as evidence the assumption holds.

| Contrast or diagnostic                                       | Estimate | SE    | 95% interval     |
|--------------------------------------------------------------|----------|-------|------------------|
| Policy $\times$ post DiD with municipality and month effects | -1.809   | 0.150 | [-2.119, -1.499] |
| Pre-period placebo DiD at month 13                           | 0.154    | 0.199 | [-0.259, 0.566]  |
| Pre-period early-adopter differential monthly slope          | 0.005    | 0.014 | [-0.025, 0.034]  |

The primary coefficient is -1.809 transports per 10,000, interval [-2.119, -1.499]. Under the stated assumptions, it estimates a lower average monthly rate after adoption among early-adopter municipalities relative to their counterfactual without the policy. It does not estimate an individual-level risk ratio, the effect of the hotline, or an effect in municipalities outside this fictional panel.

All 36 rows within a municipality share policies, levels, and serially related shocks. Treating 864 rows as independent would overstate information. The reported DiD uncertainty uses a CR1 sandwich clustered by municipality and a t reference distribution with 23 degrees of freedom. Clustering repairs the uncertainty calculation for within-municipality dependence; it does not repair confounding, anticipation, spillover, or differential trends.

## Diagnostics and falsification

The first diagnostic pretends that month 13 was the intervention date and uses only the true pre-period. Its estimate is 0.154, interval  $[-0.259, 0.566]$ . The second fits an early-adopter-by-month interaction over months 1-18. The estimated differential monthly slope is 0.005, interval  $[-0.025, 0.034]$ . Neither interval excludes zero.

These results do not reveal a clear pre-policy discontinuity or linear trend difference at the precision of this panel. They are supporting diagnostics, not a certificate of parallel trends. They cannot detect a shock that begins exactly with the policy and affects groups differently. They also summarize the pre-period in restricted forms: one fake date and one linear interaction. Always inspect the raw trajectories and explain why the comparison series should respond similarly to nonpolicy forces.

## Interrupted time series and the shared interruption

An ITS asks whether a series changes level or slope at a known intervention time relative to its modeled pre-intervention pattern. For each group, the analysis models a linear time trend, a month-19 level change, a post-month-19 slope change, and annual sine and cosine terms. Newey-West lag 3 covariance allows residual dependence across nearby months; the intervals use 30 residual degrees of freedom.

| ITS quantity                                           | Estimate | SE    | 95% interval       |
|--------------------------------------------------------|----------|-------|--------------------|
| Early-adopter level change at month 19                 | -2.813   | 0.115 | $[-3.047, -2.579]$ |
| Early-adopter post-intervention slope change per month | 0.006    | 0.008 | $[-0.010, 0.022]$  |
| Not-yet-adopter level change at month 19               | -1.008   | 0.187 | $[-1.390, -0.626]$ |

The treated-only ITS level change is -2.813. It cannot be called the mobile-policy effect because the regional hotline begins in the same month. The comparison series also changes level by -1.008, providing evidence of a shared interruption. Subtracting the two modeled level changes gives -1.805, close to the DiD coefficient; their difference is -0.004. Here the comparison series removes much of the shared hotline discontinuity. That move is credible only if the comparison series is suitable and the shared intervention affects the groups comparably.

The estimated early-adopter slope change is 0.006 per month, interval  $[-0.010, 0.022]$ . This is compatible with little additional slope change after the immediate drop. A level change is not a slope change. Serial-correlation-aware uncertainty makes the interval more defensible for this model, but it cannot determine which event caused a discontinuity.

**Simulator truth, for teaching audit only.** The generator assigns a  $-1.700$  policy effect in early adopters and a  $-1.050$  hotline effect in every municipality. Those hidden parameters test code and explain why a treated-only discontinuity mixes two causes. They are not information an analyst could use to identify the effect in a real policy evaluation.

### Claim Audit

**Supports:** a reproducible, municipality-clustered estimate of the post-month-19 difference in changes; transparent pre-period placebo and differential-trend diagnostics; serial-correlation-aware ITS level and slope estimates; and a direct demonstration that a same-month shared intervention contaminates treated-only ITS attribution.

**Does not establish:** randomized adoption; proof of parallel untreated trends; that the hotline affects both groups identically; absence of anticipation or spillover; an individual-level effect; a real municipal-policy effect; or transport beyond the fictional municipalities and implementation versions.

**Strongest defensible conclusion:** “In this synthetic panel, the estimated early-adopter-minus-not-yet-adopter change after month 19 was  $-1.809$  youth emergency transports per 10,000, with a municipality-clustered 95% interval from  $-2.119$  to  $-1.499$ . The pre-period placebo and differential-slope intervals included zero, and comparison-series ITS removed a shared month-19 interruption, but causal interpretation still depends on parallel untreated trends, comparable hotline effects, and no differential co-intervention, anticipation, or spillover.”

### Study in the Larger Evidence Program

The next study should document why municipalities entered the first wave, measure service capacity and outcome ascertainment before adoption, and collect more pre-policy months. If later adoption dates become available, use design-aware staggered-adoption methods that preserve cohort and calendar-time comparisons rather than assuming one universal two-way fixed-effects coefficient is appropriate. Add negative-control outcomes that the mobile policy should not affect, audit hotline exposure by municipality, and investigate spillovers across municipal boundaries. A stronger evidence program would connect policy timing to service-process measures and prespecify how differential regional shocks will be handled.

## Six principal problems

For each, write your answer, reason, confidence from 1 to 5, and one revision cue before reading the solutions.

**Problem 6.1 — Define the DiD estimand.** Name the intervention, target municipalities, post-policy period, comparison condition, and outcome unit. Explain why the early-adopter before-after change is not the estimand.

**Problem 6.2 — Reconstruct the difference in changes.** Use the four group-period means to calculate the early-adopter change, comparison change, and DiD. Interpret the sign.

**Problem 6.3 — Align uncertainty with the design.** Why is municipality-clustered uncertainty appropriate? What does clustering address, and which identification threats remain untouched?

**Problem 6.4 — Read the pre-period diagnostics.** Interpret the placebo-date and differential-slope intervals. Do they prove parallel trends? Name one violation they could miss.

**Problem 6.5 — Separate ITS level and slope.** Interpret the early-adopter ITS level and slope changes. Why does Newey-West uncertainty help, and why does it not identify the cause of the level change?

**Problem 6.6 — Repair a treated-only claim.** Rewrite: “The mobile response policy caused an immediate -2.813 reduction.” Use the comparison ITS and DiD results, the shared hotline, the identifying assumptions, and the synthetic-data boundary.

## Fully worked solutions

**Solution 6.1. Answer:** The estimand is the average effect of the mobile youth-crisis response policy on monthly emergency transports per 10,000 among the twelve early-adopter municipalities during months 19-36, compared with those municipalities’ counterfactual outcomes without the policy. The before-after change is -3.313, but it also contains the hotline, common time changes, and any same-date event. **Reason:** A temporal contrast does not construct its own counterfactual. The not-yet-adopter change is used to approximate the change early adopters would have experienced without the mobile policy. **Confidence:** Give 5 only if group, period, intervention, comparison, outcome, and averaging population are explicit. **Revision:** Replace “after versus before” with the untreated potential-outcome comparison.

**Solution 6.2. Answer:** Early adopters change by  $9.674 - 12.987 = -3.313$ . Not-yet adopters change by  $10.702 - 12.207 = -1.505$ . The DiD is  $-3.313 - (-1.505) = -1.809$  transports per 10,000. The negative sign means the outcome fell farther in early adopters. **Reason:** DiD subtracts a comparison change, not a comparison level. **Confidence:** Give 5 if both subtractions and the sign convention are visible. **Revision:** Write the four means before compressing the arithmetic.

**Solution 6.3. Answer:** The 36 observations within each municipality share stable conditions and serially related shocks, so they are not 36 independent experimental units. Clustering by the 24 municipalities allows arbitrary within-municipality residual dependence under the CR1 procedure and uses  $t(23)$ . It does not solve differential trends, anticipation, spillover, differential measurement, or a group-specific co-intervention. **Reason:** Variance estimation and causal identification answer different questions. **Confidence:** Give 5 only if the clustering unit and at least two untouched threats are named. **Revision:** Remove any sentence claiming clustered SEs make adoption as-if random.

**Solution 6.4. Answer:** The month-13 placebo estimate is 0.154, interval  $[-0.259, 0.566]$ ; the pre-period differential slope is 0.005 per month, interval  $[-0.025, 0.034]$ . Neither interval excludes zero, so these restricted diagnostics do not reveal a clear pre-period violation. They do not prove the unobserved post-policy counterfactual is parallel. A group-specific shock beginning at month 19 could pass both tests. **Reason:** Pre-period data can falsify some stories but cannot observe untreated early-adopter outcomes after treatment. **Confidence:** Give 5 if “no detected violation” is not rewritten as “assumption verified.” **Revision:** Add a concrete violation that starts at adoption.

**Solution 6.5. Answer:** The early series shows an immediate modeled level change of -2.813 and a post-intervention slope change of 0.006 per month. The slope interval includes zero, so the data are compatible with an immediate drop without a clear change in subsequent slope. Newey-West lag 3 uncertainty allows nearby residuals to be correlated. It cannot distinguish the mobile policy from the regional hotline or another month-19 event. **Reason:** Dependence-aware variance protects an interval under the fitted time-series model; it does not supply a counterfactual cause. **Confidence:** Give 5 if

level, slope, uncertainty, and attribution are separated. **Revision:** Delete causal attribution from any treated-only estimate.

**Solution 6.6. Answer:** “In this synthetic panel, the early-adopter series has a month-19 level change of -2.813 transports per 10,000, but a regional hotline began everywhere in the same month, so that treated-only ITS cannot isolate the mobile policy. The not-yet-adopter level change is -1.008; their difference, -1.805, is close to the municipality-and-month fixed-effects DiD estimate of -1.809, interval  $[-2.119, -1.499]$ . A policy-effect interpretation requires parallel untreated trends, comparable hotline effects, and no differential co-intervention, anticipation, or spillover; no result describes real municipalities.” **Reason:** The comparison series addresses a shared interruption while making its own comparability assumptions visible. **Confidence:** Give 5 if the statement is conditional, quantified, and explicitly synthetic. **Revision:** Name the hotline and at least one assumption before using causal language.

## Unit 7 Thresholds and Encouragements

### Purpose and objectives

Some interventions are assigned by a threshold. Others are received imperfectly after an offer, invitation, or encouragement. Both situations can support causal reasoning without universal treatment assignment, but they identify different local effects under different assumptions. This unit keeps those designs separate. Case A uses a tutoring-eligibility cutoff to introduce regression discontinuity (RD). Case B uses randomized tutoring encouragement with incomplete uptake to introduce instrumental variables (IV).

By the end, you should be able to define the running variable, cutoff, intervention, target population, and local RD estimand; explain why continuity near a cutoff is not the same as random assignment; read bandwidth, bunching, covariate-continuity, and placebo-cutoff diagnostics without treating them as proofs; distinguish encouragement assignment, tutoring receipt, first stage, reduced form, and Wald ratio; define always-takers, never-takers, compliers, and defiers; state relevance, independence, exclusion, and monotonicity; diagnose weak-instrument risk; and explain why neither an RD effect at one cutoff nor an IV effect among compliers is automatically a population average treatment effect.

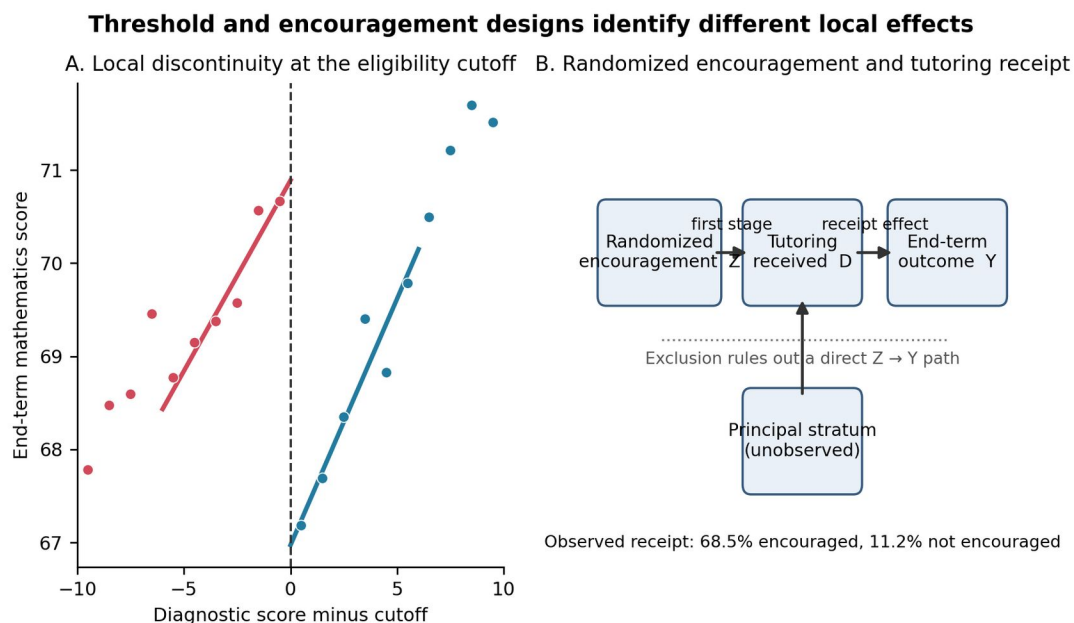


Figure 7. Two local tutoring designs: Panel A shows binned end-term mathematics scores and separate local linear fits on the two sides of the diagnostic-score cutoff at 60; Panel B shows randomized encouragement affecting tutoring receipt, receipt affecting the end-term outcome, an unobserved principal stratum affecting receipt, and the exclusion restriction ruling out a direct encouragement-to-outcome path.

The figure places the cases side by side because each uses a discontinuity in exposure. It does not make them interchangeable. RD compares applicants just below and just above a rule. IV compares randomized encouragement groups, then scales the outcome effect by the encouragement's effect on

receipt. The local population in Case A is defined by score. The local population in Case B is defined by treatment response to encouragement.

## Case A Regression discontinuity at a tutoring cutoff

### The synthetic tutoring-eligibility program

A fictional school district screens 2600 applicants for an intensive mathematics-tutoring program. The diagnostic score is recorded before tutoring. Applicants scoring below 60 become eligible for an offer; applicants scoring at or above 60 do not. Lower scores indicate greater need. The primary outcome is end-term mathematics score. Baseline GPA is recorded before the diagnostic decision and serves as a covariate-continuity diagnostic.

The rule creates a sharp RD: eligibility changes deterministically at the cutoff. It does not make score 59.999 random or score 60.001 random. The design argument is instead that, absent the eligibility rule, expected outcomes would change smoothly through 60 and applicants could not precisely sort to one side with an outcome-relevant advantage.

**Scientific question.** For applicants at the diagnostic-score threshold, what is the average effect of becoming eligible for this tutoring offer rather than not becoming eligible?

**Running variable and cutoff.** The running variable is the pre-intervention diagnostic score. The cutoff is 60. Eligibility equals one below the cutoff and zero at or above it.

**Target population and estimand.** The target is applicants whose scores are at the cutoff. The estimand is the discontinuity in mean end-term mathematics score at 60 under tutoring eligibility versus ineligibility. This is a local eligibility effect. It is not the effect for applicants far below the cutoff, the effect of tutoring receipt among recipients, or the district-wide ATE.

**Unit and timing.** A row is an applicant. Diagnostic score and baseline GPA precede eligibility. Eligibility precedes tutoring access. End-term score follows the intervention window. No score is exactly 60 in this continuous synthetic sample, so the boundary convention does not create a tied observation.

### What local linear estimation does

The primary analysis uses applicants within 6 points of the cutoff and fits separate linear outcome trends below and above 60. The fitted intercept on each side estimates the conditional mean as score approaches the cutoff. Their difference is the estimated discontinuity. Allowing separate slopes prevents the eligibility coefficient from being forced to summarize one common trend through both sides.

The bandwidth is part of the design specification. A narrower window asks the comparison to be more local but uses fewer observations and can be noisy. A wider window gains precision but relies on a linear approximation farther from the cutoff. Looking across bandwidths is a sensitivity analysis, not a search for the most favorable result.

### Pause and Decide

Before reading the estimate, write down the intervention version, running variable, cutoff, direction of eligibility, outcome time, and target population. Decide whether you would call the rule randomized. Then list three observations that would weaken the RD argument: one involving score

manipulation, one involving another change at 60, and one involving an implausible outcome trend. Finally, choose a primary bandwidth before seeing the sensitivity table and explain the bias-precision tradeoff.

## Identification assumptions for the RD case

**Continuity of untreated and treated potential outcomes.** In the absence of the eligibility jump, the expected potential outcomes should approach the same limits from the two sides. Smooth changes with diagnostic score are allowed. A second intervention, grading rule, or service threshold that also changes at 60 would threaten interpretation.

**No precise manipulation around the cutoff.** Applicants, counselors, and scorers should not be able to place outcome-relevant applicants just below 60 rather than just above it. An administrative score with a documented protocol can support this story. A density diagnostic can detect some visible sorting, but a smooth density cannot prove that manipulation is absent.

**Consistent intervention and limited interference.** “Eligibility” must correspond to a sufficiently stable offer and should not change the outcomes of nearby ineligible applicants through crowd-out or peer spillover. If the rule changes school resources or teacher behavior for everyone, the local contrast may not isolate an individual eligibility effect.

**Local functional form.** The chosen local linear specification must approximate the conditional mean adequately within the prespecified window. This is an estimation condition layered on the continuity argument. It is not evidence that eligibility was randomized.

Continuity assumptions are not random assignment. Near-cutoff applicants may be substantively similar, but their score still determines eligibility. The causal argument comes from smooth potential-outcome limits and an unmanipulated rule, not from pretending the running variable was shuffled.

## RD results and diagnostics

**Orientation.** A discontinuity estimate is a local contrast, and locality is not a caveat to be added at the end: it belongs to the estimand. Read the bandwidth table as the answer to how local the estimate is, not as a robustness ritual, and read the density count as an attempt to catch sorting across the threshold before it is presented as an effect.

| Quantity                                   | Estimate | SE    | 95% interval    | Local n |
|--------------------------------------------|----------|-------|-----------------|---------|
| Eligibility discontinuity, bandwidth 6     | 3.917    | 0.607 | [2.727, 5.106]  | 1197    |
| Baseline-GPA discontinuity at 60           | 0.016    | 0.039 | [-0.059, 0.092] | 1197    |
| Outcome discontinuity at placebo cutoff 70 | -0.924   | 0.825 | [-2.541, 0.693] | 634     |

The primary local linear estimate is 3.917 end-term points, interval [2.727, 5.106]. Under the RD assumptions, this estimates the effect of eligibility for applicants at the cutoff. The interval quantifies

sampling and local-model uncertainty under the stated HC1 procedure. It does not cover violations of continuity, manipulation, interference, or an unrecorded co-threshold.

The baseline-GPA discontinuity is 0.016, interval  $[-0.059, 0.092]$ . Because GPA precedes eligibility, a sharp jump would be hard to attribute to tutoring and would challenge local comparability or the score process. The present result does not reveal a clear jump. It cannot certify continuity of every observed or unobserved pre-intervention characteristic.

At the false cutoff of 70, where tutoring eligibility does not change, the estimated outcome discontinuity is -0.924, interval  $[-2.541, 0.693]$ . That interval includes zero. A placebo cutoff probes whether the method manufactures discontinuities in a nearby untreated score region. One null placebo cannot validate the true cutoff or justify searching many false cutoffs and reporting only a convenient one.

### Bandwidth sensitivity and locality

| Half-bandwidth  | Local n | RD estimate | SE    | 95% interval   |
|-----------------|---------|-------------|-------|----------------|
| 4               | 825     | 4.407       | 0.740 | [2.957, 5.857] |
| 6, prespecified | 1197    | 3.917       | 0.607 | [2.727, 5.106] |
| 8               | 1499    | 3.713       | 0.523 | [2.687, 4.739] |
| 10              | 1776    | 3.585       | 0.472 | [2.660, 4.510] |

The estimates remain positive across the four prespecified windows, while the standard error falls as the window widens. The point estimates are not identical because each window changes both the observations used and the approximation required. This pattern supports robustness to these particular bandwidth choices. It does not transport the estimate to low-scoring applicants far from 60. Locality is part of what the design identifies, not a defect to erase with a wider regression.

### Manipulation or bunching diagnostic

Within two score points of the cutoff, 191 applicants fall immediately below and 202 immediately above, a below-to-above ratio of 0.946. A simple prespecified binomial-style count diagnostic gives  $z = -0.555$  and two-sided  $p = 0.579$ . This check does not reveal conspicuous bunching in this realization.

The calculation is intentionally modest. It is not a full density-estimation procedure, and a large  $p$  value is not proof of no sorting. Score heaping, discretionary retesting, or manipulation that preserves aggregate counts could evade it. The strongest audit combines the histogram, institutional scoring rules, logs of retests or overrides, and continuity of predetermined covariates.

**Simulator truth, for teaching audit only.** The generator adds a 4.200-point eligibility discontinuity. The estimate need not equal that hidden value exactly because the sample contains outcome noise and the analysis uses a local linear approximation. The generator parameter tests code; an analyst in a real RD study would not know it. As the window widens, the standard error falls and the estimate moves downward; relative to the hidden 4.200-point simulator discontinuity, the wider-window estimates are farther away in this realization. That pattern illustrates the precision-versus-local-approximation tradeoff, but one realization does not estimate repeated-sampling bias, and a real study would not reveal the true discontinuity.

### Claim Audit for Case A

**Supports:** a reproducible local linear estimate at the score-60 eligibility boundary; transparent bandwidth sensitivity; a simple cutoff-count diagnostic; a baseline-covariate continuity check; and a nearby placebo cutoff.

**Does not establish:** random assignment; the effect of actually receiving tutoring; absence of all manipulation or co-thresholds; a correct functional form outside the local windows; an effect for applicants far from 60; or a real tutoring effect.

**Strongest defensible conclusion:** “In this synthetic cutoff study, the estimated discontinuity in end-term mathematics score at the tutoring-eligibility threshold was 3.917 points, with a 95% interval from 2.727 to 5.106 in the prespecified  $\pm 6$  window. The baseline-covariate, count, placebo-cutoff, and bandwidth checks did not reveal a clear contradiction, but a causal interpretation remains local and depends on continuity, no precise manipulation, no competing cutoff, consistent eligibility, and limited interference.”

## Case B Randomized encouragement and instrumental variables

### The synthetic tutoring encouragement experiment

A separate fictional study enrolls 3000 students who can choose whether to use tutoring. Exactly 1500 are randomly assigned a personalized encouragement and 1500 are assigned no encouragement. Encouragement changes the probability of tutoring receipt but does not determine it: some students use tutoring without encouragement, and some encouraged students do not use it. The outcome is end-term mathematics score.

Let  $Z$  be randomized encouragement,  $D$  be tutoring receipt, and  $Y$  be the end-term outcome. Randomization directly supports the effect of  $Z$  on  $Y$ . It does not by itself randomize  $D$ , because receipt depends on student behavior after assignment. IV reasoning uses  $Z$  to isolate the part of receipt changed by encouragement, but only under additional assumptions.

**Assignment estimand.** The intention-to-treat effect of encouragement is  $E[Y|Z=1] - E[Y|Z=0]$ . In IV terminology this same outcome contrast is the reduced form.

**First stage.** The effect of encouragement on tutoring receipt is  $E[D|Z=1] - E[D|Z=0]$ . Relevance requires this difference not to be zero.

**Wald estimand.** The ratio of reduced form to first stage is interpreted as the average effect of tutoring receipt for compliers if independence, exclusion, and monotonicity also hold. This is a local average treatment effect, or LATE. “Local” refers to a response type, not geographic proximity or closeness to a numeric cutoff.

### Pause and Decide

Before reading the IV estimates, separate four quantities on paper: encouragement assignment, tutoring receipt, outcome, and the unobserved response type that connects assignment to receipt. State which effect randomization identifies without an exclusion restriction. Predict how the Wald ratio behaves if the first stage is close to zero. Finally, describe one plausible direct pathway from a personalized tutoring encouragement to end-term score that would violate exclusion even if no tutoring were received.

## Principal strata and the monotonicity decision

Principal strata classify students by the tutoring receipt they would show under both encouragement conditions:

| Principal stratum | Receipt if not encouraged | Receipt if encouraged | Interpretation                           |
|-------------------|---------------------------|-----------------------|------------------------------------------|
| Always-taker      | 1                         | 1                     | Uses tutoring either way                 |
| Never-taker       | 0                         | 0                     | Does not use tutoring either way         |
| Complier          | 0                         | 1                     | Uses tutoring because of encouragement   |
| Defier            | 1                         | 0                     | Avoids tutoring because of encouragement |

Each student has only one observed encouragement condition, so the individual stratum is not observed. Under random assignment and monotonicity, group receipt rates imply population shares: the no-encouragement receipt rate estimates the always-taker share; the first stage estimates the complier share; and one minus the encouraged receipt rate estimates the never-taker share. Monotonicity rules out defiers. It is a substantive claim about how this encouragement affects receipt, not a fact delivered by the first-stage coefficient.

## Identification assumptions for the IV case

**Relevance.** Encouragement changes tutoring receipt. Weak relevance makes the denominator small, the Wald ratio unstable, and conventional intervals unreliable. A large first-stage statistic can make weak-instrument noise less concerning; it cannot establish the other assumptions.

**Independence.** Random encouragement assignment is independent of potential receipt and potential outcomes. This follows from the implemented assignment mechanism only if allocation was concealed, implemented as planned, and analyzed according to assignment.

**Exclusion.** Encouragement affects end-term score only through tutoring receipt. A message that changes confidence, study effort, teacher attention, or access to other resources could violate exclusion even when the randomized encouragement ITT remains valid.

**Monotonicity.** Encouragement does not cause any student to forgo tutoring that the student would otherwise have used. This rules out defiers. It can be more plausible for a gentle invitation than for a stigmatizing or burdensome intervention, but plausibility must be argued from the intervention.

**Consistent versions and limited interference.** The encouragement and tutoring received must have sufficiently stable meanings, and one student's encouragement or tutoring should not materially change another student's outcome or receipt.

These assumptions answer different questions. Relevance is partly empirical. Independence comes from design execution. Exclusion and monotonicity are not fully testable from  $Z$ ,  $D$ , and  $Y$ . Strong relevance does not make exclusion “almost true.”

## IV results

**Orientation.** The ratio below divides an intention-to-treat contrast by the share of students whom encouragement actually moved. That division rescales the effect to the compliers and to no one else, so the larger number here is not a larger effect than the intention-to-treat estimate. It is an effect for a smaller group that cannot be listed by name.

| Quantity                                      | Estimate | SE    | 95% interval   |
|-----------------------------------------------|----------|-------|----------------|
| Assignment ITT of encouragement on outcome    | 2.938    | 0.307 | [2.337, 3.540] |
| First stage: encouragement effect on receipt  | 0.573    | 0.015 | [0.544, 0.601] |
| Reduced form: encouragement effect on outcome | 2.938    | 0.307 | [2.337, 3.540] |
| Wald estimate: reduced form ÷ first stage     | 5.131    | 0.516 | [4.120, 6.142] |

Tutoring receipt is 68.5% among encouraged students and 11.2% among students not encouraged. Their difference is 0.573, the first stage. The corresponding first-stage F statistic is 1559.379. This is strong relevance in the synthetic sample; it does not test independence, exclusion, or monotonicity.

The encouraged group averages 85.729 outcome points and the no-encouragement group averages 82.791. Their difference, 2.938, is both the assignment ITT and reduced form. Under random assignment, that contrast is the effect of this encouragement package on the outcome. Calling it the reduced form does not change the estimand.

The Wald calculation is

$$2.938 \div 0.573 = 5.131.$$

Its joint delta-method standard error is 0.516, producing interval [4.120, 6.142]. Under relevance, independence, exclusion, monotonicity, consistent versions, and limited interference, this is the average tutoring-receipt effect among compliers: students whose receipt changes because of encouragement. The Wald estimand is a complier effect, not automatically the population ATE.

Under the monotonicity interpretation, the observed receipt rates imply shares of 11.2% always-takers, 57.3% compliers, and 31.5% never-takers. These are population-share estimates under the assumptions, not observed labels for individual students. The data do not tell us which particular student is a complier.

### Weak-instrument warning

If encouragement barely changed receipt, the reduced form would be divided by a number near zero. A small chance imbalance in outcomes could then become a large and unstable Wald estimate. Standard normal approximations can perform poorly in that setting. Report the first stage and its uncertainty before the ratio, prespecify how weak relevance will be handled, and consider weak-instrument-robust methods rather than interpreting a dramatic ratio. This synthetic first stage is

deliberately strong so the unit can focus on identification. Strong instruments reduce one problem; they do not license broader claims.

**Simulator truth, for teaching audit only.** The generator gives tutoring receipt a 5.200-point effect and contains always-takers, compliers, and never-takers but no defiers. It gives encouragement no direct effect. These hidden choices make the Wald assumptions true in the simulator. An analyst in a real encouragement study could not verify exclusion or every response type from the observed data.

### Claim Audit for Case B

**Supports:** a randomized assignment ITT for this encouragement; a reproducible first stage and reduced form; a strong relevance diagnostic; and, conditionally, a Wald estimate for compliers.

**Does not establish:** that tutoring receipt was randomized; exclusion or monotonicity from the data alone; each student's principal stratum; the effect among always-takers or never-takers; a population ATE; transport to another encouragement or tutoring version; or a real tutoring effect.

**Strongest defensible conclusion:** "In this synthetic randomized-encouragement study, encouragement increased tutoring receipt by 0.573 and increased end-term mathematics score by 2.938 points. Their Wald ratio was 5.131, interval [4.120, 6.142]. The ratio has a complier-effect interpretation only if encouragement is independent, relevant, affects outcomes only through tutoring receipt, produces no defiers, and satisfies the stated version and interference conditions."

## Diagnostics and falsification across the two caselets

The RD count, baseline-covariate, placebo-cutoff, and bandwidth checks probe the score process, predetermined composition, nearby discontinuities, and local-model dependence. The IV first stage probes relevance. None directly verifies the counterfactual continuity of outcomes, the absence of every co-threshold, exclusion, or monotonicity. A diagnostic earns its place by naming the threat it could reveal and the threat it leaves unresolved. The two caselets therefore retain separate diagnostic ledgers even when their estimates are discussed together.

## Comparing what the two designs identify

The RD estimate concerns applicants at a score threshold and the effect of eligibility. Its local population is defined by the running variable. The IV estimate concerns students whose receipt responds to encouragement and the effect of receipt. Its local population is defined by principal stratum. Those populations need not overlap, even though both cases involve tutoring.

Neither result should be rewritten as "tutoring raises scores by about four or five points for students." The RD intervention is eligibility, the IV intervention in the numerator is encouragement, and the IV receipt effect is conditional on untestable assumptions and applies to compliers. The designs can contribute complementary evidence only after their interventions, populations, outcomes, and assumptions are aligned. Agreement between two local numbers would not by itself prove a universal effect; disagreement could reflect different populations, treatment versions, or biases rather than a contradiction.

### Study in the Larger Evidence Program

For the cutoff program, preserve the scoring protocol, record retests and overrides, inspect finer score distributions, test additional predetermined covariates chosen before looking at outcomes, and follow

cohorts under stable eligibility rules. If effects away from 60 matter, do not extrapolate mechanically; design another comparison for those applicants.

For the encouragement study, document message content, delivery, contamination, alternative services, and any direct psychological or behavioral effects of the invitation. Prespecify weak-instrument procedures and measure intermediate pathways that could challenge exclusion without conditioning the primary ITT on post-assignment variables. Replicate with materially different encouragements whose plausible direct effects differ.

Across both cases, compare local effects only after defining the target decision. A future synthesis should ask whether the same tutoring version reaches the same students and outcome scale.

Matching, Unit 4 clustering diagnostics, and a genuine Unit 6 event-study analysis remain separate hardening decisions; none is a casual add-on to these RD and IV caselets.

## Six principal problems

For each, write your answer, reason, confidence from 1 to 5, and one revision cue before reading the solutions.

**Problem 7.1 — Define the RD estimand.** Name the running variable, cutoff, eligibility rule, intervention contrast, target population, and outcome time. Explain why the effect is local and why the design is not literal random assignment.

**Problem 7.2 — Audit the cutoff diagnostics.** Interpret the local count comparison, baseline-GPA continuity estimate, and placebo-cutoff estimate. What does each probe, and what can all three still miss?

**Problem 7.3 — Interpret bandwidth sensitivity and locality.** Compare the estimates and standard errors across half-bandwidths 4, 6, 8, and 10. Why is the prespecified bandwidth primary? Rewrite a claim that applies the estimate to every applicant.

**Problem 7.4 — Reconstruct the IV estimate.** Use the two receipt rates to calculate the first stage, use the two outcome means to calculate the reduced form, and divide to obtain the Wald estimate. State the assignment ITT separately.

**Problem 7.5 — Audit IV assumptions and principal strata.** Define always-takers, never-takers, compliers, and defiers. Apply relevance, independence, exclusion, and monotonicity to this encouragement. Explain what the first-stage F statistic can and cannot establish.

**Problem 7.6 — Compare designs and repair an overbroad claim.** Rewrite: “Tutoring improves mathematics scores by about five points for all students.” Use the RD and IV estimands, populations, assumptions, intervals, and synthetic-data boundary.

## Fully worked solutions

**Solution 7.1. Answer:** The running variable is the pre-tutoring diagnostic score, the cutoff is 60, and applicants below it become eligible. The estimand is the discontinuity in mean end-term mathematics score under eligibility versus ineligibility for applicants at 60. It is local because identification compares limits as scores approach one cutoff, not outcomes across the whole score distribution. It is not literal randomization because score deterministically selects eligibility. **Reason:** Continuity of potential-outcome regressions and no precise manipulation can make adjacent limits comparable without a randomized score. **Confidence:** Give 5 only if the intervention is eligibility, the target is at

the cutoff, and continuity is not called randomization. **Revision:** Replace “students near the cutoff were randomized” with the continuity argument.

**Solution 7.2. Answer:** The local count check finds 191 applicants below and 202 above, ratio 0.946,  $p = 0.579$ ; it does not reveal conspicuous aggregate bunching. The baseline-GPA discontinuity is 0.016, interval  $[-0.059, 0.092]$ ; it does not reveal a clear jump in this predetermined covariate. The placebo-cutoff estimate is -0.924, interval  $[-2.541, 0.693]$ ; it does not reveal a clear discontinuity where eligibility stays unchanged. All three can miss a second intervention exactly at 60, subtle or balanced manipulation, discontinuity in an unmeasured factor, or functional-form error specific to the true cutoff. **Reason:** Diagnostics can contradict parts of the design story but cannot observe all counterfactual continuity conditions. **Confidence:** Give 5 if every null-compatible result is described as “not detected,” not “proved absent.” **Revision:** Add one concrete violation that would pass the three checks.

**Solution 7.3. Answer:** The estimates are 4.407, 3.917, 3.713, and 3.585 for half-bandwidths 4, 6, 8, and 10. The corresponding SEs decline from 0.740 to 0.472 as local  $n$  grows from 825 to 1776. The  $\pm 6$  window remains primary because it was chosen before comparing answers; the other windows test sensitivity. A defensible rewrite is: “For applicants at the score-60 threshold in this synthetic program, the estimated eligibility discontinuity was 3.917 points, interval  $[2.727, 5.106]$ , under the continuity and no-manipulation assumptions.” **Reason:** Wider windows buy precision by asking a stronger approximation to hold farther from the cutoff. **Confidence:** Give 5 if the claim keeps the cutoff population and intervention. **Revision:** Delete “all applicants” and state the primary window.

**Solution 7.4. Answer:** Receipt is 68.5% when encouraged and 11.2% when not encouraged, so the first stage is  $68.5\% - 11.2\% = 0.573$ . Outcome means are 85.729 and 82.791, so the reduced form and assignment ITT are  $85.729 - 82.791 = 2.938$  points. The Wald estimate is  $2.938 \div 0.573 = 5.131$  points. **Reason:** Randomization identifies the assignment contrast; dividing by the first stage rescales that contrast to the subgroup whose receipt is changed under the IV assumptions. **Confidence:** Give 5 if the numerator, denominator, and two estimands are labeled. **Revision:** Do not call the Wald estimate an ITT or the reduced form a receipt effect.

**Solution 7.5. Answer:** Always-takers receive tutoring under either encouragement condition; never-takers under neither; compliers only when encouraged; defiers only when not encouraged. Relevance is supported by a 0.573 first stage and  $F = 1559.379$ . Independence comes from valid random assignment of encouragement. Exclusion requires no route from encouragement to outcome except tutoring receipt. Monotonicity rules out defiers. The  $F$  statistic addresses weak relevance, not randomization implementation, exclusion, monotonicity, versions, or interference. **Reason:** IV identification is an intersection of design and substantive restrictions; one strong empirical diagnostic cannot substitute for the others. **Confidence:** Give 5 only if all four assumptions have distinct meanings. **Revision:** Name a direct encouragement-to-outcome path that would violate exclusion.

**Solution 7.6. Answer:** “In the synthetic cutoff case, tutoring eligibility at score 60 is associated with an estimated local discontinuity of 3.917 points, interval  $[2.727, 5.106]$ , under continuity and no-manipulation assumptions. In the separate randomized-encouragement case, encouragement changes receipt by 0.573 and outcome by 2.938 points; the conditional Wald estimate is 5.131, interval  $[4.120, 6.142]$ , for compliers under relevance, independence, exclusion, and monotonicity. Neither local estimate is automatically an effect for all students, and neither study describes real people.” **Reason:** The two designs change different interventions and identify effects in differently defined

local populations. **Confidence:** Give 5 if the rewrite preserves both estimands, both assumption sets, and the synthetic boundary. **Revision:** Remove any universal tutoring claim and name eligibility versus receipt.

## Unit 8 Mechanisms, Mediation, and Effect Heterogeneity

### Purpose and objectives

A randomized experiment can identify a total effect while leaving its mechanism uncertain. It can also reveal that the total effect differs across a characteristic measured before assignment without making that characteristic a treatment. These are separate tasks. This unit uses one fictional cognitive-training trial to keep three claims apart: the total effect of assignment, effect heterogeneity across a prespecified baseline-anxiety stratum, and mediation through strategy use measured after assignment.

By the end, you should be able to define total, controlled-direct, and natural-effect questions without treating them as synonyms; explain why randomizing training does not randomize a post-assignment mediator; distinguish a randomized first stage from an unrandomized mediator-outcome association; identify mediator-outcome confounding and exposure-induced confounding as threats; interpret a prespecified baseline interaction on its stated scale; avoid the “significant here, not significant there” fallacy; and write a mechanism claim that stops at the evidence ceiling.

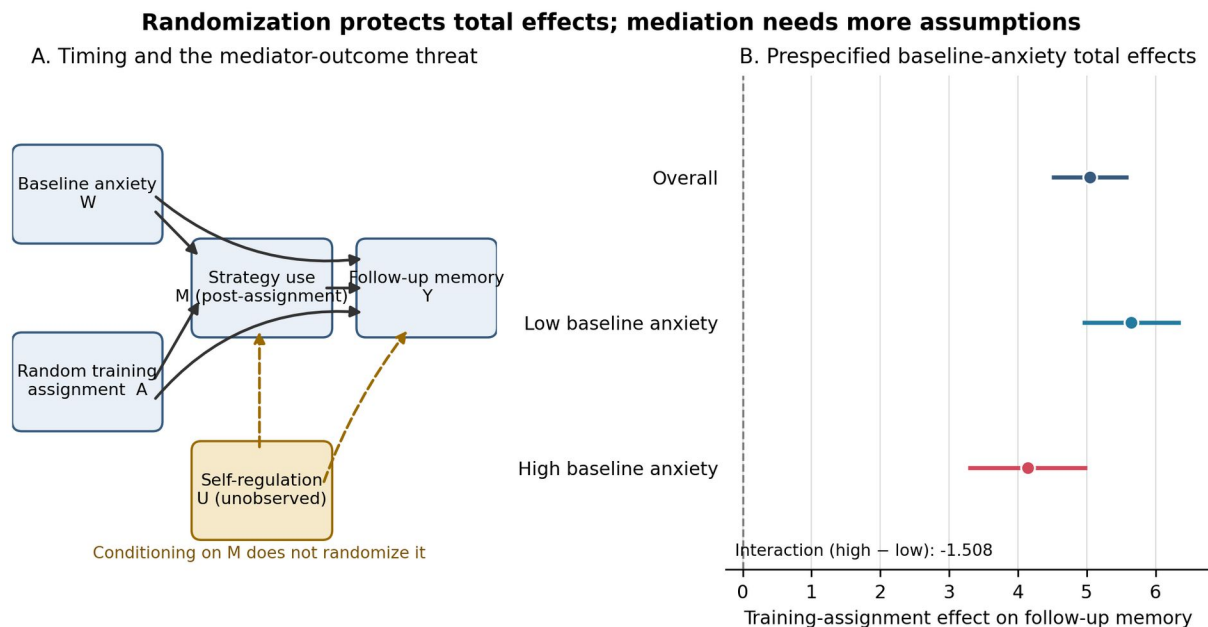


Figure 8. Cognitive-training mechanism and heterogeneity audit: Panel A shows randomized training assignment affecting post-assignment strategy use and follow-up memory, baseline anxiety as a prespecified effect modifier, and unobserved self-regulation as a common cause of strategy use and memory; Panel B shows overall, low-anxiety, and high-anxiety total assignment-effect estimates with 95% intervals.

The panels answer different questions. Panel B uses randomization and a characteristic measured before assignment to estimate total assignment effects. Panel A shows why a post-assignment mechanism requires more than randomization of A: strategy use M is caused by assignment, but it

also shares an unobserved cause  $U$  with outcome  $Y$ . Conditioning on  $M$  neither randomizes  $M$  nor erases  $U$ .

Randomization protects total effects; mediation needs more assumptions. That distinction organizes the unit.

## The synthetic cognitive-training trial

A fictional trial enrolls 3200 adults. Before assignment, each participant completes a memory assessment and an anxiety screen. The protocol prespecifies the binary high-anxiety screen as a potential effect modifier and randomization stratum. Exactly half of the 1920 low-anxiety participants and half of the 1280 high-anxiety participants are assigned cognitive training; the remainder receive time-matched comparison activities. Strategy use is measured after the training period. Follow-up memory is measured afterward.

Let  $A$  denote randomized training assignment,  $W$  the baseline anxiety stratum,  $M$  post-assignment strategy use, and  $Y$  follow-up memory. Baseline memory  $X$  is a prognostic covariate. The primary question is the average total effect of assignment in the eligible trial cohort. The prespecified heterogeneity question asks whether that total effect differs between  $W=0$  and  $W=1$ . The mechanism question asks how much of the effect operates through a defined intervention on  $M$ . Those questions do not share one automatic estimator.

**Overall total-effect estimand.**  $E[Y(1) - Y(0)]$  in the eligible cohort under the two assignment versions.

**Stratum-specific total effects.**  $E[Y(1) - Y(0) \mid W=w]$  for the prespecified baseline-anxiety strata.  $W$  is not manipulated; it indexes populations over which randomized assignment effects are averaged.

**Controlled direct effect.** For a specified strategy-use level  $m$ ,  $E[Y(1,m) - Y(0,m)]$ . This compares training conditions while intervening to hold strategy use to the same level. A controlled direct effect is not the coefficient on  $A$  after automatically adding  $M$  to a regression.

**Natural direct and indirect effects.** These compare nested counterfactuals involving the mediator value that would arise under one assignment condition. They require carefully defined intervention versions and stronger identification conditions. This unit names them but does not claim to identify them from the trial.

## Pause and Decide

Before reading the results, draw a timeline for  $X$  and  $W$ ,  $A$ ,  $M$ , and  $Y$ . Circle every variable randomized by the protocol. Then write three predictions: the sign of  $A$ 's total effect on  $Y$ , the sign of  $A$ 's effect on  $M$ , and whether the total effect will be smaller in the high-anxiety stratum. Finally, decide whether adding  $M$  to the outcome regression would estimate a direct effect. Give a reason involving timing and one involving the unobserved self-regulation variable in Figure 8.

## Identification assumptions and claim boundaries

### Total effects and prespecified heterogeneity

Valid stratified random assignment identifies the total assignment effect overall and within  $W$  strata if allocation was concealed and implemented, outcomes are observed under sufficiently stable

intervention versions, and interference is limited. Baseline adjustment can improve precision. It is not needed to transform a randomized contrast into a causal one.

The interaction coefficient compares two stratum-specific total effects on the additive memory-score scale. It is not “the effect of anxiety,” because anxiety was not randomized. It also does not establish that the same interaction would appear on a binary, ratio, or other outcome scale. Prespecification protects interpretation from an undisclosed search over many moderators; it does not make the subgroup sample infinite or transport the interaction to another population.

### Mechanism and mediation

Randomization identifies the effect of A on M: the training-versus-comparison contrast in strategy use. Randomization also identifies the total effect of A on Y. Those two facts do not identify the effect of M on Y. Participants with greater strategy use can differ in self-regulation, motivation, fatigue, opportunity, or other causes of memory, even after A and measured baseline covariates are held constant.

A common mediation identification argument requires no unmeasured exposure-outcome confounding, no unmeasured mediator-outcome confounding after appropriate pre-exposure adjustment, and no mediator-outcome confounder caused by exposure, together with consistency, positivity, and limited interference. Natural-effect interpretations add cross-world structure that should be justified rather than hidden in software output. These are scientific assumptions, not consequences of a small p value.

In Figure 8, latent self-regulation U raises both M and Y. Because A also causes M, conditioning on M can compare participants with different mixtures of U across assignment groups. In graph terms, conditioning on M opens  $A \rightarrow M \leftarrow U \rightarrow Y$ —the same collider-opening logic introduced in Unit 3—while U remains an unmeasured common cause of M and Y. The mediator-conditioned assignment coefficient therefore need not equal a controlled direct effect. The difference between the total-effect estimate and that coefficient is not automatically an indirect effect.

### Total-effect and heterogeneity results

**Orientation.** The total-effect estimates inherit their credibility from randomization and need no further assumption. The subgroup contrast that follows was prespecified, and that is what separates it from a search through the data for a difference that happens to be large.

| Assignment-effect quantity                          | Estimate | SE    | 95% interval   |
|-----------------------------------------------------|----------|-------|----------------|
| Unadjusted randomized ITT                           | 4.966    | 0.354 | [4.271, 5.660] |
| Baseline-adjusted, cohort-standardized total effect | 5.047    | 0.271 | [4.516, 5.577] |
| Total effect, low baseline anxiety                  | 5.650    | 0.351 | [4.963, 6.337] |
| Total effect, high baseline anxiety                 | 4.142    | 0.426 | [3.307, 4.977] |

| Assignment-effect quantity               | Estimate | SE    | 95% interval     |
|------------------------------------------|----------|-------|------------------|
| Interaction: high minus low total effect | -1.508   | 0.552 | [-2.589, -0.426] |

The assigned group averages 71.980 follow-up memory points and the comparison group 67.015, giving an unadjusted ITT of 4.966. The baseline-adjusted model includes assignment, baseline memory, baseline anxiety, and the prespecified assignment-by-anxiety interaction. Standardizing its assignment contrast over the cohort's 40.0% high-anxiety share gives 5.047, interval [4.516, 5.577]. The close agreement of the two overall estimates is reassuring about implementation, not a license to choose whichever is more favorable.

The estimated total effect is 5.650 points in the low-anxiety stratum and 4.142 points in the high-anxiety stratum. The direct comparison is their interaction, -1.508, interval [-2.589, -0.426]. The interval excludes zero in this realization. That answers a difference-in-effects question on the memory-score scale; two separate subgroup p values would not.

The protocol fixed W, the interaction, and the additive scale before analysis. If dozens of baseline variables and cut points had been searched, the same printed interval would omit the selection process. Heterogeneity analysis should record the moderator definition, scale, model, multiplicity plan, missing-data handling, and whether the question is confirmatory or exploratory.

## Mechanism results and the negative example

**Orientation.** Nothing in this section is identified by the randomization. Conditioning on a variable that the treatment caused answers a different question from the one the section headings suggest, and the numbers are shown so that the failure has a definite size instead of remaining a vague warning.

| Mechanism quantity                           | Estimate | SE    | 95% interval    |
|----------------------------------------------|----------|-------|-----------------|
| Assignment effect on strategy use            | 7.051    | 0.226 | [6.608, 7.494]  |
| Mediator-conditioned assignment contrast     | 0.298    | 0.257 | [-0.205, 0.801] |
| Strategy-use/outcome conditional association | 0.673    | 0.017 | [0.639, 0.708]  |

Strategy use averages 46.338 in the training group and 39.311 in the comparison group. The baseline-adjusted assignment effect on M is 7.051, interval [6.608, 7.494]. Because A is randomized and M is measured after assignment, this supports the claim that this training package changes the strategy-use measure in this trial.

When M is added to the outcome model, the cohort-standardized assignment coefficient becomes 0.298, interval [-0.205, 0.801]. The M coefficient is 0.673, interval [0.639, 0.708]. Both are reproducible regressions. Neither becomes a causal direct or mediator effect merely because baseline X and W are included.

The drop from the total-effect estimate 5.047 to the mediator-conditioned contrast 0.298 is visually tempting as “the mediated amount.” Reject that arithmetic. It conditions on a variable caused by

assignment, and the mediator-outcome relation remains confounded by latent self-regulation. A numerically large first stage, a strong M-Y association, and an attenuated A coefficient are mechanism predictions; together they can be compatible with mediation without identifying a mediated effect.

A traditional causal-steps/product-of-coefficients calculation commonly associated with Baron and Kenny multiplies the first-stage coefficient, displayed as 7.051, by the mediator coefficient, displayed as 0.673. Computed from the unrounded fitted values the product is 4.748738, approximately 4.749, and subtracting the mediator-conditioned assignment contrast from the total-effect estimate gives the same six-decimal result. Multiplying displayed values reproduces that product only approximately: even the six-decimal coefficients 7.051006 and 0.673484 multiply to 4.748740. In the hidden generator, the descriptive calculation is 1.598738 points, or about 50.8%, larger than the 3.150-point strategy-mediated component. This is a discrepancy from a hidden structural quantity, not a formal estimate of estimator bias or an identified indirect effect. The required mediator assumptions are violated here, and a real analyst would not know the hidden component.

Three features of that calculation deserve statement rather than inference. First, the agreement between the product of coefficients and the total-minus-conditioned difference is a property of these fitted coefficients in this realization, computed from unrounded values. It holds to six decimals here and is not offered as an algebraic identity that would survive a change of model or a different sample.

Second, the stored record carries the unrounded coefficients, and every published product in this unit is computed from those rather than from the values printed on the page.

Third, the first-stage model used here contains no assignment-by-anxiety interaction, and that is not an omission the data contradict: the hidden generator gives assignment a constant effect on strategy use, so a first stage without that interaction is compatible with the mechanism that produced these rows. The mediation failure demonstrated in this section comes from mediator-outcome confounding by latent self-regulation, not from the functional form of the first stage. Reporting the failure precisely matters more than reporting it emphatically.

**Simulator truth, for teaching audit only.** The hidden generator gives total effects of 5.650 points in the low-anxiety stratum and 4.150 in the high-anxiety stratum, averaging 5.050 in this cohort. Its average controlled direct effect is 1.900, and its strategy-mediated component is 3.150 points. The observed total-effect estimates recover the randomized target closely. The mediator-conditioned coefficient does not recover the hidden controlled direct effect because latent self-regulation confounds M and Y. These hidden values audit the lesson; an analyst in a real trial would not know them.

## Diagnostics and falsification

**Randomization audit.** Confirm allocation counts within the prespecified W strata, protocol deviations, treatment versions, and outcome observation. Exact 1:1 allocation within W supports the implemented design; it does not prove concealment or no interference outside the data record.

**Temporal audit.** Verify that X and W precede A, M follows A, and Y follows M. Correct order is necessary for the proposed mechanism, but temporal order does not remove mediator-outcome confounding.

**Mechanism predictions.** Ask whether A changes M and whether M predicts Y after measured baseline adjustment. Failure of either prediction would challenge this specific mechanism story. Success cannot rule out U, exposure-induced mediator-outcome confounding, alternative pathways, measurement error, or reverse influence if measurement windows overlap.

**Moderator audit.** Confirm that W and the interaction scale were prespecified, estimate the interaction directly, show both stratum estimates with uncertainty, and report the search universe. A significant effect in one stratum and a nonsignificant effect in another is not evidence that the effects differ.

**Measurement audit.** A strategy-use questionnaire may reflect reporting style as well as behavior. Randomization can identify an effect on the recorded scale while leaving the construct interpretation uncertain. A mechanism program should vary measurement methods, timing, and intervention components rather than trusting one mediator instrument.

## Claim Audit

**Supports:** an overall randomized assignment ITT; a baseline-adjusted standardized total-effect estimate; prespecified low- and high-anxiety total effects and their additive interaction; a randomized assignment effect on measured strategy use; and reproducible mediator-conditioned descriptive regressions.

**Does not establish:** that anxiety itself causes effect differences; a universal interaction on every outcome scale; that strategy use causes memory; a controlled direct effect; a natural direct or indirect effect; the proportion mediated; absence of latent or exposure-induced mediator-outcome confounding; transport to another training program or population; or a real cognitive-training effect.

**Strongest defensible conclusion:** “In this synthetic stratified randomized trial, cognitive-training assignment increased follow-up memory by an estimated 5.047 points, interval [4.516, 5.577]. The prespecified additive effect was -1.508 points smaller in the high-anxiety stratum than in the low-anxiety stratum. Assignment also increased measured strategy use by 7.051 points. Because strategy use was post-assignment and shared an unmeasured cause with the outcome, the mediator-conditioned coefficient is not interpreted as a direct effect or the remaining difference as an indirect effect.”

## Study in the Larger Evidence Program

Preserve the randomized total effect as the primary result. Replicate the baseline-anxiety interaction with the same screen, cut point, outcome scale, and analysis before claiming a stable moderator. To study mechanism, improve temporal resolution, measure plausible mediator-outcome confounders before assignment, validate strategy-use measurement, and consider a second randomized intervention that changes the proposed mediator without duplicating every other component of training. A component or factorial design can sometimes test a pathway more directly than conditioning on observed behavior.

Mechanism evidence should accumulate across interventions and measurements. If changing a strategy component shifts M and Y as predicted, if alternative mediators do not explain the pattern, and if results replicate under different biases, the causal story strengthens. No single mediator regression carries that burden. Unit 10 will return to this cross-case triangulation logic.

## Six principal problems

For each, write your answer, reason, confidence from 1 to 5, and one revision cue before reading the solutions.

**Problem 8.1 — Define the three claim types.** Define the cohort total effect, the two baseline-anxiety stratum total effects, and a controlled direct effect. State which are identified by this assignment design and which require additional assumptions.

**Problem 8.2 — Reconstruct the moderator result.** Use the low- and high-anxiety total-effect estimates to calculate the additive interaction. Explain why comparing their two separate significance labels would answer the wrong question.

**Problem 8.3 — Audit the mediation claim.** Interpret the assignment effect on strategy use, the mediator-conditioned assignment contrast, and the strategy-use/outcome association. Reconstruct the traditional product-of-coefficients calculation, identify the collider path, and explain why neither the product nor total minus conditioned is an identified indirect effect.

**Problem 8.4 — Separate direct-effect concepts.** Explain the intervention represented by a controlled direct effect. Contrast it with a natural direct effect and with the coefficient on A in the regression containing M.

**Problem 8.5 — Repair subgroup language.** Rewrite: “Training works for low-anxiety participants but not for high-anxiety participants.” Use the interaction, both intervals, prespecification, and the outcome scale.

**Problem 8.6 — Write the evidence-program claim.** Produce one paragraph that preserves the randomized total-effect conclusion, treats the moderator appropriately, limits the mechanism claim, states the synthetic boundary, and proposes a next study.

## Fully worked solutions

**Solution 8.1. Answer:** The cohort total effect is  $E[Y(1)-Y(0)]$  averaged over eligible participants. The stratum effects are  $E[Y(1)-Y(0)|W=0]$  and  $E[Y(1)-Y(0)|W=1]$ . Stratified randomization identifies all three assignment effects under valid implementation, stable versions, outcome observation, and limited interference. A controlled direct effect  $E[Y(1,m)-Y(0,m)]$  additionally imagines setting strategy use to one common m; assignment of A alone does not identify that intervention. **Reason:** Randomization protects contrasts defined by the randomized treatment, not every post-treatment variable. **Confidence:** Give 5 if W remains a baseline population index and M is not called randomized. **Revision:** Add the intervention on M explicitly.

**Solution 8.2. Answer:** High minus low is  $4.142 - 5.650 = -1.508$  points, with interval  $[-2.589, -0.426]$ . This direct interaction tests whether the additive assignment effects differ. “Significant in one subgroup but not the other” compares each effect with zero under different standard errors and does not test their difference. **Reason:** Heterogeneity is a contrast of causal contrasts. **Confidence:** Give 5 if the answer states the additive memory-score scale and prespecified W. **Revision:** Replace two significance labels with the interaction estimate and interval.

**Solution 8.3. Answer:** Assignment raises strategy use by 7.051 points, interval  $[6.608, 7.494]$ , a randomized first-stage result. The M-containing regression gives an assignment contrast of 0.298 and an M coefficient of 0.673. Their traditional product-of-coefficients calculation multiplies the unrounded versions of those two coefficients and gives 4.748738, approximately 4.749, the same six-

decimal result as  $5.047 - 0.298$ ; multiplying the displayed values gives only an approximation. Conditioning on  $M$  opens  $A \rightarrow M \leftarrow U \rightarrow Y$ , the Unit 3 collider logic, and leaves the mediator-outcome relation confounded by  $U$ . The calculation therefore does not identify an indirect effect. Reason: A randomized cause of a mediator does not randomize the mediator. Confidence: Give 5 if the arithmetic, timing, and  $U$  appear. Revision: Label the product “traditional descriptive calculation,” not “indirect effect,” and the second regression “mediator-conditioned,” not “direct effect.”

**Solution 8.4. Answer:** A controlled direct effect compares assignment while forcing strategy use to the same specified level  $m$  under both assignments. A natural direct effect instead holds  $M$  to the value it would naturally take under a reference assignment, creating nested counterfactuals and stronger identification demands. The  $A$  coefficient after adding observed  $M$  is an associational conditional contrast unless the relevant mediator-outcome exchangeability, positivity, consistency, and interference conditions hold; it is not a definition of either causal direct effect. **Reason:** Estimands are intervention contrasts, not regression-column names. **Confidence:** Give 5 if the answer states what is set and to which value. **Revision:** Replace “control for the mediator” with the counterfactual intervention.

**Solution 8.5. Answer:** “In this synthetic trial, the estimated total assignment effect was 5.650 points among participants in the prespecified low-anxiety stratum and 4.142 among those in the high-anxiety stratum. Their additive interaction was -1.508, interval  $[-2.589, -0.426]$ , indicating a smaller effect in the high-anxiety stratum on this memory-score scale.” **Reason:** Both stratum effects are positive here; the scientific heterogeneity question is their difference, not whether each separately crosses a significance threshold. **Confidence:** Give 5 if no claim is made that anxiety was randomized or that the modifier transports. **Revision:** State the scale and remove “works/does not work.”

**Solution 8.6. Answer:** “In this entirely synthetic stratified randomized trial, training assignment increased follow-up memory by 5.047 points, interval  $[4.516, 5.577]$ . A prespecified additive interaction indicated a -1.508-point smaller total effect in the high-anxiety stratum. Assignment increased post-training strategy use, but strategy use was not randomized and shared an unmeasured cause with memory, so the analysis does not identify a direct, indirect, or proportion-mediated effect. A next study should preserve the primary randomized comparison while manipulating a well-defined strategy component or otherwise strengthening identification of the mediator-outcome relation. No result describes real participants.” **Reason:** The paragraph separates what assignment identifies from what the mechanism model assumes. **Confidence:** Give 5 if the total, moderator, mechanism ceiling, next design, and synthetic boundary all remain. **Revision:** Remove any causal verb attached to  $M$  unless a mediator intervention is defined.

## Unit 9 Selection, Missingness, Measurement, and Generalization

### Purpose and objectives

A randomized assignment can protect a causal comparison inside a study while outcome observation, measurement, and population selection still limit the claim. This unit uses a fictional multisite replication with a binary primary outcome to ask four linked questions: Which randomized contrast is identified among recruited students? What must be assumed when the primary outcome is missing? Does a complete self-report measure answer the same question as a blinded assessment? What extra assumptions transport the study to a differently composed population?

By the end, you should be able to define marginal risks, risk differences, and risk ratios; distinguish the enrolled population from the target population; explain why complete-case analysis needs an observation assumption even after randomization; standardize cell-specific risks over recruited and target compositions; separate construct validity from missingness and precision; use worst- and best-case bounds as an assumption-free diagnostic; and write a transport claim that names both its support and its limits.



Figure 9. Selection, measurement, and transport audit: Panel A plots the proportion of students whose blinded outcome was observed at Cedar, Lake, Ridge, and Harbor, with a paired marker and a printed value for comparison and program assignment at each site; Panel B shows risk-difference estimates with 95 percent intervals for crude complete cases, the recruited sample after site-by-distress standardization, the external target after standardization, and the complete self-report outcome.

Figure 9 does not offer one preferred number without a question. Panel A concerns whether the primary outcome was observed. Panel B changes one element at a time: adjustment for observed-cell composition, the population used for averaging, and the outcome measurement process. Similarity between two points can be reassuring about arithmetic; it cannot make their estimands or assumptions identical.

## The synthetic multisite replication

Four fictional sites recruit 4000 students for a brief digital coping program. Baseline distress is recorded before assignment. Within every site-by-distress cell, exactly half of the students are assigned the program and half comparison resources. The primary outcome is clinically meaningful improvement at 12 weeks, coded 1 or 0 by a blinded evaluator. That assessment is observed for 3506 students (87.7%). A complete self-report improvement indicator is also recorded, but it is a different measurement process.

Let  $A$  be randomized assignment,  $X$  the combination of site and baseline-distress stratum,  $R$  indicate observation of the blinded primary outcome,  $Y$  the blinded binary outcome, and  $Y^*$  the self-report indicator. Let  $S=1$  denote recruitment into this study and  $S=0$  denote membership in the external target population. The intervention versions and outcome time are fixed by the fictional protocol.

**Recruited-sample estimands.** The marginal risks are  $E[Y(1)|S=1]$  and  $E[Y(0)|S=1]$ . Their risk difference is  $E[Y(1)-Y(0)|S=1]$ ; their risk ratio is  $E[Y(1)|S=1]/E[Y(0)|S=1]$ . Both are intention-to-treat effects of assignment. They express the same pair of marginal risks on additive and relative scales.

**Target-population estimands.** Replace  $S=1$  with the stated external target distribution. The target differs in its shares of sites and baseline distress. Transport therefore changes the averaging weights, not the observed treatment assigned to any student.

**Measurement estimand.** Replacing  $Y$  with  $Y^*$  defines a different outcome. A self-report effect is not automatically a noisy version of the blinded-assessment effect; it is an effect on the response process encoded by that measure.

## Pause and Decide

Before reading the results, write four statements. First, name the intervention, comparison, outcome time, and recruited population. Second, decide whether a risk difference of 0.14 and a risk ratio of 1.43 would be competing answers or two scales for the same marginal risks. Third, state what randomization does and does not guarantee when  $Y$  is unobserved. Fourth, predict whether transporting toward more high-distress and Harbor students will change the effect when cell-specific effects differ.

## Identification assumptions and claim boundaries

### Four filters between assignment and a population claim

#### Randomization and observation are different mechanisms

Exact within-cell randomization balances assignment probabilities. It does not force every assigned student to complete a blinded follow-up. The program and comparison observation rates are 88.0% and 87.3%, a difference of 0.7 percentage points. A small overall difference does not establish harmless missingness: observation also varies by site and baseline distress, and outcome values among missing students are unknown.

The standardized complete-case analysis assumes that, within each assignment-by-site-by-distress cell, observation is independent of the potential blinded outcome. This is a conditional missing-at-random identification statement for this estimator, not something randomization proves. It also

requires positivity: every cell used in standardization must contain observed outcomes under both assignments. Little et al. (2012) emphasize that preventing and documenting missingness is preferable to treating one unverifiable analysis assumption as a cure.

### Measurement is not just a standard-error problem

The blinded assessment and self-report indicator are both binary, but their labels do not make them interchangeable. A complete outcome can be systematically assignment-sensitive; a blinded but incomplete outcome can face selection through R. More observations improve precision only for the outcome actually measured. They do not establish that a questionnaire and blinded evaluation represent the same construct, threshold, reporting process, or intervention version.

Construct validity asks what the measure means. Missing-data validity asks why values are unobserved. Precision asks how variable an estimator would be under its stated model. These are three questions. A larger n cannot answer the first two by shrinking a standard error.

### Generalization changes the averaging population

All eight site-by-distress cells appear in both the recruited and target compositions. The largest target-to-sample cell-weight ratio is 2.267. That supports empirical overlap for this coarse standardization, but it does not establish conditional transportability. Transport requires that the cell-specific potential-outcome risks relevant to the target equal those represented by the trial, after conditioning on X, with sufficiently stable intervention and outcome versions and no unmeasured effect modifier whose distribution changes between  $S=1$  and  $S=0$ . Cole and Stuart (2010) show how target-population weights can make this population change explicit.

### Effect scale belongs to the estimand

For a binary outcome, the additive and relative scales answer different decision questions. A risk difference directly gives the change in outcome probability and depends on baseline risk. A risk ratio expresses proportional change and also depends on which marginal risks are averaged. Neither is a conditional odds ratio. This unit reports both marginal scales and never substitutes a logistic-regression coefficient for the population contrast.

## Read the selection table before the effect table

| Site and baseline distress | Recruited share | Target share | Blinded outcome observed: comparison | Blinded outcome observed: program |
|----------------------------|-----------------|--------------|--------------------------------------|-----------------------------------|
| Cedar, lower               | 25.0%           | 12.0%        | 93.2%                                | 94.6%                             |
| Cedar, high                | 10.0%           | 10.0%        | 81.5%                                | 88.0%                             |
| Lake, lower                | 20.0%           | 15.0%        | 90.0%                                | 89.8%                             |
| Lake, high                 | 10.0%           | 13.0%        | 85.5%                                | 83.0%                             |
| Ridge, lower               | 12.5%           | 10.0%        | 86.0%                                | 88.0%                             |
| Ridge, high                | 10.0%           | 15.0%        | 77.5%                                | 80.5%                             |
| Harbor, lower              | 5.0%            | 8.0%         | 91.0%                                | 93.0%                             |

| Site and baseline distress | Recruited share | Target share | Blinded outcome observed: comparison | Blinded outcome observed: program |
|----------------------------|-----------------|--------------|--------------------------------------|-----------------------------------|
| Harbor, high               | 7.5%            | 17.0%        | 83.3%                                | 74.7%                             |

The overall observation-rate difference is only 0.7 percentage points, yet Harbor high-distress students show lower observation under program assignment while Cedar high-distress students show higher observation. These differences can be sampling variation or selection signals; the table cannot decide. It prevents a near-equal pooled rate from being mistaken for a missingness mechanism.

The two share columns answer a separate question. Cedar lower-distress students are 25.0% of the recruited cohort but 12.0% of the target; Harbor high-distress students move from 7.5% to 17.0%. Standardization keeps each cell's two observed arm risks fixed and changes only these averaging weights. That arithmetic is transparent. Its causal transport interpretation still depends on whether the cells are sufficient to align trial and target potential-outcome risks.

## Recruited-sample results

**Orientation.** Four filters sit between assignment and a population claim: who was recruited, who was observed, what was measured, and which population the average refers to. The sections that follow change one filter at a time, so the differences between them, rather than any single number, are the lesson.

| Recruited-sample blinded outcome                    | Comparison risk | Program risk | Risk difference (95% interval)   | Risk ratio (95% interval) |
|-----------------------------------------------------|-----------------|--------------|----------------------------------|---------------------------|
| Crude complete cases                                | 0.324           | 0.459        | 0.135 (SE 0.016), [0.103, 0.167] | 1.416 [1.301, 1.541]      |
| Standardized over recruited site-by-distress shares | 0.320           | 0.457        | 0.138 (SE 0.016), [0.106, 0.169] | 1.431 [1.315, 1.557]      |

Among complete cases, improvement risks are 0.459 under program assignment and 0.324 under comparison, for a crude risk difference of 0.135 and risk ratio 1.416. The cell-standardized analysis first computes the two observed risks in each site-by-distress cell, then averages both arms using the recruited cell shares. It gives risks 0.457 and 0.320, risk difference 0.138, and risk ratio 1.431.

The crude and standardized estimates are close here. That agreement says the realized mixture among complete cases did not substantially change this contrast after the prespecified cells were restored. It does not test whether the missing outcomes resemble observed outcomes within cells. Both analyses still rely on an observation assumption.

The normal interval for the standardized risk difference is [0.106, 0.169]. The risk-ratio interval is calculated on the log scale and is [1.315, 1.557]. These intervals condition on fixed cell weights. They omit uncertainty about the observation rule, construct, target composition, and transport assumptions.

## Transporting the marginal effect

**Orientation.** Transport changes the estimand before it changes any number. Unit 2 introduced this move with site shares; the same arithmetic runs here on site-by-distress cells, and the interval widens because the target leans on cells that the recruited sample filled less heavily.

| Population used for averaging | Comparison risk | Program risk | Risk difference (95% interval)   | Risk ratio (95% interval) |
|-------------------------------|-----------------|--------------|----------------------------------|---------------------------|
| Recruited sample              | 0.320           | 0.457        | 0.138 (SE 0.016), [0.106, 0.169] | 1.431 [1.315, 1.557]      |
| External target               | 0.283           | 0.449        | 0.166 (SE 0.018), [0.131, 0.201] | 1.585 [1.433, 1.753]      |

Applying the external target shares changes the marginal comparison risk to 0.283 and program risk to 0.449. The target-standardized risk difference is 0.166 and the risk ratio is 1.585. The target contains more students in high-distress cells and in sites where the generator’s additive assignment effect is larger, so its marginal risk difference exceeds the recruited-sample value in this realization.

That is not proof that the program “works better elsewhere.” The transported estimate combines trial cell risks with an externally specified composition under a conditional transportability assumption. A target with an unrepresented site, a different program implementation, a different blinded threshold, or an unmeasured effect modifier would require more than these eight weights.

## Measurement comparison

**Orientation.** The self-report contrast is not a noisier version of the blinded contrast. It is a contrast of a different variable whose error rates differ by arm. Compare the two rows for what they estimate, not for how close they happen to land.

| Recruited-sample measure                | Comparison risk | Program risk | Risk difference (95% interval)   | Risk ratio (95% interval) |
|-----------------------------------------|-----------------|--------------|----------------------------------|---------------------------|
| Blinded assessment, cell standardized   | 0.320           | 0.457        | 0.138 (SE 0.016), [0.106, 0.169] | 1.431 [1.315, 1.557]      |
| Complete self-report, cell standardized | 0.345           | 0.493        | 0.148 (SE 0.015), [0.117, 0.178] | 1.428 [1.325, 1.538]      |

The complete self-report yields a risk difference of 0.148 and risk ratio 1.428. Its completeness removes no missing self-report values; it does not validate the construct or repair missing blinded assessments. Agreement with the blinded estimate would not prove equal classification, and disagreement would not reveal which measure is correct. A measurement audit needs evidence about content, thresholds, blinding, reliability, assignment awareness, and relation to decisions—not merely another p value.

### What the self-report comparison is expected to show

The realized self-report risk difference, 0.1475 before rounding, sits close to the hidden blinded value of 0.145. That closeness is a property of this sample, not a property of the measure. The generator's

sensitivity and specificity imply what a cell-standardized self-report contrast estimates in expectation, and the target is not what it estimates.

| Quantity                                              | Value    |
|-------------------------------------------------------|----------|
| Expected self-report risk under comparison assignment | 0.341920 |
| Expected self-report risk under program assignment    | 0.520750 |
| Expected self-report risk difference                  | 0.178830 |
| Hidden blinded risk difference in this sample         | 0.145000 |
| Expected excess of the self-report contrast           | 0.033830 |
| Realized self-report contrast in this sample          | 0.147500 |

Read the last two rows together. In expectation the self-report contrast overstates the target by 0.034, roughly a quarter of the true difference, because assignment to the program raises sensitivity from 0.82 to 0.91 while lowering specificity from 0.90 to 0.84. The realized value lands near the truth because sampling variation happened to offset most of that expected excess.

A favorable agreement in one simulated sample is therefore not evidence that differential misclassification is harmless here, and an analyst without access to the generator could not tell the two situations apart. This is the same discipline applied to bias that the book applies to effects: one realization does not estimate a repeated-sampling quantity.

Verification note. These expected values are computed from the generator's cell-specific risks together with its stated sensitivity and specificity, standardized by the same sample cell weights used elsewhere in this unit. They are hidden quantities disclosed for teaching audit only.

## Bounds before an assumption

An intentionally severe audit fills every missing program outcome with 0 and every missing comparison outcome with 1, then reverses those assignments. The resulting recruited-sample risk-difference bounds are  $[-0.006, 0.241]$ . They are wide and include zero.

These are not plausible-value estimates or confidence intervals. They show what the observed file alone permits without restricting missing outcomes. A useful sensitivity analysis narrows this region with explicit, scientifically argued departures from the primary observation assumption. Unit 10 will place such departures beside other design vulnerabilities; it will not declare the most convenient assumption true.

## Simulator truth, for teaching audit only

The hidden generator gives a recruited-sample primary risk difference of 0.145 and risk ratio 1.432. Under the external target composition, the corresponding values are 0.164 and 1.543. The hidden observation rule depends only on assignment, site, and baseline distress, so the cell-standardized blinded analysis is aligned with the simulator. The self-report classification process differs by assignment.

Those hidden choices explain why the computational exercise behaves as designed. An analyst in a real trial would not know the unobserved outcomes, the true cell effects, or exact classification

parameters. The truth audit therefore checks code and teaching logic; it is not evidence that the assumptions are empirically verified.

## Diagnostics and falsification

**Assignment audit.** Confirm exact 1:1 allocation inside all eight cells and inspect deviations, versions, interference, and baseline timing. Randomization protects assignment contrasts before later selection; it does not guarantee outcome observation or external relevance.

**Observation audit.** Report R by assignment and prognostic baseline cells, describe follow-up procedures and reasons, preserve outcomes that can be collected after discontinuation, and compare primary analyses with prespecified departures. Similar arm-level rates can conceal opposite cell patterns. A model for R cannot test independence from missing Y.

**Support audit.** List recruited and target shares, flag empty or sparse cells, and report large target-to-sample ratios. Here every target cell is represented and the maximum ratio is 2.267. Support is necessary, not sufficient, for transport.

**Measurement audit.** Describe who measures Y, what constitutes improvement, whether assessors know assignment, and how self-report differs. Measurement invariance across assignment, site, language, or distress cannot be inferred from a complete column.

**Scale audit.** Present the two marginal risks before any difference or ratio. Then report both effect scales if each serves a decision. Avoid translating a conditional odds coefficient into a population risk claim.

## Claim Audit

**Supports:** exact randomized assignment within site-by-distress cells; reproducible complete-case and cell-standardized marginal risks; recruited-sample and externally weighted risk differences and risk ratios under stated observation and transport assumptions; observation diagnostics; a separate complete self-report contrast; and assumption-free extreme bounds for the missing blinded outcome.

**Does not establish:** that missing outcomes are conditionally exchangeable with observed outcomes; that self-report and blinded assessment measure the same construct; that a complete secondary outcome repairs missing primary outcomes; transport beyond the eight stated cells; absence of unmeasured effect modification or version differences; a universal effect on every scale; or any effect in real students.

**Strongest defensible conclusion:** “In this synthetic multisite randomized replication, the site-by-distress standardized blinded-assessment risk difference in the recruited cohort was 0.138, interval [0.106, 0.169], with risk ratio 1.431, interval [1.315, 1.557]. Under conditional outcome-observation and transportability assumptions, reweighting to the specified target gave risk difference 0.166 and risk ratio 1.585. The complete self-report result concerns a different measurement process, and extreme missing-outcome bounds include zero. All data are synthetic.”

## Study in the Larger Evidence Program

A stronger replication would reduce missing blinded outcomes by design, record reasons and post-discontinuation outcomes, validate the improvement threshold, and prespecify sensitivity parameters linked to plausible departures. Target-person covariates should come from a documented sampling

frame, with measurement and intervention versions aligned across trial and target. Replication across sites can reveal effect modification only when the same constructs and treatments are preserved well enough to compare.

Selection, measurement, and transport should not be collapsed into one “real-world validity” label. Different studies can fail differently. A program of evidence becomes stronger when one design’s vulnerable step is challenged by another design, measurement mode, population, or data source. Unit 10 turns that cross-case logic into a sensitivity and triangulation workflow.

## Six principal problems

For each, write your answer, reason, confidence from 1 to 5, and one revision cue before reading the solutions.

**Problem 9.1 — Define binary-outcome estimands.** Define recruited-sample and target-population risks under each assignment. Construct their marginal risk differences and risk ratios. Explain why neither is a conditional log-odds coefficient.

**Problem 9.2 — Audit outcome observation.** Use the two overall observation rates and the Figure 9 site pattern. State the identifying observation assumption for the cell-standardized primary analysis and two things the rates cannot prove.

**Problem 9.3 — Reconstruct the recruited effect.** Use the standardized program and comparison risks to calculate the risk difference and risk ratio. Interpret both without converting one into the other.

**Problem 9.4 — Transport to the target.** Explain how the target estimate is formed, why it differs from the recruited estimate, and what positivity and conditional transportability require.

**Problem 9.5 — Separate measurement from missingness.** Compare the blinded and self-report results. Rewrite: “Because every student self-reported, the self-report estimate validates the primary result and solves attrition.”

**Problem 9.6 — Bound and synthesize.** Interpret the extreme missing-outcome bounds, then write a one-paragraph claim that includes the randomized result, missingness, measurement, target, scale, and synthetic boundary.

## Fully worked solutions

**Solution 9.1. Answer:** In population  $P$ , the two marginal risks are  $E[Y(1)|P]$  and  $E[Y(0)|P]$ . Their difference is  $E[Y(1)-Y(0)|P]$ ; their ratio is  $E[Y(1)|P]/E[Y(0)|P]$ . Use  $P=S=1$  for recruited students and  $P=\text{target}$  for the external composition. These quantities average potential-outcome risks over a named population. A conditional log-odds coefficient conditions on covariates and uses a noncollapsible scale; it is not either marginal contrast by definition. **Reason:** Outcome scale and averaging population are parts of the estimand. **Confidence:** Give 5 if both marginal risks appear before the contrast. **Revision:** Add the population and outcome time.

**Solution 9.2. Answer:** Blinded outcomes are observed for 88.0% of program students and 87.3% of comparison students, a 0.7-point difference, but site patterns vary. The analysis assumes  $R$  is independent of the potential blinded outcome within assignment, site, and baseline distress, with observed outcomes under both assignments in every weighted cell. Similar overall rates cannot prove independence from missing  $Y$  or rule out offsetting cell-specific selection; they also say nothing

about measurement validity. **Reason:** Randomization governs A, not R. **Confidence:** Give 5 if the statement conditions on assignment and X and refuses to infer from rate similarity alone. **Revision:** Name the missing potential outcome explicitly.

**Solution 9.3. Answer:**  $0.457 - 0.320 = 0.138$ . Dividing gives  $0.457/0.320 = 1.431$ . The difference is an estimated 13.8-percentage-point increase in improvement probability; the ratio says the program risk is about 1.431 times the comparison risk. **Reason:** Both derive from the same two standardized marginal risks but answer additive and relative questions. **Confidence:** Give 5 if the ratio is not described as percentage points. **Revision:** Restore the two risks and units.

**Solution 9.4. Answer:** Compute the two observed risks in every site-by-distress cell and average them using target shares rather than recruited shares. This gives target risk difference 0.166 rather than 0.138 because the target emphasizes cells with different baseline risks and additive effects. Positivity requires target-relevant cells to be represented under both assignments with observed outcomes; conditional transportability requires the cell-specific potential-outcome risks or effects to apply across recruitment status, with aligned versions and no omitted modifier that changes distribution. **Reason:** Reweighting changes composition but cannot manufacture an absent cell or repair incompatible versions. **Confidence:** Give 5 if overlap and exchangeability remain separate. **Revision:** Name one unmeasured modifier that could defeat transport.

**Solution 9.5. Answer:** “The complete self-report outcome gives a recruited-sample standardized risk difference of 0.148, while the incomplete blinded assessment gives 0.138 under a conditional observation assumption. The measures use different response processes. Completeness improves the self-report analysis’s available n but neither validates its construct against the blinded outcome nor imputes missing primary assessments.” **Reason:** Measurement validity and missingness are distinct filters. **Confidence:** Give 5 if the rewrite does not call either measure truth. **Revision:** Add who assessed each outcome and whether assignment awareness could matter.

**Solution 9.6. Answer:** The extreme bounds  $[-0.006, 0.241]$  assign missing outcomes in opposite worst and best directions. They show that the observed file alone permits both a null/harmful and a beneficial risk difference; they are not a confidence interval or a claim that extremes are plausible. A calibrated synthesis is: “In this synthetic randomized replication, the recruited-sample standardized blinded-outcome risk difference was 0.138 and risk ratio 1.431 under conditional observation assumptions. Reweighting to the specified target gave 0.166 and 1.585 under additional transportability assumptions. A complete self-report measure produced a separate contrast, while unrestricted missing-outcome bounds included zero. No result describes real students.” **Reason:** The conclusion keeps design, selection, measure, population, scale, and uncertainty visible. **Confidence:** Give 5 if no single estimate is allowed to erase the others’ assumptions. **Revision:** State the next sensitivity or replication that would most reduce decision uncertainty.

## Unit 10 Sensitivity, Triangulation, and Reproducible Causal Analysis

One estimate rarely finishes a causal argument. A result can survive a second regression because both regressions inherit the same unmeasured confounding. A script can reproduce every decimal while the intervention remains ill defined. A large study can be too small for its assignment unit. This final unit turns those limits into a workflow: quantify selected departures from assumptions, compare evidence whose vulnerabilities differ, preserve a computational audit trail, and plan the next design around a decision-relevant effect.

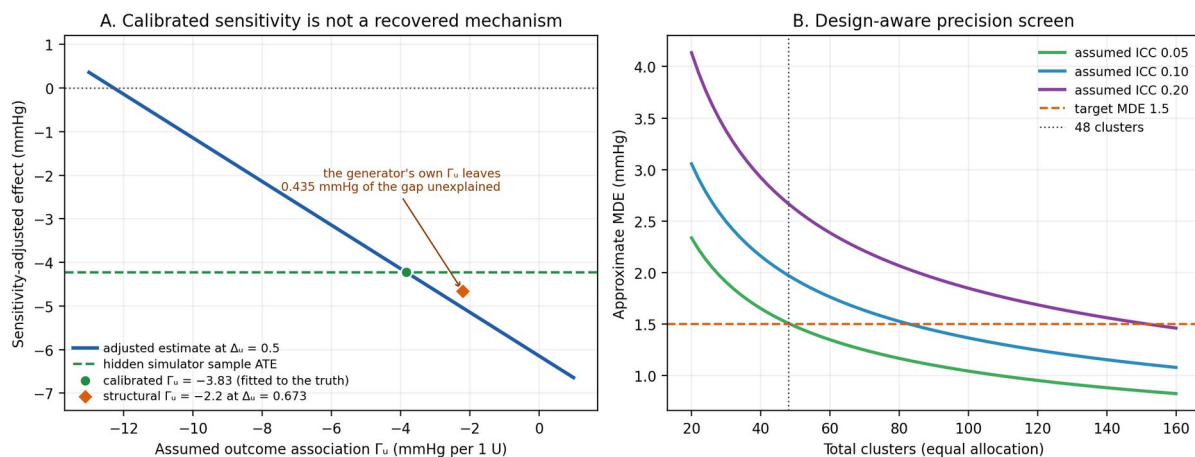


Figure 10. Calibrated sensitivity and precision planning: Panel A shows how the Unit 5 outcome-regression estimate changes across assumed residual-confounder outcome associations when the standardized confounder mean difference is fixed at 0.5, marking the hidden simulator effect, the calibrated association that reaches it, and the generator's own structural coefficient of -2.2 paired with its replayed adjusted gap of 0.673, which together still leave 0.435 mmHg unexplained; Panel B shows approximate minimum detectable effects across total cluster counts for assumed intraclass correlations of 0.05, 0.10, and 0.20, with 48 clusters and a 1.5-mmHg target marked.

### Return to the Unit 5 observational question

The voluntary hypertension-coaching case estimated the eligible-cohort average effect on follow-up systolic blood pressure. Outcome-regression standardization reported -6.142 mmHg, SE 0.268, interval [-6.668, -5.616]. IPW gave a nearby answer and measured balance was strong. Yet the negative-control association was 0.309, warning that measured adjustment had not made the participation groups exchangeable.

That warning did not say how much residual confounding remained, in which direction it operated, or what the causal estimate should be. Sensitivity analysis must therefore add explicit quantities rather than silently treating a diagnostic as a correction.

## Identification assumptions for sensitivity

### A two-parameter bias function

**Orientation.** Sensitivity analysis does not test an assumption. It states what an unmeasured confounder would have to look like for a conclusion to change, and then asks whether anyone would defend those values. Both parameters come from outside the data, so a table of them is an argument to be judged rather than a result to be reported.

Imagine an omitted baseline factor  $U$ , scaled to one standard-deviation units after the measured covariates  $X$  have been considered. Define:

- **Delta-U** as the residual mean difference in  $U$  between coaching participants and nonparticipants after adjustment for  $X$ ;
- **Gamma-U** as the conditional outcome difference in mmHg associated with a one-unit increase in  $U$ , holding participation and  $X$  fixed.

For this deliberately simple exercise, assume the omitted-confounding bias is

$$B = \text{Delta-U} \times \text{Gamma-U}$$

and define the sensitivity-adjusted contrast as the reported contrast minus  $B$ . This is a bias function, not a newly identified estimator. Its usefulness comes from exposing which departures are being contemplated.

Fix Delta-U at 0.500 and vary Gamma-U:

| Delta-U | Gamma-U | Assumed bias B | Sensitivity-adjusted estimate | Shifted 95% interval |
|---------|---------|----------------|-------------------------------|----------------------|
| 0.500   | 0.000   | 0.000          | -6.142                        | [-6.668, -5.616]     |
| 0.500   | -2.000  | -1.000         | -5.142                        | [-5.668, -4.616]     |
| 0.500   | -3.000  | -1.500         | -4.642                        | [-5.168, -4.116]     |
| 0.500   | -6.000  | -3.000         | -3.142                        | [-3.668, -2.616]     |

When Delta-U=0.500 and Gamma-U=-3.000, the assumed bias is -1.500 mmHg. Subtracting it moves the estimate to -4.642, with the mechanically shifted interval [-5.168, -4.116]. The interval is conditional on the two sensitivity parameters. It does not include uncertainty about Delta-U or Gamma-U, and its shift does not validate the linear bias model.

The table is not a menu from which to select the answer one prefers. Plausible values should be argued from subject-matter knowledge, validation data, negative controls, or external studies. A two-dimensional grid would normally vary both parameters. Fixing Delta-U at 0.500 keeps the arithmetic visible for this first exercise.

### Hidden calibration, not analyst evidence

The simulator's hidden sample ATE is -4.227 mmHg. At Delta-U=0.500, Gamma-U would need to be -3.830 for this bias function to land on that hidden value; it would need to be -12.284 to move the estimate to zero. The first number calibrates this bias function to the synthetic realization rather than explaining it, and the second locates a tipping point. Neither is observable in a real study.

Unit 5 disclosed that measured adjustment removed offsetting positive confounding while latent engagement continued to pull the contrast downward. The sensitivity analysis now encodes a family of departures without pretending to recover engagement. The negative control detected incompatibility with a simple no-residual-confounding story, but it could not calculate Delta-U, Gamma-U, their product, or the causal answer.

### A calibrated parameter is not a recovered mechanism

The value Gamma-U=-3.830 is calibrated, not estimated. It is the number that forces this bias function, at Delta-U=0.500, to land exactly on the hidden sample average treatment effect, so by construction it always lands there. Separating that from the generator's own structure is worth doing once, in the open, because the distinction is the main limitation of two-parameter sensitivity analysis.

| Quantity                                                                                         | Value     |
|--------------------------------------------------------------------------------------------------|-----------|
| Generator's coefficient on latent engagement (structural Gamma-U)                                | -2.200000 |
| Replayed standardized engagement gap between arms after measured adjustment (structural Delta-U) | 0.672803  |
| Implied structural bias contribution                                                             | -1.480166 |
| Reported outcome-regression estimate                                                             | -6.141777 |
| Estimate after removing the structural contribution                                              | -4.661611 |
| Hidden sample average treatment effect                                                           | -4.226561 |
| Remaining discrepancy                                                                            | -0.435050 |

Substituting the generator's true parameters does not reproduce the hidden answer. It leaves 0.435 mmHg unexplained, because the gap between the reported estimate and the hidden effect also contains finite-sample variation and the fitted model's approximation of the generator, not confounding alone. The calibrated value absorbs all three into one number, and it therefore overstates how strong the confounder-outcome association would have to be.

State the conclusion in the defensible direction: a sensitivity analysis describes how large an omitted-bias term would have to be before a conclusion changes, and it does not identify the mechanism that produced the data. An analyst who reports the calibrated number as an estimate of the confounder's effect has quietly converted an assumption into a finding.

Verification note. The structural row is obtained by replaying the Unit 5 generator, recovering each participant's latent engagement, and standardizing the program-minus-comparison engagement difference with the same outcome model used for the effect. The replayed data reproduce the published Unit 5 file to within 5e-09, and these hidden values are disclosed for teaching audit only.

**Pause and Decide.** Which claim is more defensible: “the effect is robust to unmeasured confounding” or “the conclusion remains negative under the stated parameter combinations”? Record the parameter range, the scientific argument for it, and one departure the two-parameter model does not cover.

## Diagnostics and falsification across cases

### Triangulation is not a pile of analyses

Triangulation becomes informative when evidence bearing on a compatible question has meaningfully different bias structures. Agreement between Unit 5's outcome-regression and IPW estimates checks implementation and some model dependence, but both use the same participants, measured covariates, outcome, and conditional-exchangeability assumption. That is useful robustness work, not independent triangulation.

Cross-case lessons suggest a program of studies:

| Evidence setting          | Vulnerable causal step                              | Existing stress test                                          | Design that would change the bias structure                       |
|---------------------------|-----------------------------------------------------|---------------------------------------------------------------|-------------------------------------------------------------------|
| Unit 5 voluntary coaching | Unmeasured participation–outcome confounding        | Negative control, overlap, balance, explicit bias parameters  | Randomized rollout or a defensible encouragement                  |
| Unit 6 policy timing      | Parallel trends and group-specific co-interventions | Pre-period placebo, differential trend, comparison-series ITS | Later-adopting jurisdictions or a separately reviewed event study |
| Unit 7 cutoff             | Continuity, manipulation, and locality              | Density, covariate continuity, placebo cutoff, bandwidths     | Replication at another cutoff or institution                      |
| Unit 7 encouragement      | Exclusion, monotonicity, and complier locality      | First stage and design audit                                  | A different encouragement or direct randomization of receipt      |
| Unit 9 multisite trial    | Outcome observation, measurement, and transport     | Bounds, alternate measure, target weights                     | Lower-loss replication in the target population                   |

These estimates should not be averaged merely because they appear in one book. They concern different interventions, outcomes, populations, and estimands. Even studies of the same intervention should be combined only after checking version, outcome, time, population, and effect-scale compatibility. Triangulation asks whether an explanation must survive evidence that would fail differently. It does not make incompatible targets interchangeable.

A disagreement is not automatic failure. It may reveal effect heterogeneity, different intervention versions, measurement changes, or one design's bias. The useful response is to map the discrepancy to the estimands and vulnerable steps, then select a study that distinguishes those explanations.

### A reproducibility ladder

The word *reproducible* is often asked to carry several different promises. Keep them separate:

1. **Rerun.** The same code on the same data produces the same artifacts.

2. **Independent recomputation.** A separately written program reproduces the quantities from the same data. This resource uses NumPy, standard-library Python, and base R paths.
3. **Data-generation audit.** A seeded generator, dictionary, provenance note, expected-results record, and checksums preserve the synthetic input and its lineage.
4. **Direct replication.** A new sample uses a closely aligned intervention, outcome, population, and analysis.
5. **Conceptual replication.** A different operationalization or design tests the causal explanation under another bias structure.

Passing levels 1–3 makes arithmetic and provenance inspectable. It does not prove randomization occurred, parallel trends held, an IV exclusion restriction was true, a mediator was unconfounded, missingness was benign, or the target population was represented. Replication adds new empirical information; rerunning a checksum does not.

### The minimum audit trail

For each quantitative claim, preserve the scientific question, estimand, design and timing, input data and dictionary, analysis source, software environment, expected result, and a statement of assumptions and exclusions. Record exploratory choices, deviations, failed diagnostics, and changes to the outcome or population. A polished final model without its decision history is reproducible only in the narrowest computational sense.

This book’s three computation paths are intentionally asymmetric. The primary path creates the result record; the standard-library path is written without NumPy; base R independently reconstructs the same quantities. Continuous integration checks each path, the rendered numbers, figure alternatives, metadata, page structure, and checksums. Agreement is evidence against certain coding errors. It is not a vote that turns an assumption into a fact.

## Design-aware precision planning

**Orientation.** This section plans a study that does not exist, so every input is an assumption. The intraclass correlation dominates the rest: across the values screened here, the clusters required for the same detectable effect run from 16 to 152.

“We will recruit 2,400 people” is not a precision plan when 48 clusters are assigned. The assignment unit, cluster size, intraclass correlation, outcome scale, analysis, attrition, and decision-relevant effect all matter.

Consider a hypothetical future cluster-randomized coaching trial with 50 participants per cluster, equal allocation, a two-sided alpha of 0.05, and 80% power. Use a planning standard deviation of 7.1 mmHg and ask what minimum detectable effect (MDE) follows from different *assumed* ICCs. These ICCs are not estimated from Unit 4 or any other case.

For  $J$  total clusters, cluster size  $m$ , and ICC  $\rho$ , use the screening design effect  $1+(m-1)\rho$  and effective sample size  $Jm/\text{design effect}$ . Under a normal approximation,

$$\text{MDE} = 2(z_{.975} + z_{.80}) \times \text{SD} / \sqrt{\text{effective sample size}}.$$

At 48 clusters:

| Assumed ICC                       | Design effect | Effective sample size | Approximate MDE | Even clusters needed for MDE $\leq 1.5$ |
|-----------------------------------|---------------|-----------------------|-----------------|-----------------------------------------|
| 0.00 (individual-level reference) | 1.000         | 2400.000              | 0.812           | 16                                      |
| 0.05                              | 3.450         | 695.652               | 1.508           | 50                                      |
| 0.10                              | 5.900         | 406.780               | 1.972           | 84                                      |
| 0.20                              | 10.800        | 222.222               | 2.669           | 152                                     |

Ignoring clustering produces an MDE of 0.812 mmHg. With ICC 0.10, the design effect is 5.900, effective sample size is about 406.780, and MDE is 1.972. To make the screening MDE no larger than 1.5 under that ICC requires 84 total clusters, rounded up to preserve equal allocation.

This table is a planning sensitivity analysis, not a power certificate. It uses normal critical values and equal cluster sizes, and omits cluster degrees of freedom, attrition, treatment nonreceipt, covariate-adjustment gains, uncertainty in SD and ICC, multiplicity, and implementation variation. A final protocol should simulate the prespecified assignment, observation, and analysis—including realistic cluster sizes and loss—over justified planning distributions.

Precision cannot repair identification. A large, tightly estimated biased contrast remains biased. Nor should an MDE be chosen by asking only what the budget can detect. Begin with the smallest effect that would change a decision, then assess whether the design can estimate it with adequate precision and credible identification.

## Claim Audit

**Supports:** reproducible algebraic sensitivity results for explicitly fixed Delta-U and Gamma-U; hidden-simulator calibration clearly fenced from analyst evidence; a cross-case map of distinct causal vulnerabilities; a reproducibility ladder that separates computation from replication; and transparent screening MDEs under stated cluster-design assumptions.

**Does not establish:** the true amount or direction of Unit 5 residual confounding; that the negative control identifies a correction; that unlike estimands can be pooled; that computational agreement proves identification; that the planning ICCs describe a real population; or that the normal-approximation MDEs replace protocol-specific simulation.

**Strongest defensible conclusion:** “In the synthetic Unit 5 case, the standardized estimate changes in a transparent way under stated two-parameter residual-confounding scenarios; the negative control motivates but does not calibrate those scenarios. Across the book, stronger evidence would combine compatible questions studied with designs whose vulnerable steps differ. Independent computation makes the arithmetic auditable, while future precision must be planned around assignment units, plausible correlation, missingness, analysis, and a decision-relevant effect.”

## Study in the Larger Evidence Program

The next study should be chosen by the uncertainty most likely to change a decision. If unmeasured participation confounding dominates, randomize access or a defensible encouragement. If transport dominates, recruit the target or measure richer effect modifiers. If outcome observation dominates,

invest in follow-up and prespecify departures. If precision dominates, add independent assignment units rather than only more people inside existing clusters.

Triangulation is strongest when the studies share enough of the causal question to be compared and differ enough in vulnerability to be informative. That balance requires design work before data collection, not a post hoc list of analyses after a preferred estimate appears.

## Six principal problems

For each, write your answer, reason, confidence from 1 to 5, and one revision cue before reading the solutions.

**Problem 10.1 — Calculate a sensitivity scenario.** Starting from -6.142 with interval [-6.668, -5.616], use Delta-U=0.500 and Gamma-U=-3.000. Calculate B, the sensitivity-adjusted estimate, and the shifted interval. State what remains conditional.

**Problem 10.2 — Use a negative control without overclaiming.** Explain what the 0.309 negative-control association contributes, why it cannot supply Delta-U or Gamma-U, and why the hidden value -4.227 cannot be used in a real analysis.

**Problem 10.3 — Audit triangulation.** Decide which is stronger triangulation for the Unit 5 question: another propensity model on the same file, or a randomized rollout with aligned intervention and outcome. Explain why the book's unlike unit estimates should not be averaged.

**Problem 10.4 — Separate reproducibility and replication.** Classify: rerun the same notebook; reproduce the number in base R; regenerate the seeded CSV; repeat the study in new clinics; test a randomized access design. State what the first three cannot establish.

**Problem 10.5 — Plan cluster-aware precision.** For 48 clusters of 50, planning SD 7.1, and assumed ICC 0.10, calculate the design effect, effective sample size, and MDE. Compare it with the ICC-zero reference and state two omissions.

**Problem 10.6 — Close the claim and choose the next study.** Rewrite: "Multiple models reproduce the effect, so the causal conclusion is robust and the future trial is adequately powered." Include sensitivity parameters, triangulation, computational scope, and a design-aware next step.

## Fully worked solutions

**Solution 10.1. Answer:**  $B = 0.500 \times (-3.000) = -1.500$ . The adjusted value is  $-6.142 - (-1.500) = -4.642$ . Shifting both endpoints by +1.500 gives [-5.168, -4.116]. The result is conditional on the chosen parameters, the product bias function, and the original analysis; it does not estimate uncertainty in the parameters. **Reason:** Sensitivity analysis exposes a departure rather than identifying it. **Confidence:** Give 5 if the sign of B is checked before subtraction. **Revision:** Write one scientific source for each parameter.

**Solution 10.2. Answer:** The negative-control association is evidence against an uncomplicated claim that measured adjustment eliminated shared causes, provided the negative-control assumptions are credible. It does not map uniquely to residual imbalance in U, U's outcome association, their sign, or their product. The hidden -4.227 value is known only because this case is simulated; using it to choose -3.830 would reverse the scientific task by calibrating assumptions to an unavailable answer. **Reason:** A diagnostic can reveal incompatibility without identifying the correction. **Confidence:**

Give 5 if detection and quantification remain separate. **Revision:** Name one alternative explanation for the negative control.

**Solution 10.3. Answer:** The randomized rollout is stronger triangulation if its intervention, outcome, time, and population align well enough, because assignment changes the main confounding vulnerability. Another propensity model can assess implementation and model dependence but inherits the same data and exchangeability problem. Unit estimates should not be averaged because they concern different treatments, outcomes, populations, times, and estimands. **Reason:** Triangulation needs compatible questions and different bias structures. **Confidence:** Give 5 if neither “different” nor “similar” alone is treated as sufficient. **Revision:** Map one disagreement to effect heterogeneity versus bias.

**Solution 10.4. Answer:** The notebook is a rerun; base R is independent recomputation; regenerating the seeded CSV is a data-generation audit; new clinics are a direct replication if the protocol aligns; randomized access is a conceptual replication with a changed bias structure. The first three check computation and provenance on the same synthetic evidence. They cannot establish exchangeability, measurement validity, transportability, or performance in a new sample. **Reason:** Reproducible arithmetic and new empirical evidence are different achievements. **Confidence:** Give 5 if the base-R path is not called a new study. **Revision:** Name the metadata needed to reproduce the environment.

**Solution 10.5. Answer:** The design effect is  $1+(50-1)(0.10)=5.900$ . Effective  $n$  is  $48 \times 50 / 5.900 = 406.780$ . The screening MDE is 1.972 mmHg, compared with 0.812 when ICC is set to zero. Omissions include cluster-based degrees of freedom, unequal sizes, attrition, uncertain planning inputs, and adjustment gains. **Reason:** Correlated observations contribute less independent information than 2,400 independent assignments. **Confidence:** Give 5 if ICC is labeled assumed. **Revision:** Simulate the planned analysis under a distribution of cluster sizes.

**Solution 10.6. Answer:** “Independent code reproduces the calculations on this synthetic dataset, but does not verify the identification assumptions. The Unit 5 conclusion remains negative under the stated Delta-U/Gamma-U scenarios; the negative control warns of residual confounding without calculating it. A randomized rollout with aligned intervention and outcome would change the principal bias structure. Its precision should be simulated using plausible ICC, cluster-size, attrition, and analysis assumptions and a decision-relevant target effect.” **Reason:** The rewrite gives computation, sensitivity, triangulation, and planning distinct jobs. **Confidence:** Give 5 if “robust” and “powered” are replaced by conditional statements. **Revision:** Add the population, outcome time, and smallest decision-relevant effect.

## Assumptions index

Every claim in this book rests on assumptions of three kinds. Design assumptions are created by the way a study is run. Identification assumptions are what turn observed contrasts into causal quantities, and most of them cannot be tested with the data at hand. Modelling inputs are numbers supplied by the analyst. Each entry below states what this book can actually check, which is usually less than readers expect.

**Randomization.** Assignment is independent of potential outcomes by design. Kind of assumption: design. What this book can actually check: Allocation counts and baseline balance; balance cannot prove independence. Units: 1, 4, 7, 9.

**Consistency and a well-defined intervention.** A unit's observed outcome is its potential outcome under the condition received. Kind of assumption: identification. What this book can actually check: Description of versions and protocol; not testable from the file. Units: All.

**No interference.** One unit's outcome is unaffected by other units' assignments. Kind of assumption: identification. What this book can actually check: Design reasoning; spillover argument in the clustered and municipal cases. Units: 4, 6.

**Positivity and overlap.** Both conditions occur at every covariate stratum used. Kind of assumption: identification. What this book can actually check: Propensity distributions and weight diagnostics. Units: 5, 9.

**Conditional exchangeability.** No unmeasured confounding within adjustment strata. Kind of assumption: identification. What this book can actually check: Negative controls and sensitivity analysis; never provable from the data. Units: 3, 5, 10.

**Transportability.** Conditional effects carry to the target given the covariates used. Kind of assumption: identification. What this book can actually check: Composition tables and an explicitly changed estimand; not testable here. Units: 2, 9.

**Missing at random given the conditioning set.** Observation does not depend on the missing outcome within cells. Kind of assumption: identification. What this book can actually check: Observation-rate tables, plus bounds that assume nothing. Units: 9.

**Non-differential measurement.** Error rates do not differ by arm. Kind of assumption: identification. What this book can actually check: Blinded-versus-self-report comparison and its expected-value audit. Units: 9.

**Parallel untreated trends.** Groups would have changed alike without the policy. Kind of assumption: identification. What this book can actually check: Placebo adoption date and pre-period slope; both can pass while the assumption fails. Units: 6.

**No anticipation.** Behavior does not change before adoption. Kind of assumption: identification. What this book can actually check: Inspection of the pre-period. Units: 6.

**Stable interruption.** Only the named change occurs at the interruption. Kind of assumption: identification. What this book can actually check: The shared regional hotline is discussed rather than assumed away. Units: 6.

**Continuity at the cutoff.** Without the program, outcomes vary smoothly through the threshold. Kind of assumption: identification. What this book can actually check: Bandwidth sensitivity and a density or bunching count. Units: 7.

**No manipulation of the running variable.** Units cannot sort precisely across the threshold. Kind of assumption: identification. What this book can actually check: Count of observations just each side of the cutoff, in the spirit of a density test. Units: 7.

**Instrument relevance.** Encouragement actually shifts receipt. Kind of assumption: testable. What this book can actually check: The first-stage estimate and its precision. Units: 7.

**Exclusion restriction.** Encouragement affects outcomes only through receipt. Kind of assumption: identification. What this book can actually check: Design features and argument only. Units: 7.

**Monotonicity.** No unit is moved away from treatment by encouragement. Kind of assumption: identification. What this book can actually check: Implied stratum shares; a negative share would be a contradiction. Units: 7.

**Sequential ignorability and cross-world conditions.** No confounding of treatment-mediator, mediator-outcome, and their cross-world counterpart. Kind of assumption: identification. What this book can actually check: An explicit refusal to report an identified indirect effect, with the failure quantified. Units: 8.

**Stated sensitivity parameters.** The assumed confounder gap between arms and its outcome association. Kind of assumption: modeling input. What this book can actually check: Provenance of the two numbers and comparison with a structural replay. Units: 10.

**Planning inputs.** Assumed standard deviation, intraclass correlation, cluster size, allocation. Kind of assumption: modeling input. What this book can actually check: A scenario range rather than a single number. Units: 10.

## Scope and deliberate omissions

This primer teaches a repeatable path from question to estimand, design, assumptions, estimation, sensitivity, and claim. It does not attempt to be an encyclopedia of causal inference. Important methods left for advanced study include longitudinal g-methods for time-varying confounding, survival and competing-risk estimands, interference and network spillovers, modern staggered-adoption and event-study estimators, formal mediation sensitivity, matching and doubly robust or targeted-learning estimators, Bayesian causal models, synthetic controls and regression kink designs, multiple imputation and MNAR selection or pattern-mixture models, measurement-error and latent-variable causal models, and formal meta-analysis.

Omission is not dismissal. Each topic requires its own intervention, identification assumptions, diagnostics, and uncertainty treatment; compressing one into a paragraph would recreate the model-first habits this book is meant to correct. Matching, Unit 4 ICC/design-effect and cluster-robust diagnostics, and a genuine Unit 6 event-study remain separate hardening decisions rather than implied parts of the ten-unit manuscript.

The practical boundary is equally important. The cases use deliberately structured synthetic data, moderate-dimensional covariates, and one primary analysis at a time. Real studies may require data linkage, measurement validation, complex sampling, protocol deviations, privacy protection, stakeholder decisions, and regulatory or ethical review. Readers should use the workflow to recognize when specialist methods and collaborators are needed, then consult advanced sources rather than treating this book as a substitute for design-specific expertise.

## Reproduce and extend this book

All ten analysis datasets are plain-text CSV files and can be opened in any statistics package; identify the row count, measurement levels, exposure or assignment variable, timing, and unit before fitting a model. Unit 7 deliberately uses two separate files because its cutoff and encouragement caselets are different studies. Unit 10 adds no new participant sample: it reuses the verified Unit 5 case, records prospective planning assumptions, and contributes only the replayed latent-engagement column, an eleventh CSV that exists so the structural audit in that unit can be rechecked independently. The causal questions, estimands, diagrams, target weights, panel models, sandwich analyses, missingness bounds, transport standardization, sensitivity function, and precision screen are provided as source code rather than treating any software menu as an identification procedure. Each completed unit has a primary NumPy analysis, independent standard-library Python verification, and an independent base-R cross-check; data-generating units also have deterministic generators, and every one of those paths must pass before a change to the book is accepted. The eleven base-R paths were run on the author's Ubuntu workstation with R 4.1.2 (2021-11-01) and again on a current R release in the public repository's continuous integration. No claim is made here about output from any menu-driven package, on any platform. Numerical agreement validates calculations on these CSVs and the Unit 10 reuse and planning record, not intervention definitions, real-world exchangeability, continuity at a cutoff, IV exclusion, parallel trends, mediation assumptions, missingness, transport, measurement, sensitivity parameters, or planning inputs.

The companion materials are public. The directory `open-materials/causal-reasoning-with-statistics` at [github.com/nicholaskarlson/data](https://github.com/nicholaskarlson/data) holds the ten analysis datasets and the derived Unit 10 replay column, a dictionary and provenance note for each dataset, one verified record per case together with a record for the two hidden-simulator addenda, and the three computation paths. From that directory, `python3 run_all.py` regenerates the data, recomputes every record, and cross-checks the result with a standard-library implementation, and `Rscript run_all.R` repeats the checks in base R. Both leave the files exactly as downloaded. The state described here is commit `0cd74cbb1226dfca13a959643aa4b74eda861428`; `SOURCE.json` records the authoring commit the export was taken from, and `SHA256SUMS` lists every file with its hash, so a reader can confirm that the files they have are the files this book was written against.

Re-running a generator changes its packaged dataset only if the deterministic environment or the script changes; compare the recorded checksum first. `SHA256SUMS`, the dictionaries, the provenance notes, the result records and the source files identify their inputs, so a disagreement can always be traced to a named file rather than to an unspecified difference in setup.

## Interval conventions, rounding, and typography

Units 1 through 5 form 95 percent intervals with the rounded multiplier 1.96 and say so in each verified-results record. Units 6 through 10 use the exact standard normal quantile 1.959963985, and the clustered, time-series, and discontinuity cases use a  $t$  multiplier on the degrees of freedom named beside their tables. The difference is deliberate rather than an oversight: the early units keep the arithmetic checkable by hand, and the later units report the multiplier their code actually applied. Editing the earlier records to say 1.959963985 would make them describe something other than the computation they document.

Displayed values are rounded for reading while stored values carry six decimals. Recomputing a published interval from the rounded numbers printed in the text will therefore sometimes differ in the final digit, and the published interval is the one computed from unrounded inputs.

Negative numbers inside interval brackets are set with a typographic minus sign so that a line break cannot separate the sign from its digits. Any validator that compares rendered text against stored numbers should normalize that character to a hyphen before comparing.

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The nine sources above name methods that the units either use or deliberately decline to use: the density test behind the manipulation check in Unit 7, the cluster-robust and autocorrelation-consistent variance estimators in Units 4 and 6, partial identification behind the bounds in Unit 9, matching as the alternative to weighting discussed in Unit 5, the risk-ratio-scale E-value as a relative of the bias function in Unit 10, and the staggered-adoption difference-in-differences literature that Unit 6's single common adoption date does not require but that a reader extending the case would need. Naming them is not a claim that this book implements them.

## Where to find the companion

The companion to this book is public and free. Everything described below is at <https://github.com/nicholaskarlson/data>, in the directory open-materials/causal-reasoning-with-statistics.

Ten synthetic datasets as plain-text CSV files, one per case, with a data dictionary and a provenance note for each. An eleventh CSV holds the replayed latent-engagement column that Unit 10 audits.

One verified result record per case as readable JSON, plus a record for the two hidden-simulator addenda in Units 9 and 10. These files hold every published estimate, standard error and interval at six decimals.

Three independent computation paths: the primary analysis in Python with NumPy, a second implementation in the Python standard library that imports nothing, and a third in base R that needs no contributed packages. Agreement between them at a tolerance of two parts in a million is what stands behind the numbers you have been reading.

The seeded generators that produce the datasets, so you can confirm the data were not adjusted after the fact, and a checksum file listing every file with its SHA-256 hash.

To use it: download the repository as a ZIP from the green Code button, unzip it, open a terminal in the companion directory, and run `python3 run_all.py` and then, if you have R installed, `Rscript run_all.R`. Python 3.10 or newer with NumPy is the only requirement for the first command; `pip install -r requirements.txt` installs it. Both commands print a single summary line when everything agrees, and name the offending file when anything does not.

The licence is the same one that covers this book. The data, dictionaries, records and documentation are CC BY 4.0, and the scripts are additionally available under the MIT License, so code can be reused without the attribution mechanics that CC BY imposes on software. Both are stated in the companion's own licence file.

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